```
module ieee802-ethernet-interface-half-duplex {
  yang-version 1.1;
  namespace
    "urn:ieee:std:802.3:yang:ieee802-ethernet-interface-half-duplex";
  prefix ieee802-eth-half-duplex;

  import ietf-yang-types {
    prefix yang;
    reference
      "IETF RFC 6991";
  }
  import ietf-interfaces {
    prefix if;
    reference
      "IETF RFC 8343";
  }
  import iana-if-type {
    prefix ianaift;
    reference
      "http://www.iana.org/assignments/yang-parameters/
       iana-if-type@2023-01-26.yang";
  }
  import ieee802-ethernet-interface {
    prefix ieee802-eth-if;
  }

  organization
    "IEEE Std 802.3 Ethernet Working Group
     Web URL: http://www.ieee802.org/3/";
  contact
    "Web URL: http://www.ieee802.org/3/";
  description
    "This module contains YANG definitions for ~~configuring Ethernet~~half
duplex
     Ethernet interfaces~~ that are deprecated, and are no longer~~
~~     widely used in the industry. The definitions are maintained~~
~~     for backwards compatibility purposes, but the general~~
~~     expectation is that this module is not anticipated to be~~
~~     widely implemented.";~~
, e.g., 10BASE-T1S.";

  revision 2024-~~07-17~~12-23 {
    description
      "Updates under IEEE Std 802.3.2-202x, Draft D3.~~0~~1";
    reference
      "IEEE Std 802.3-2022, unless dated explicitly";
  }

  feature dynamic-rate-control {
    status obsolete;
    description
      "This feature indicates that the device supports Ethernet
       interfaces lowering the average data rate of the MAC
       sublayer, with frame granularity, by using Rate Control to
       dynamically increase the inter-packet gap for some types of
       Ethernet interface.
       Only valid for Ethernet interfaces operating at speeds
       (data rates)above 1000 Mb/s.";
```

```
    reference
      "IEEE Std 802.3, 30.3.1.1.33 aRateControlAbility";
  }

  feature csma-cd {
    description
      "This feature indicates that the device supports Ethernet
       interfaces running at half-duplex using CSMA/CD.";
  }

  typedef dynamic-rate-control-type {
    type enumeration {
      enum disabled {
        description
          "Dynamic rate control is disabled";
      }
      enum sonet-oc192 {
        value 2;
        description
          "Dynamic rate control is enabled for a 10 Gb/s Ethernet
           interface to SONET/SDH OC192/STM64.";
      }
    }
    default "disabled";
    status obsolete;
    description
      "Allowed values for dynamic-rate-control.";
    reference
      "IEEE Std 802.3, 4.4.2 ipgStretchRatio and 30.3.1.1.34
       aRateControlStatus";
  }

  augment "/if:interfaces/if:interface/ieee802-eth-if:ethernet" {
    when "derived-from-or-self(../if:type, 'ianaift:ethernetCsmacd')
          and ieee802-eth-if:duplex = 'half'" {
      description
        "Applies to half-duplex Ethernet interfaces.";
    }
    status obsolete;
    description
      "Augment with Ethernet interface configuration parameters
       for half-duplex operation.";
    leaf dynamic-rate-control {
      if-feature "dynamic-rate-control";
      type dynamic-rate-control-type;
      status obsolete;
      description
        "Enables dynamic rate control and specifies what speed
         (data rate) the dynamic rate control is operating at.
         The value of this attribute is constrained by the MAC
         data rate and hardware support.
         The default value is implementation-dependent.";
      reference
        "IEEE Std 802.3, 30.3.1.1.34 aRateControlStatus";
    }
  }

  augment "/if:interfaces/if:interface/ieee802-eth-if:ethernet/"
```

```
        + "ieee802-eth-if:capabilities" {
    when "derived-from-or-self(../../if:type,
      "derived-from-or-self(../../if:type,
       'ianaift:ethernetCsmacd')
           and ../ieee802-eth-if:duplex = 'half'" {
      description
        "Applies to half-duplex Ethernet interfaces";
    }
    status obsolete;
    description
      "Augment with configuration capabilities for half-duplex
       Ethernet interface.";
    leaf dynamic-rate-control-supported {
      if-feature "dynamic-rate-control";
      type boolean;
      default "false";
      status obsolete;
      description
        "Indicates whether the Ethernet interface supports lowering
         the average data rate of the MAC sublayer, with frame
         granularity, by using Rate Control to dynamically increase
         the inter-packet gap.
         Only valid for Ethernet interfaces operating at speeds
         (data rates) above 1000 Mb/s.";
      reference
        "IEEE Std 802.3, 30.3.1.1.33 aRateControlAbility";
    }
  }

  augment "/if:interfaces/if:interface/ieee802-eth-if:ethernet/"
        + "ieee802-eth-if:statistics/ieee802-eth-if:frame" {
    when "derived-from-or-self(../../../if:type,
      "derived-from-or-self(../../../if:type,
       'ianaift:ethernetCsmacd') and
              ../../ieee802-eth-if:duplex = 'half'" {
      description
        "Applies to half-duplex Ethernet interfaces.";
    }
    description
      "Augment with statistics for half-duplex Ethernet interface.";
    container csma-cd {
      if-feature "csma-cd";
      description
        "Holds counters that are specific to CDMA/CD half-duplex
         operation of Ethernet interfaces.
         This counter does not increment on Ethernet interfaces
         operating at speeds (data rates) greater than 10 Mb/s, or
         on Ethernet interfaces operating in full-duplex mode.
         Discontinuities in the value of this counter can occur at
         re-initialization of the management system, and at other
         times as indicated by the value of the
         'discontinuity-time' leaf defined in the ietf-interfaces
         YANG module (IETF RFC 8343).";
      leaf in-errors-sqe-test {
        type yang:counter64;
        units "errors";
        description
          "A count of times that the SQE TEST ERROR is received on
```

```
       a particular interface. The SQE TEST ERROR is set in
       accordance with the rules for verification of the SQE
       detection mechanism in the PLS Carrier Sense Function as
       described in IEEE Std 802.3, 7.2.4.6.";
    reference
      "IEEE Std 802.3, 7.2.4.6, and 30.3.2.1.4 aSQETestErrors";
  }
  leaf out-frames-collision-single {
    type yang:counter64;
    units "frames";
    description
      "A count of frames that are involved in a single
       collision, and are subsequently transmitted
       successfully. A frame that is counted by an instance of
       this object is also counted by the corresponding
       instance of either 'out-unicast-frames',
       'out-broadcast-frames', or 'out-multicast-frames',
       and is not counted by the corresponding instance of the
       'out-frames-collision-multiple'.";
    reference
      "IEEE Std 802.3, 30.3.1.1.3 aSingleCollisionFrames";
  }
  leaf out-frames-collision-multiple {
    type yang:counter64;
    units "frames";
    description
      "A count of frames that are involved in multiple
       collisions, and are subsequently transmitted
       successfully. A frame that is counted by an instance of
       this object is also counted by the corresponding
       instance of either 'out-unicast-frames',
       'out-broadcast-frames', or 'out-multicast-frames', and
       is not counted by the corresponding instance of the
       'out-frames-collision-single'.";
    reference
      "IEEE Std 802.3, 30.3.1.1.4 aMultipleCollisionFrames";
  }
  leaf out-frames-deferred {
    type yang:counter64;
    units "frames";
    description
      "A count of frames for which the first transmission
       attempt on a particular Ethernet interface is delayed
       because the medium is busy.
       A deferred frame that is not subject to any number of
       collisions is not counted by an instance of
       'out-frames-collision-single' or
       'out-frames-collision-multiple' objects.";
    reference
      "IEEE Std 802.3, 30.3.1.1.9
       aFramesWithDeferredXmissions";
  }
  leaf out-frames-collisions-excessive {
    type yang:counter64;
    units "frames";
    description
      "A count of frames for which transmission on a particular
       Ethernet interface fails due to excessive collisions.";
```

```
      reference
        "IEEE Std 802.3, 30.3.1.1.11 aFramesAbortedDueToXSColls";
    }
    leaf out-collisions-late {
      type yang:counter64;
      units "collisions";
      description
        "The number of times that a collision is detected on a
         particular Ethernet interface later than one slotTime
         into the transmission of a packet.
         A (late) collision included in a count represented by an
         instance of this object is also considered as a
         (generic) collision for purposes of other
         collision-related statistics.";
      reference
        "IEEE Std 802.3, 30.3.1.1.10 aLateCollisions";
    }
    leaf out-errors-carrier-sense {
      type yang:counter64;
      units "errors";
      description
        "The number of times that the carrier sense condition was
         lost or never asserted when attempting to transmit a
         frame on a particular Ethernet interface.
         The count represented by an instance of this object is
         incremented at most once per transmission attempt, even
         if the carrier sense condition fluctuates during a
         transmission attempt.";
      reference
        "IEEE Std 802.3, 30.3.1.1.13 aCarrierSenseErrors";
    }
    list collision-histogram {
      key "collision-count";
      description
        "A collection of collision histograms for a particular
         interface.";
      reference
        "IEEE Std 802.3, 30.3.1.1.30 aCollisionFrames";
      leaf collision-count {
        type yang:counter64;
        units "collisions";
        description
          "The number of per-frame media collisions for which a
           particular collision histogram cell represents the
           frequency on a particular interface.";
      }
      leaf collision-count-frames {
        type yang:counter64;
        units "frames";
        description
          "A count of individual MAC frames for which the
           transmission (successful or otherwise) on a particular
           interface occurs after the frame has experienced
           exactly the number of collisions in the associated
           dot3CollCount object.
           For example, a frame which is transmitted on an
           interface after experiencing exactly 4 collisions
           would be indicated by incrementing only
```

```
              collision-count-frames object associated with the
              collision-count value of 4. No other instance of
              collision-count-frames would be incremented in this
              example.";
         }
       }
     }
   }
 }
module ieee802-ethernet-interface {
  yang-version 1.1;
  namespace "urn:ieee:std:802.3:yang:ieee802-ethernet-interface";
  prefix ieee802-eth-if;

  import ietf-yang-types {
    prefix yang;
    reference
      "IETF RFC 6991";
  }
  import ietf-interfacesiana-if-type {
    prefix ifianaift;
    reference
      "IETF RFC 8343";
  }
  import iana-if-type {
    prefix ianaift;
    reference
      "http://www.iana.org/assignments/yang-parameters/
       iana-if-type@2023-01-26.yang";
  }

  //
  import ietf-interfaces {
    prefix if;
    reference
      "IETF RFC 8343";
  }
  import ieee802-ethernet-mauphy-type {
  //  prefix ieee802-mauphy;
  //  reference
      //           "IEEE Std 802.3-2022";
  //}
}

  organization
    "IEEE Std 802.3 Ethernet Working Group
     Web URL: http://www.ieee802.org/3/";
  contact
    "Web URL: http://www.ieee802.org/3/";
  description
    "This module contains YANG definitions for configuring IEEE Std
     802.3 Ethernet Interfaces.
     In this YANG module, 'Ethernet interface' can be interpreted
     as referring to 'IEEE Std 802.3 compliant Ethernet
     interfaces'.";

  revision 2024-07-1712-23 {
    description
```

```
        "Updates under IEEE Std 802.3.2-202x, Draft D3.~~0~~1";
      reference
        "IEEE Std 802.3-2022 and IEEE Std 802.3.1-202X, unless dated
         explicitly";
    }

    feature ethernet-pfc {
      description
        "This device supports Ethernet priority flow-control.";
    }

    feature ethernet-pause {
      description
        "This device supports Ethernet PAUSE.";
    }

    typedef eth-if-speed-type {
      type decimal64 {
        fraction-digits 3;
      }
      units "Gb/s";
      status obsolete;
      description
        "Used to represent the configured, negotiated, or actual
         speed of an Ethernet interface in Gigabits per second
         (Gb/s), accurate to 3 decimal places (i.e., accurate to 1
         Mb/s).";
    }

    typedef duplex-type {
      type enumeration {
        enum full {
          description
            "Full duplex.";
        }
        enum half {
          description
            "Half duplex.";
        }
        enum unknown {
          description
            "Link is currently disconnected or initializing.";
        }
      }
      default "full";
      description
        "Used to represent the configured, negotiated, or actual
         duplex mode of an Ethernet interface.";
      reference
        "IEEE Std 802.3, 30.3.1.1.32, aDuplexStatus";
    }

    typedef pause-fc-direction-type {
      type enumeration {
        enum disabled {
          description
            "Flow-control disabled in both ingress and egress
             directions~~, i.e., PAUSE frames are not transmitted and~~.";
```

```
              ~~PAUSE frames received in the ingress direction are~~
              ~~discarded without processing.";~~
        }
        enum ingress-only {
          description
            "PAUSE frame based flow control is enabled in the ingress
             direction only~~, i.e., PAUSE frames may be transmitted to~~.";
             ~~reduce the ingress traffic flow, but PAUSE frames~~
             ~~received in the ingress direction are discarded without~~
             ~~reducing the egress traffic rate.";~~
        }
        enum egress-only {
          description
            "PAUSE frame based flow control is enabled in the egress
             direction only~~, i.e., PAUSE frames are not transmitted,~~.";
             ~~but PAUSE frames received in the ingress direction are~~
             ~~processed to reduce the egress traffic rate.";~~
        }
        enum bi-directional {
          description
            "PAUSE frame based flow control is enabled in both
             ingress and egress directions~~, i.e., PAUSE frames may~~.";
             ~~be transmitted to reduce the ingress traffic flow, and~~
             ~~PAUSE frames received on ingress are processed to~~
             ~~reduce the egress traffic rate.";~~
        }
        enum undefined {
          description
            "Link is currently disconnected or initializing.";
        }
      }
      description
        "Used to represent the configured, negotiated, or actual
         PAUSE frame-based flow control setting.";
      reference
        "IEEE Std 802.3.1, dot3PauseAdminMode and dot3PauseOperMode";
    }

    augment "/if:interfaces/if:interface" {
      when "derived-from-or-self(if:type, 'ianaift:ethernetCsmacd')" {
        description
          "Applies to all ~~P2P ~~Ethernet interfaces.";
      }
      description
        "Augment interface model with Ethernet interface
         specific configuration nodes.";
      container ethernet {
        description
          "Contains all Ethernet interface related configuration.";
        container auto-negotiation {
          presence "The presence of this container indicates that
                    auto-negotiation is supported on this Ethernet
                    interface.";
          description
            "Contains auto-negotiation transmission parameters

             This container contains a data node that allows the
             advertised duplex value in the negotiation to be
```

```
          restricted.

          If not specified then the default behavior for the
          duplex data node is to negotiate all available values for
          the particular type of Ethernet PHY associated with the
          interface~~.~~

~~          If auto-negotiation is enabled, and PAUSE frame based~~
~~          flow control has not been explicitly configured, then~~
~~          the default PAUSE frame based flow control capabilities~~
~~          that are negotiated allow for bi-directional or~~
~~          egress-only PAUSE frame based flow control.~~

~~          If auto-negotiation is enabled, and PAUSE frame based~~
~~          flow control has been explicitly configured, then the~~
~~          configuration settings restrict the values that may be~~
~~          negotiated. However, it should be noted that the~~
~~          protocol does not allow only egress PAUSE frame based~~
~~          flow control to be negotiated without also allowing~~
~~          bi-directional PAUSE frame based flow control.";~~
.";
        reference
          "IEEE Std 802.3, Clause 28 and Annexes 28A-D";
        leaf enable {
          type boolean;
          default "true";
          description
            "Controls whether auto-negotiation is enabled or
             disabled.
             For interface types that support auto-negotiation then
             it defaults to being enabled.

             For interface types that do not support
             auto-negotiation, the related configuration data is
             ignored.";
        }
        leaf negotiation-status {
          when "../enable = 'true'";
          type enumeration {
            enum in-progress {
              description
                "The auto-negotiation protocol is running and
                 negotiation is currently in-progress.";
            }
            enum complete {
              description
                "The auto-negotiation protocol has completed
                 successfully.";
            }
            enum failed {
              description
                "The auto-negotiation protocol has failed.";
            }
            enum unknown {
              description
                "The auto-negotiation status is not currently known,
                 this could be because it is still negotiating or the
                 protocol cannot run
```

```
                 (e.g., if no medium is present).";
            }
            enum no-negotiation {
              description
                "No auto-negotiation is executed.
                 The auto-negotation function is either not
                 supported on this interface or has not been
                 enabled.";
            }
          }
          config false;
          description
            "The status of the auto-negotiation protocol.";
          reference
            "IEEE 802.3, 30.6.1.1.4, aAutoNegAutoConfig";
        }
      }
      ~~/*~~
      leaf ~~mau~~phy-type {
        type identityref {
          base ieee802-~~mau:mau~~phy:phy-type;
        }
        config false;
        description
          "A value that uniquely identifies the IEEE 802.3 ~~MAU~~PHY type
           of the interface.";
        reference
          "IEEE Std 802.3, 30~~.????";~~.3.2.1.2 aPhyType";
      }
      leaf ~~mau~~pmd-type~~-list~~ {
        type identityref {
          base ieee802-~~mau:~~phy:pmd-type~~-list-bits~~;
        }
        config false;
        description
          "A value that uniquely identifies the ~~set of~~
           ~~possible~~ IEEE 802.3 ~~types that~~PMD type
           of the ~~MAU could be~~interface.";
        reference
          "IEEE Std 802.3, 30~~.????";~~.5.1.1.2 aMAUType";
      }
       ~~*/~~
      leaf duplex {
        type duplex-type;
        description
          "Operational duplex mode of the Ethernet interface.";
        reference
          "IEEE Std 802.3, 30.3.1.1.32 aDuplexStatus";
      }
      leaf speed {
        type eth-if-speed-type;
        units "Gb/s";
        status obsolete;
        description
          "Operational speed (data rate) of the Ethernet interface.
           The default value is implementation-dependent~~.";~~.
      ~~}~~
```

```
        container                 Supersceeded by speed in
/if:interfaces/if:interface";
        }
        /* deprecated flow-control {container */
        container flow-control {
          status deprecated;
          description
            "Holds the different types of Ethernet PAUSE frame based
             flow control that can be enabled.";.
             Supersceeded by new container - pause.";
          container pause {
            if-feature "ethernet-pause";
            description
              "IEEE Std 802.3 PAUSE frame based PAUSE frame based
               flow control.";
            reference
              "IEEE Std 802.3, Annex 31B";
            leaf direction {
              type pause-fc-direction-type;
              description
                "Indicates which direction PAUSE frame based flow
                 control is enabled in, or whether it is disabled.
                 The default flow-control settings are vendor
                 specific. If auto-negotiation is enabled, then PAUSE
                 based flow-control is negotiated by default.
                 The default value is implementation-dependent.";
            }
            container statistics {
              config false;
              description
                "Contains the number of PAUSE frames received or
                 transmitted.

                 Discontinuities in the values of counters in
                 this container can occur at re-initialization of
                 the management system, and at other times as
                 indicated by the value of the 'discontinuity-time'
                 leaf defined in the ietf-interfaces YANG module
                 (IETF RFC 8343).";
              leaf in-frames-pause {
                type yang:counter64;
                units "frames";
                description
                  "A count of PAUSE MAC Control frames transmitted on
                   this Ethernet interface.";
                reference
                  "IEEE Std 802.3, 30.3.4.3
                   aPAUSEMACCtrlFramesReceived";
              }
              leaf out-frames-pause {
                type yang:counter64;
                units "frames";
                description
                  "A count of PAUSE MAC Control frames transmitted on
                   this Ethernet interface.";
                reference
                  "IEEE Std 802.3, 30.3.4.2
                   aPAUSEMACCtrlFramesTransmitted";
```

```
        }
      }
    }
    container pfc {
      if-feature "ethernet-pfc";
      description
        "IEEE Std 802.3 Priority-based flow control.";
      reference
        "IEEE Std 802.3, Annex 31D";
      leaf enable {
        type boolean;
        description
          "True indicates that IEEE Std 802.3 priority-based
           flow control is enabled, false indicates that
           IEEE Std 802.3 priority-based flow control is
           disabled. For interfaces that have auto-negotiation,
           the priority-based flow control is enabled by
           default.";
      }
      container statistics {
        config false;
        status deprecated;
        description
          "This container collects all statistics for
           Ethernet interfaces.

           Discontinuities in the values of counters in
           this container can occur at re-initialization of
           the management system, and at other times as
           indicated by the value of the 'discontinuity-time'
           leaf defined in the ietf-interfaces YANG module
           (IETF RFC 8343).";
        leaf in-frames-pfc {
          type yang:counter64;
          units "frames";
          status deprecated;
          description
            "Deprecated in-frames-pfc as not defined in base
             standard. A count of PFC MAC Control frames
             received on this Ethernet interface.";
          reference
            "IEEE Std 802.3.1, dot3HCInPFCFrames";
        }
        leaf out-frames-pfc {
          type yang:counter64;
          units "frames";
          status deprecated;
          description
            "Deprecated out-frames-pfc as not defined in base
             standard.A count of PFC MAC Control frames
             transmitted on this interface.";
          reference
            "IEEE Std 802.3.1, dot3HCInPFCFrames";
        }
      }
    }
    leaf force-flow-control {
      type boolean;
```

```
    default "false";
    description
      "Explicitly forces the local PAUSE frame based flow
       control settings regardless of what has been
       negotiated.

       Since the auto-negotiation of flow-control settings
       does not allow all sane combinations to be negotiated
       (e.g., consider a device that is only capable of
       sending PAUSE frames connected to a peer device that
       is only capable of receiving and acting on PAUSE
       frames) and failing to agree on the flow-control
       settings does not cause the auto-negotiation to fail
       completely, then it is sometimes useful to be able to
       explicitly enable particular PAUSE frame based flow
       control settings on the local device regardless of
       what is being advertised or negotiated.";
    reference
      "IEEE Std 802.3, Table 28B-3";
  }
}
leaf max-frame-length {
  type uint16;
  units "octets";
  config false;
  description
    "This indicates the MAC frame length (including FCS bytes)
     at which frames are dropped for being too long.";
  reference
    "IEEE Std 802.3, 30.3.1.1.37 aMaxFrameLength";
}
leaf mac-control-extension-control {
  type boolean;
  config false;
  description
    "A value that identifies the current EXTENSION
     MAC Control function, as specified in
     IEEE Std 802.3, Annex 31C.";
  reference
    "IEEE Std 802.3, 30.3.8.3 aEXTENSIONMACCtrlStatus
     IEEE Std 802.3.1, dot3ExtensionMacCtrlStatus ";
}
leaf frame-limit-slow-protocol {
  type uint64;
  units "f/s";
  default "10";
  config false;
  description
    "The maximum number of Slow Protocol frames of a given
     subtype that can be transmitted in a one second
     interval.
     The default value is 10.";
  reference
    "IEEE Std 802.3, 30.3.1.1.38 aSlowProtocolFrameLimit";
}
container capabilities {
  config false;
  description
```

```
            "Container all Ethernet interface specific capabilities.";
          leaf auto-negotiation {
            type boolean;
            description
              "Indicates whether auto-negotiation may be configured on
               this interface.";
          }
        }
        container ethernet-pause {
          if-feature "ethernet-pause";
          description
            "IEEE Std 802.3 PAUSE flow control.

             Discontinuities in the values of counters in
             this container can occur at re-initialization of
             the management system, and at other times as
             indicated by the value of the 'discontinuity-time'
             leaf defined in the ietf-interfaces YANG module
             (IETF RFC 8343).";
          reference
            "IEEE Std 802.3,30.3.4 PAUSE entity managed object class and
             Clause 31,Annex 31B)";
          container control-and-status {
            description
              "PAUSE function control and status objects.";
            leaf pause-admin-control {
              type pause-fc-direction-type;
              default "disabled";
              description
                "What PAUSE functionality will run on this interface when
                 Auto-Negotiation is not implemented or disabled.";
              reference
                "IEEE Std 802.3, ????";
            }
            leaf pause-oper-status {
              type pause-fc-direction-type;
              config false;
              description
                "What PAUSE functionality is running on this
                 interface, as a result of  Auto-Negotiation or
                 setting pause-admin-control.";
              reference
                "IEEE Std 802.3, ????";
            }
            leaf link-delay-allowance {
              type uint32;
              description
                "The value in bit-times of the allowance made by the
                 MAC Control PAUSE entity for round-trip propagation
                 delay.
                 The default value is implementation-dependent.";
            }
            leaf pfc-enable-status {
              type boolean;
              config false;
              description
                "Is IEEE 802.1 Priority-based Flow Control(PFC)
                 enabled on the interface. If PFC is enabled, then
```

```
               802.3 PAUSE flow control is disabled.";
            reference
              "IEEE Std 802.3, 30.3.3.6 aPFCEnableStatus";
          }
        }
        container statistics {
          config false;
          description
            "PAUSE frame counters.";
          leaf in-frames-pause {
            type yang:counter64;
            units "frames";
            description
              "A count of MAC Control PAUSE frames transmitted on
               this Ethernet interface.";
            reference
              "IEEE Std 802.3, 30.3.4.3
               aPAUSEMACCtrlFramesReceived";
          }
          leaf out-frames-pause {
            type yang:counter64;
            units "frames";
            description
              "A count of MAC Control PAUSE frames transmitted on
               this Ethernet interface.";
            reference
              "IEEE Std 802.3, 30.3.4.2
               aPAUSEMACCtrlFramesTransmitted";
          }
        }
      }
      container statistics {
        config false;
        description
          "Contains statistics specific to Ethernet interfaces.

           Discontinuities in the values of counters in the
           container can occur at re-initialization of the
           management system, and at other times as indicated by
           the value of the 'discontinuity-time' leaf defined in
           the ietf-interfaces YANG module (IETF RFC 8343).";
        container frame {
          description
            "Contains frame statistics specific to Ethernet
             interfaces.

             All octet frame lengths include the 4 byte FCS.

             Error counters are only reported once. The count
             represented by an instance of this object is
             incremented when the frameCheckError status is
             returned by the MAC service to the MAC Client.
             Received frames for which multiple error conditions
             pertain are, according to the conventions of
             IEEE Std 802.3 Layer Management, counted exclusively
             according to the error status presented to the MAC
             Client.
```

```
                A frame that is counted by an instance of this object
                is also counted by the corresponding instance of
                'in-errors' leaf defined in the ietf-interfaces YANG
                module (IETF RFC 8343~~).~~

~~                Discontinuities in the values of counters in the~~
~~                container can occur at re-initialization of the~~
~~                management system, and at other times as indicated by~~
~~                the value of the 'discontinuity-time' leaf defined in~~
~~                the ietf-interfaces YANG module (IETF RFC 8343).";~~
).";
          leaf in-total-frames {
            type yang:counter64;
            units "frames";
            description
              "The total number of frames (including bad frames)
               received on the Ethernet interface.

               This counter is calculated by summing the following
               IEEE Std 802.3, Clause 30 counters:
               aFramesReceivedOK +
               aFrameCheckSequenceErrors +
               aAlignmentErrors +
               aFrameTooLongErrors +
               aFramesLostDueToIntMACRcvError";
            reference
              "IEEE Std 802.3, Clause 30 counters, as specified
               in the description above.";
          }
          leaf in-total-octets {
            type yang:counter64;
            units "octets";
            description
              "The total number of octets of data (including those
               in bad frames) received on the Ethernet interface.

                Includes the 4-octet FCS.";
            reference
              "IETF RFC 2819, etherStatsOctets";
          }
          leaf in-frames {
            type yang:counter64;
            units "frames";
            description
              "A count of frames (including unicast, multicast and
               broadcast) that have been successfully received on
               the Ethernet interface.

               This count does not include frames received with
               frame-too-long, FCS, length or alignment errors, or
               frames lost due to internal MAC sublayer error.";
            reference
              "IEEE Std 802.3, 30.3.1.1.5 aFramesReceivedOK";
          }
          leaf in-multicast-frames {
            type yang:counter64;
            units "frames";
            description
```

```
              "A count of multicast frames that have been
               successfully received on the Ethernet interface.

                This counter represents a subset of the frames
                counted by in-frames.

                This count does not include frames received with
                frame-too-long, FCS, length or alignment errors, or
                frames lost due to internal MAC sublayer error.";
            reference
              "IEEE Std 802.3, 30.3.1.1.21
               aMulticastFramesReceivedOK";
          }
          leaf in-broadcast-frames {
            type yang:counter64;
            units "frames";
            description
              "A count of broadcast frames that have been
               successfully received on the Ethernet interface.

               This counter represents a subset of the frames
               counted by in-frames.

               This count does not include frames received with
               frame-too-long, FCS, length or alignment errors, or
               frames lost due to internal MAC sublayer error.";
            reference
              "IEEE Std 802.3, 30.3.1.1.22
               aBroadcastFramesReceivedOK";
          }
          leaf in-error-fcs-frames {
            type yang:counter64;
            units "frames";
            description
              "A count of receive frames that are of valid length,
                        but do not pass the FCS check, regardless of
whether
                        or not the frames are an integral number of
octets
                        in length.

                        This counter is calculated by summing the
following
                        counters:
                          aFrameCheckSequenceErrors +
                          aAlignmentErrors";
            reference
              "IEEE Std 802.3, 30.3.1.1.6 aFrameCheckSequenceErrors;
               IEEE Std 802.3, 30.3.1.1.7 aAlignmentErrors";
          }
          leaf in-error-undersize-frames {
            type yang:counter64;
            units "frames";
            status deprecated;
            description
              "Deprecated in-error-undersize-frames as not defined
               in base standard. A count of frames received on a
               particular Ethernet interface that are less than
```

```
       64 bytes in length, and are discarded.

       This counter is incremented regardless of whether
       the frame passes the FCS check.";
    reference
      "IETF RFC 2819, etherStatsUndersizePkts and
       etherStatsFragments";
  }
  leaf in-error-oversize-frames {
    type yang:counter64;
    units "frames";
    description
      "A count of frames received on a particular Ethernet
       interface that exceed the maximum permitted frame
       size, that is specified in max-frame-length, and are
       discarded.

       This counter is incremented regardless of whether
       the frame passes the FCS check.";
    reference
      "IEEE Std 802.3, 30.3.1.1.25
       aFrameTooLongErrors";
  }
  leaf in-error-mac-internal-frames {
    type yang:counter64;
    units "frames";
    description
      "A count of frames for which reception on a
       particular Ethernet interface fails due to an
       internal MAC sublayer receive error.

       A frame is only counted by an instance of this
       object if it is not counted by the corresponding
       instance of either the in-error-fcs-frames,
       in-error-undersize-frames, or
       in-error-oversize-frames.  The precise meaning of
       the count represented by an instance of this object
       is implementation-specific.

       In particular, an instance of this object may
       represent a count of receive errors on a particular
       Ethernet interface that are not otherwise counted.";
    reference
      "IEEE Std 802.3, 30.3.1.1.15
       aFramesLostDueToIntMACRcvError";
  }
  leaf out-frames {
    type yang:counter64;
    units "frames";
    description
      "A count of frames (including unicast, multicast and
       broadcast) that have been successfully transmitted
       on the Ethernet interface.";
    reference
      "IEEE Std 802.3, 30.3.1.1.2 aFramesTransmittedOK";
  }
  leaf out-multicast-frames {
    type yang:counter64;
```

```
          units "frames";
          description
            "A count of multicast frames that have been
             successfully transmitted on the Ethernet interface.

             This counter represents a subset of the frames
             counted by out-frames.";
          reference
            "IEEE Std 802.3, 30.3.1.1.18
              aMulticastFramesXmittedOK";
        }
        leaf out-broadcast-frames {
          type yang:counter64;
          units "frames";
          description
            "A count of broadcast frames that have been
             successfully transmitted on the Ethernet interface.

             This counter represents a subset of the frames
             counted by out-frames.";
          reference
            "IEEE Std 802.3, 30.3.1.1.19
             aBroadcastFramesXmittedOK";
        }
        leaf out-error-mac-internal-frames {
          type yang:counter64;
          units "frames";
          description
            "A count of frames for which transmission on a
             particular Ethernet interface fails due to an
             internal MAC sublayer transmit error.

             The precise meaning of the count represented by an
             instance of this object is implementation-specific.
             In particular, an instance of this object may
             represent a count of transmission errors on a
             particular Ethernet interface that are not otherwise
             counted.";
          reference
            "IEEE Std 802.3, 30.3.1.1.12
             aFramesLostDueToIntMACXmitError";
        }
      }
      container phy {
        description
          "Ethernet statistics related to the PHY layer.

           Discontinuities in the values of counters in the
           container can occur at re-initialization of the
           management system, and at other times as indicated by
           the value of the 'discontinuity-time' leaf defined in
           the ietf-interfaces YANG module (IETF RFC 8343).";
.";
        leaf in-error-symbol {
          type yang:counter64;
          units "errors";
          description
            "A count of the number of symbol errors that have
```

```
                    occurred.

                    For the precise definition of when the symbol error
                    counter is incremented, please see the 'description'
                    text associated with aSymbolErrorDuringCarrier,
                    specified in IEEE Std 802.3, 30.3.2.1.5.";
                 reference
                   "IEEE Std 802.3, 30.3.2.1.5
                    aSymbolErrorDuringCarrier";
               }
               container lpi {
                 description
                   "Physical Ethernet statistics for the energy
                    efficiency related low power idle indications.";
                 leaf in-lpi-transitions {
                   type yang:counter64;
                   units "transitions";
                   description
                     "~~A count~~The number of ~~occurrences of~~times the ~~transition
~~from~~
                      ~~DEASSERT~~link partner transitioned to~~ ASSERT of the
LPI_INDICATE~~
                      ~~parameter. The indication reflects the state~~Low Power
Idle.
                      This counter has a maximum per second increment rate
                      of:
                      ~~the PHY according to the requirements of the RS~~   50
thousand for 100 Mb/s;
                      ~~(see IEEE Std 802.3, 22.7, 35.4, and 46.4).";~~   90
thousand for 1000 Mb/s
                         230 thousand for 10 Gb/s.";
                   reference
                     "IEEE Std 802.3, 30.3.2.1.11
                      aReceiveLPITransitions";
                 }
                 leaf in-lpi-time {
                   type decimal64 {
                     fraction-digits 6;
                   }
                   units "seconds";
                   description
                     "~~A count reflecting the total ~~The amount of time ~~(in
                      seconds) that the LPI_REQUEST parameter~~the link partner
has~~ the~~
                      ~~value ASSERT.  The request is indicated to the PHY~~been
in Low Power Idle.";
                      ~~according to the requirements of the RS (see
                      IEEE Std 802.3, 22.7, 35.4, and 46.4).";~~
                   reference
                     "IEEE Std 802.3, 30.3.2.1.9
                      aReceiveLPIMicroseconds";
                 }
                 leaf out-lpi-transitions {
                   type yang:counter64;
                   units "transitions";
                   description
                     "~~A count~~The number of ~~occurrences~~times this port
transitioned to
```

```
                 Low Power Idle.
                 This counter has a maximum per second increment rate
                 of~~ the transition from~~:
                 ~~state LPI_DEASSERTED to state LPI_ASSERTED in the~~    50
thousand for 100 Mb/s;
                 ~~LPI transmit state diagram of the RS. The state~~    90
thousand for 1000 Mb/s
                 ~~transition corresponds to the assertion of the~~    230
thousand for 10 Gb/s.";
                 ~~LPI_REQUEST parameter. The request is indicated to~~
                 ~~the PHY according to the requirements of the RS~~
                 ~~(see IEEE Std 802.3, 22.7, 35.4, 46.4.)";~~
              reference
                "IEEE Std 802.3, 30.3.2.1.10
                 aTransmitLPITransitions";
            }
            leaf out-lpi-time {
              type decimal64 {
                fraction-digits 6;
              }
              units "seconds";
              description
                "~~A count reflecting the total~~The  amount of time ~~(~~in
                 ~~seconds) that the LPI_INDICATION parameter~~ this port has
~~the~~
                 ~~value ASSERT.  The request is indicated to the PHY~~been
in Low Power Idle.";
                 ~~according to the requirements of the RS (see IEEE~~
                 ~~802.3, 22.7, 35.4, and 46.4).";~~
              reference
                "IEEE Std 802.3, 30.3.2.1.8
                 aTransmitLPIMicroseconds";
            }
          }
        }
        container mac-control {
          description
            "A group of statistics specific to MAC Control
             operation of selected Ethernet interfaces~~.~~

             ~~Discontinuities in the values of counters in the~~
             ~~container can occur at re-initialization of the~~
             ~~management system, and at other times as indicated~~
             ~~by the value of the 'discontinuity-time' leaf defined~~
             ~~in the ietf-interfaces YANG module (IETF RFC 8343).";~~
.";
          reference
            "IEEE Std 802.3.1, dot3ExtensionTable";
          leaf in-frames-mac-control-unknown {
            type yang:counter64;
            units "frames";
            description
              "A count of MAC Control frames with an unsupported
               opcode received on this Ethernet interface.

               Frames counted against this counter are also counted
               against in-discards defined in the ietf-interfaces
               YANG module (IETF RFC 8343).";
```

```
            reference
              "IEEE Std 802.3, 30.3.3.5
               aUnsupportedOpcodesReceived";
          }
          leaf in-frames-mac-control-extension {
            type yang:counter64;
            units "frames";
            description
              "The count of Extension MAC Control frames received
               on this Ethernet interface.";
            reference
              "IEEE Std 802.3, 30.3.8.2
               aEXTENSIONMACCtrlFramesReceived";
          }
          leaf out-frames-mac-control-extension {
            type yang:counter64;
            units "frames";
            description
              "The count of Extension MAC Control frames
               transmitted on this Ethernet interface.";
            reference
              "IEEE Std 802.3, 30.3.8.1
               aEXTENSIONMACCtrlFramesTransmitted";
          }
        }
      }
    }
  }
}
module ieee802-ethernet-link-oam {
  yang-version 1.1;
  namespace "urn:ieee:std:802.3:yang:ieee802-ethernet-link-oam";
  prefix ieee802-link-oam;

  import ieee802-types {
    prefix ieee;
    reference
      "IEEE 802 types";
  }
  import ietf-yang-types {
    prefix yang;
    reference
      "IETF RFC 6991";
  }
  import iana-if-type {
    prefix ianaift;
    reference
      "http://www.iana.org/assignments/yang-parameters/
       iana-if-type@2023-01-26.yang";
  }
  import ietf-interfaces {
    prefix if;
    reference
      "IETF RFC 8343";
  }

  organization
    "IEEE 802.3 Ethernet Working Group
```

```
    Web URL: http://www.ieee802.org/3/";
contact
  "Web URL: http://www.ieee802.org/3/";
description
  "This module contains a collection of YANG definitions
   for managing the Ethernet Link OAM feature defined by IEEE
   802.3. It provides functionality roughly equivalent to that of
   the DOT3-OAM-MIB defined in IETF RFC 4878.";

revision 2024-07-1712-23 {
  description
    "Updates under IEEE Std 802.3.2-202x, Draft D3.01";
  reference
    "IEEE Std 802.3-2022, unless dated explicitly";
}

feature uni-directional-link-fault {
  description
    "This feature means the device supports Uni Directional Link
     Fault detection.";
  reference
    "IEEE Std 802.3, 57.1.2:a, 30.3.6.1.6 aOAMLocalConfiguration
     and 30.3.6.1.7 aOAMRemoteConfiguration";
}

feature remote-loopback-initiate {
  description
    "This feature means the device supports being the initiator
     of remote loopback.";
  reference
    "IEEE Std 802.3, 57.1.2:b, 30.3.6.1.6
     aOAMLocalConfiguration";
}

feature remote-loopback-respond {
  description
    "This feature means the device supports responding to remote
     loopback control OAMPDUs received from the peer";
  reference
    "IEEE Std 802.3, 57.1.2:b, 30.3.6.1.7
     aOAMRemoteConfiguration";
}

feature link-monitoring-local {
  description
    "This feature means the device monitors the link at the local
     side and can generate Link Event OAMPDUs to the peer
     device.";
  reference
    "IEEE Std 802.3, 57.1.2:c:1, 30.3.6.1.6
     aOAMLocalConfiguration, and 30.3.6.1.7
     aOAMRemoteConfiguration";
}

feature link-monitoring-remote {
  description
    "This feature means the device can process Link Event OAMPDUs
     received from the peer device and report itself about this
```

```
      event on its own management interface.";
    reference
      "IEEE Std 802.3, 57.1.2:c:1,
       30.3.6.1.6 aOAMLocalConfiguration,
       and 30.3.6.1.7 aOAMRemoteConfiguration";
  }

  feature remote-mib-retrieval-initiate {
    status deprecated;
    description
      "remote-mib-retrieval-initiate is deprecated and changed name
       to feature remote-data-retrieval-initiate. This feature
       means the device supports data retrieval from the peer
       device. I.e. the device can send Variable Requests OAMPDUs
       to the peer side and process the received Variable Response
       OAMPDUs.";
    reference
      "IEEE Std 802.3, 57.1.2:c:2,
       30.3.6.1.6 aOAMLocalConfiguration,
       and 30.3.6.1.7 aOAMRemoteConfiguration";
  }

  feature remote-data-retrieval-initiate {
    description
      "This feature means the device supports data retrieval from
       the peer device. I.e. the device can send Variable Requests
       OAMPDUs to the peer side and process the received Variable
       Response OAMPDUs.";
    reference
      "IEEE Std 802.3, 57.1.2:c:2,
       30.3.6.1.6 aOAMLocalConfiguration,
       and 30.3.6.1.7 aOAMRemoteConfiguration";
  }

  feature remote-mib-retrieval-respond {
    description
      "This feature means the device allows the peer device to
       retrieve data from the managed device. I.e. the device can
       process received Variable Requests OAMPDUs and respond with
       Variable Response OAMPDUs.";
    reference
      "IEEE Std 802.3, 57.1.2:c:2,
       30.3.6.1.6 aOAMLocalConfiguration,
       and 30.3.6.1.7 aOAMRemoteConfiguration";
  }

  identity event-type {
    description
      "Base identity for all Link OAM event types.";
  }

  identity threshold-event-type {
    base event-type;
    description
      "Event type for a Link Monitoring threshold event.";
  }

  identity link-fault-event {
```

```
    if-feature "uni-directional-link-fault";
    base event-type;
    description
      "Event type for a uni-directional link fault event.";
    reference
      "IEEE Std 802.3, 57.2.10.1";
  }

  identity dying-gasp-event {
    base event-type;
    description
      "Event type for a dying gasp event.";
    reference
      "IEEE Std 802.3, 57.2.10.1";
  }

  identity critical-event {
    base event-type;
    description
      "Event type for a critical event.";
    reference
      "IEEE Std 802.3, 57.2.10.1";
  }

  typedef threshold-event-enum {
    type enumeration {
      enum symbol-period-event {
        value 1;
        description
          "Errored symbol period event.";
      }
      enum frame-period-event {
        value 2;
        description
          "Errored frame period event.";
      }
      enum frame-event {
        value 3;
        description
          "Errored frame event";
      }
      enum frame-seconds-event {
        value 4;
        description
          "Errored frame seconds event.";
      }
    }
    description
      "Enumeration of the valid threshold event types.";
    reference
      "IEEE Std 802.3, 57.5.3";
  }

  typedef mode {
    type enumeration {
      enum passive {
        value 0;
        description
```

```
          "Ethernet Link OAM Passive mode.";
      }
      enum active {
        value 1;
        description
          "Ethernet Link OAM Active mode.";
      }
    }
    description
      "Enumeration of the valid modes in which Link OAM may run.";
    reference
      "IEEE Std 802.3, 57.2.9 and 30.3.6.1.3.";
  }

  typedef event-location {
    type enumeration {
      enum event-location-local {
        value 1;
        description
          "A local event.";
      }
      enum event-location-remote {
        value 2;
        description
          "A remote event.";
      }
    }
    description
      "The location of the event that caused a log entry.";
  }

  typedef loopback-status {
    type enumeration {
      enum none {
        value 1;
        description
          "Loopback is not being performed.";
      }
      enum initiating {
        value 2;
        description
          "Initiating master loopback.";
      }
      enum master-loopback {
        value 3;
        description
          "In master loopback mode.";
      }
      enum terminating {
        value 4;
        description
          "Terminating master loopback mode.";
      }
      enum local-loopback {
        value 5;
        description
          "In slave loopback mode.";
      }
```

```
      enum unknown {
        value 6;
        description
          "Parser and multiplexer combination unexpected.";
      }
    }
    description
      "The loopback mode of an OAM interface.";
    reference
      "IEEE Std 802.3, 57.2.11";
  }

  typedef operational-state {
    type enumeration {
      enum disabled {
        value 1;
        description
          "IEEE Std 802.3 OAM is disabled.";
      }
      enum link-fault {
        value 2;
        description
          "IEEE Std 802.3 OAM has encountered a link fault.";
      }
      enum passive-wait {
        value 3;
        description
          "Passive OAM entity waiting to see if peer is
           OAM capable.";
      }
      enum active-send-local {
        value 4;
        description
          "Active OAM entity trying to determine if peer
           is OAM capable.";
      }
      enum send-local-and-remote {
        value 5;
        description
          "OAM discovered peer but still to accept or
           reject peer configuration.";
      }
      enum send-local-and-remote-ok {
        value 6;
        description
          "OAM peering is allowed by local device.";
      }
      enum peering-locally-rejected {
        value 7;
        description
          "OAM peering rejected by local device.";
      }
      enum peering-remotely-rejected {
        value 8;
        description
          "OAM peering rejected by remote device.";
      }
      enum operational {
```

```
      value 9;
      description
        "IEEE Std 802.3 OAM is operational.";
    }
    enum operational-half-duplex {
      value 10;
      description
        "IEEE Std 802.3 OAM is operating in half-duplex mode.";
    }
  }
  description
    "Operational state of an interface.";
  reference
    "IEEE Std 802.3, 30.3.6.1.4,
     30.3.6.1.10, and 30.3.6.1.11";
}

typedef vendor-oui {
  type string {
    length "6";
  }
  description
    "24-bit  MAC addresses – large(MA-L). Previously know as
     Organizationally Unique Identifier (OUI).";
  reference
    "IEEE Std 802-2014, Clause 8.2.2";
}

typedef admin-state {
  type enumeration {
    enum enabled {
      value 1;
      description
        "IEEE Std 802.3, Clause 57 OAM is in the
         enabled admin state.";
    }
    enum disabled {
      value 2;
      description
        "IEEE Std 802.3, Clause 57 OAM is in the
         disabled admin state.";
    }
  }
  description
    "Admin state of the OAM function on an interface.";
  reference
    "IEEE Std 802.3, 30.3.6.1.2 and 30.3.6.2";
}

grouping event-details {
  description
    "Nodes describing an event, used in the event log and in
     notifications.";
  reference
    "IETF RFC 4878, Dot3OamEventLogEntry";
  leaf oui {
    type vendor-oui;
    mandatory true;
```

```
    description
      "Organizationally Unique Identifier for the device that
       generated the event.";
  }
  leaf timestamp {
    type uint64;
    units "milliseconds";
    mandatory true;
    description
      "Timestamp in milliseconds since Unix epoch for when the
       event occurred.";
  }
  leaf location {
    type event-location;
    mandatory true;
    description
      "Where the event occurred (local or remote).";
  }
  leaf event-type {
    type identityref {
      base event-type;
    }
    mandatory true;
    description
      "Type of event that occurred.";
    reference
      "IEEE Std 802.3, 30.3.6.1.10 and 30.3.6.11";
  }
  leaf running-total {
    type yang:counter64;
    mandatory true;
    description
      "The running total number of errors seen since OAM was
       enabled on the interface. For threshold events, this is
       the total number of times that particular type of error
       (e.g. symbol error) has occurred, which may be greater
       than the number of threshold-crossing event notifications
       of that type generated during that time (which is conveyed
       by the event-total leaf).";
  }
  leaf event-total {
    type yang:counter64;
    mandatory true;
    description
      "Total number of times this event has occurred since OAM
       was enabled on the interface. For threshold events this is
       the number of events generated of this type (as opposed to
       the total number of errors of that type, which may be
       greater, and is conveyed by the running-total leaf.";
  }
}

grouping threshold-event-details {
  description
    "Nodes describing a threshold event, used in the event
     log and in notifications";
  reference
    "IETF RFC 4878, Dot3OamEventLogEntry";
```

```
    container threshold {
      when "../event-type = 'threshold-event-type'" {
        description
          "These nodes only apply to threshold event types";
      }
      if-feature
        "link-monitoring-local or
         link-monitoring-remote";
      description
        "Nodes specific to threshold (link monitoring) events";
      leaf threshold-event-type {
        type threshold-event-enum;
        mandatory true;
        description
          "The type of threshold event";
        reference
          "IEEE Std 802.3, 57.5.3";
      }
      leaf window {
        type uint64;
        mandatory true;
        description
          "Size of the window in which the event was generated.
           Units are dependent on the threshold event type.";
      }
      leaf threshold {
        type uint64;
        mandatory true;
        description
          "Size of the threshold that was breached during the
           window. Units are dependent on the threshold
           event type.";
      }
      leaf value {
        type uint64;
        mandatory true;
        description
          "Breaching value. Units are dependent on the threshold
           event type, and match that of the threshold.";
      }
    }
  }

  grouping statistics-common {
    description
      "Collection of Link OAM event/packet counters.";
    reference
      "IETF RFC 4878, Dot3OamStatsEntry";
    leaf out-information {
      type yang:counter64;
      mandatory true;
      description
        "Number of information OAMPDUs transmitted.";
      reference
        "IEEE Std 802.3, 30.3.6.1.20";
    }
    leaf in-information {
      type yang:counter64;
```

```
    mandatory true;
    description
      "Number of information OAMPDUs received.";
    reference
      "IEEE Std 802.3, 30.3.6.1.21";
  }
  leaf out-unique-event-notification {
    if-feature "link-monitoring-local";
    type yang:counter64;
    mandatory true;
    description
      "Number of unique event notification OAMPDUs transmitted.";
    reference
      "IEEE Std 802.3, 30.3.6.1.22";
  }
  leaf in-unique-event-notification {
    if-feature "link-monitoring-remote";
    type yang:counter64;
    mandatory true;
    description
      "Number of unique event notification OAMPDUs received.";
    reference
      "IEEE Std 802.3, 30.3.6.1.24";
  }
  leaf out-duplicate-event-notification {
    if-feature "link-monitoring-local";
    type yang:counter64;
    mandatory true;
    description
      "Number of duplicate event notification OAMPDUs
               transmitted.";
    reference
      "IEEE Std 802.3, 30.3.6.1.23";
  }
  leaf in-duplicate-event-notification {
    if-feature "link-monitoring-remote";
    type yang:counter64;
    mandatory true;
    description
      "Number of duplicate event notification OAMPDUs
       received.";
    reference
      "IEEE Std 802.3, 30.3.6.1.25";
  }
  leaf out-loopback-control {
    if-feature "remote-loopback-initiate";
    type yang:counter64;
    mandatory true;
    description
      "Number of loopback control OAMPDUs transmitted.";
    reference
      "IEEE Std 802.3, 30.3.6.1.26";
  }
  leaf in-loopback-control {
    if-feature "remote-loopback-respond";
    type yang:counter64;
    mandatory true;
    description
```

```
    "Number of loopback control OAMPDUs received.";
  reference
    "IEEE Std 802.3, 30.3.6.1.27";
}
leaf out-variable-request {
  if-feature "remote-data-retrieval-initiate";
  type yang:counter64;
  mandatory true;
  description
    "Number of variable request OAMPDUs transmitted.";
  reference
    "IEEE Std 802.3, 30.3.6.1.28";
}
leaf in-variable-request {
  if-feature "remote-mib-retrieval-respond";
  type yang:counter64;
  mandatory true;
  description
    "Number of variable request OAMPDUs received.";
  reference
    "IEEE Std 802.3, 30.3.6.1.29";
}
leaf out-variable-response {
  if-feature "remote-mib-retrieval-respond";
  type yang:counter64;
  mandatory true;
  description
    "Number of variable response OAMPDUs transmitted.";
  reference
    "IEEE Std 802.3, 30.3.6.1.30";
}
leaf in-variable-response {
  if-feature "remote-data-retrieval-initiate";
  type yang:counter64;
  mandatory true;
  description
    "Number of variable response OAMPDUs received.";
  reference
    "IEEE Std 802.3, 30.3.6.1.31";
}
leaf out-org-specific {
  type yang:counter64;
  mandatory true;
  description
    "Number of organization specific OAMPDUs transmitted.";
  reference
    "IEEE Std 802.3, 30.3.6.1.32";
}
leaf in-org-specific {
  type yang:counter64;
  mandatory true;
  description
    "Number of organization specific OAMPDUs received.";
  reference
    "IEEE Std 802.3, 30.3.6.1.33";
}
leaf out-unsupported-codes {
  type yang:counter64;
```

```
      mandatory true;
      description
        "Number of OAMPDUs with unsupported codes transmitted.";
      reference
        "IEEE Std 802.3, 30.3.6.1.18";
    }
    leaf in-unsupported-codes {
      type yang:counter64;
      mandatory true;
      description
        "Number of OAMPDUs with unsupported codes received.";
      reference
        "IEEE Std 802.3, 30.3.6.1.19";
    }
    leaf frames-lost-due-to-oam {
      type yang:counter64;
      mandatory true;
      description
        "A count of the number of frames that were dropped by the
         OAM multiplexer. Since the OAM multiplexer has multiple
         inputs and a single output, there may be cases where
         frames are dropped due to transmit resource contention.
         This counter is incremented whenever a frame is dropped by
         the OAM layer.";
      reference
        "IEEE Std 802.3, 30.3.6.1.46";
    }
  }

  grouping discovery-remote {
    description
      "Nodes describing the discovery process remote end of a
           link.";
    leaf mode {
      type mode;
      description
        "Mode (passive/active).";
      reference
        "IEEE Std 802.3, 30.3.6.1.3";
    }
    container functions-supported {
      description
        "The Link OAM functions supported by this interface.";
      reference
        "IEEE Std 802.3, 30.3.6.1.7";
      leaf uni-directional-link-fault {
        type boolean;
        description
          "Unidirectional link fault support.";
      }
      leaf loopback {
        type boolean;
        description
          "Remote Loopback support.";
      }
      leaf link-monitoring {
        type boolean;
        description
```

```
          "Link monitoring support.";
      }
      leaf mib-retrieval {
        type boolean;
        description
          "MIB variable retrieval support.";
      }
    }
    leaf revision {
      type uint64;
      config false;
      description
        "Configuration revision.";
      reference
        "IEEE Std 802.3, 30.3.6.1.12 and 30.3.6.1.13";
    }
    leaf oammtu {
      type uint16;
      units "octets";
      config false;
      description
        "The maximum OAMPDU size for the remote node.
         The peer OAM entities exchange the maximum size they can
         support and negotiate to use the smaller of the two maximum
         OAMPDU sizes.";
      reference
        "IEEE Std 802.3, 30.3.6.1.8 and 30.3.6.1.9";
    }
  }

  grouping discovery-local {
    description
      "Nodes describing the local end discovery process of a
           link.";
    leaf mode {
      type mode;
      description
        "This object configures the mode of OAM operation as active
         or passive. Active mode provides capabilities to initiate
         monitoring activities with the remote OAM peer eneity,
         while passive mode waits for the peer to initiate actions
         with it. Changing this value results in incrementing the
         revision field of locally generated OAMPDUs
         (see IEEE Std 802.3, 30.3.6.1.12) and triggers the
         OAM discovery process if the operational state was
         already 'operational'.  The default value is
         implementation-dependent.";
      reference
        "IEEE Std 802.3, 30.3.6.1.3";
    }
    container functions-supported {
      description
        "The Link OAM functions supported by this interface.";
      reference
        "IEEE Std 802.3, 30.3.6.1.7";
      leaf uni-directional-link-fault {
        if-feature "uni-directional-link-fault";
        type boolean;
```

```
  description
    "Unidirectional link fault support.
     This affects the setting of the 'Unidirectional Support'
     bit in the OAM configuration field put in the
     Information OAMPDU.
     This bit indicates to the peer device that it can send
     OAM PDUs on links that are operating in unidirectional
     mode (traffic flowing in one direction only).";
}
leaf loopback {
  if-feature "remote-loopback-initiate";
  type boolean;
  default "true";
  description
    "Remote Loopback support.";
}
container link-monitor {
  if-feature
    "link-monitoring-remote or
     link-monitoring-local";
  description
    "Configure link monitor parameters.";
  reference
    "IEEE Std 802.3, 57.1.2:c";
  leaf link-monitoring {
    type boolean;
    default "true";
    description
      "Enable or disable monitoring.
       This affects the setting of the 'Link Events' bit in
       the OAM configuration field put in the Information
       OAMPDU. This bit indicates to the peer device that the
       OAM entity can send and receive Event Notification
       OAMPDUs.";
  }
  list event-type {
    if-feature "link-monitoring-local";
    key "threshold-type";
    description
      "A list containing at most one entry for each of the
       threshold event types. If there is no entry for a
       particular event type, the default values are used for
       both window size and threshold.";
    leaf threshold-type {
      type threshold-event-enum;
      description
        "The type of threshold event for which this list
         entry is specifying the configuration.";
      reference
        "IEEE Std 802.3, 57.5.3";
    }
    leaf window {
      type uint64;
      description
        "The size of the window to use when monitoring for
         this threshold event. The units, default and upper
         and lower bounds depend on the threshold type as
         follows:
```

```
         Symbol Period:
           Units:   number of symbols
           Default: number of symbols in one second for the
                    underlying physical layer
           Min:     number of symbols in one second for the
                    underlying physical layer
           Max:     number of symbols in one minute for the
                    underlying physical layer

         Frame:
           Units:   deciseconds
           Default: 1 second
           Min:     1 second
           Max:     1 minute

         Frame Period:
           Units:   number of frames
           Default: number of minFrameSize frames in one
                    second for the underlying physical layer
           Min:     number of minFrameSize frames in one
                    second for the underlying physical layer
           Max:     number of minFrameSize frames in one
                    minute for the underlying physical layer

         Frame Seconds:
           Units:   deciseconds
           Default: 60 seconds
           Min:     10 seconds
           Max:     900 seconds";
      reference
        "IEEE Std 802.3, 30.3.6.1.34, 30.3.6.1.36, 30.3.6.1.38,
         and 30.3.6.1.40";
    }
    leaf threshold {
      type uint64 {
        range "1..max";
      }
      default "1";
      description
        "The threshold value to use when determining whether
            to generate an event given the number of errors that
         occurred in a given window. The units depend on the
         threshold type as follows:

         Symbol Period: number of errored symbols
         Frame:         number of errored frames
         Frame Period:  number of errored frames
         Frame Seconds: number of seconds containing at least
         1 frame error";
      reference
        "IEEE Std 802.3, 30.3.6.1.34, 30.3.6.1.36, 30.3.6.1.38,
         and 30.3.6.1.40";
    }
  }
}
leaf mib-retrieval {
  if-feature
```

```
        "remote-mib-retrieval-initiate or
         remote-mib-retrieval-respond";
      type boolean;
      status deprecated;
      description
        "leaf mib-retrieval is deprecated and changed name to
         data-retrieval.
         MIB variable retrieval support.
         This affects the setting of the 'Variable Retrieval' bit
         in the OAM configuration field put in the Information
         OAMPDU. This bit indicates to the peer device that the
         OAM entity can send and receive Variable Request and
         Response OAMPDUs.";
    }
    leaf data-retrieval {
      if-feature
        "remote-data-retrieval-initiate or
         remote-mib-retrieval-respond";
      type boolean;
      description
        "Variable retrieval support.
         This affects the setting of the 'Variable Retrieval' bit
         in the OAM configuration field put in the Information
         OAMPDU. This bit indicates to the peer device that the
         OAM entity can send and receive Variable Request and
         Response OAMPDUs.";
    }
  }
  leaf revision {
    type uint64;
    config false;
    description
      "Configuration revision.";
    reference
      "IEEE Std 802.3, 30.3.6.1.12 and 30.3.6.1.13";
  }
  leaf oammtu {
    type uint16;
    units "octets";
    config false;
    description
      "The maximum OAMPDU size for the local node. The peer OAM
       entities exchange the maximum size they can support and
       negotiate to use the smaller of the two maximum OAMPDU
       sizes.";
    reference
      "IEEE Std 802.3, 30.3.6.1.8 and 30.3.6.1.9";
  }
}

grouping discovery-info {
  description
    "Information relating to the discovery process.";
  container local {
    description
      "Properties of the local device.";
    leaf operational-status {
      type operational-state;
```

```
    config false;
    mandatory true;
    description
      "Operational status.";
    reference
      "IETF RFC 4878, dot3OamOperStatus; IEEE Std 802.3,
       30.3.6.1.4, 30.3.6.1.10, and 30.3.6.1.11";
  }
  leaf loopback-mode {
    if-feature
      "remote-loopback-initiate or
       remote-loopback-respond";
    type loopback-status;
    config false;
    mandatory true;
    description
      "The loopback mode the interface is in.";
    reference
      "IEEE Std 802.3, 30.3.6.1.14";
  }
  uses discovery-local;
}
container remote {
  config false;
  description
    "Properties of the remote (peer) device.";
  leaf mac-address {
    type ieee:mac-address;
    description
      "Remote MAC address.";
    reference
      "IEEE Std 802.3, 30.3.6.1.5";
  }
  leaf vendor-oui {
    type vendor-oui;
    description
      "Remote vendor OUI.";
    reference
      "IEEE Std 802.3, 30.3.6.1.16";
  }
  leaf vendor-info {
    type uint64;
    description
      "Remote vendor info. The semantics of this value are
       proprietary and specific to the vendor.";
    reference
      "IEEE Std 802.3, 30.3.6.1.17";
  }
  leaf loopback-mode {
    type loopback-status;
    mandatory true;
    description
      "The loopback mode the interface is in.";
    reference
      "IEEE Std 802.3, 30.3.6.1.15";
  }
  uses discovery-remote;
}
```

```
}

augment "/if:interfaces/if:interface" {
  when "derived-from-or-self(if:type, 'ianaift:ethernetCsmacd') or
        derived-from-or-self(if:type, 'ianaift:ptm') " {
    description
      "Augments the interface model with nodes
       specific to Ethernet Link OAM.";
  }
  description
    "Augments Ethernet interface model with nodes
     specific to Ethernet Link OAM.";
  container link-oam {
    presence "Implies Link OAM is configured on the interface.";
    description
      "Interface operational state for Ethernet Link OAM.";
    action remote-loopback {
      if-feature "remote-loopback-initiate";
      description
        "Start/stop remote loopback on the specified interface.";
      reference
        "IEEE Std 802.3, 57.1.2:b";
      input {
        leaf enable {
          type boolean;
          mandatory true;
          description
            "Whether to enable or disable remote loopback.";
        }
      }
      output {
        leaf success {
          type boolean;
          mandatory true;
          description
            "True if the operation was successful,
             false otherwise.";
        }
        leaf error-message {
          type string;
          description
            "If the operation failed, optionally used to
             provide extra details.";
        }
      }
    }
    action reset-stats {
      description
        "Reset Ethernet Link OAM statistics on this interface.";
      output {
        leaf success {
          type boolean;
          mandatory true;
          description
            "True if the operation was successful,
             false otherwise.";
        }
        leaf error-message {
```

```
      type string;
      description
        "If the operation failed, optionally used to provide
         extra details.";
    }
  }
}
notification non-threshold-event {
  description
    "This notification is sent when a local or remote
     non-threshold crossing event is detected.";
  uses event-details {
    refine "event-type" {
      must ". != 'threshold-event-type'" {
        description
          "This leaf is not set to
           'threshold-event-type'.";
      }
    }
  }
}
notification threshold-event {
  if-feature
    "link-monitoring-local or
     link-monitoring-remote";
  description
    "This notification is sent when a local or remote
     threshold crossing event is detected.";
  uses event-details {
    refine "event-type" {
      must ". = 'threshold-event-type'" {
        description
          "This leaf is set to 'threshold-event-type'.";
      }
    }
  }
  uses threshold-event-details;
}
leaf admin {
  type admin-state;
  default "disabled";
  description
    "This object is used to provision the default
     administrative OAM mode for this interface. This object
     represents the desired state of OAM for this interface.
     It starts in the disabled state until an explicit
     management action or configuration information retained
     by the system causes a transition to the enabled(1)
     state. When enabled(1), Ethernet OAM will attempt to
     operate over this interface. The default value is
     implementation-dependent.";
}
container discovery-info {
  description
    "Information relating to the discovery process.";
  uses discovery-info;
}
container event-log {
```

```
    config false;
    description
      "List of Ethernet Link OAM event log entries on the
       interface.";
    list event-log-entry {
      key "index";
      description
        "Ethernet Link OAM event log entry.";
      leaf index {
        type uint64;
        description
          "Index of this event in the event log.";
      }
      uses event-details;
      uses threshold-event-details;
    }
  }
  container statistics {
    config false;
    description
      "Statistics for an 802.3 OAM interface.";
    uses statistics-common;
    leaf local-error-symbol-period-log-entries {
      type yang:counter64;
      mandatory true;
      description
        "Number of local error symbol period log entries.";
    }
    leaf local-error-frame-log-entries {
      type yang:counter64;
      mandatory true;
      description
        "Number of local error frame log entries.";
    }
    leaf local-error-frame-period-log-entries {
      type yang:counter64;
      mandatory true;
      description
        "Number of local error frame period log entries.";
    }
    leaf local-error-frame-second-log-entries {
      type yang:counter64;
      mandatory true;
      description
        "Number of local error frame second log entries.";
    }
    leaf remote-error-symbol-period-log-entries {
      if-feature "link-monitoring-remote";
      type yang:counter64;
      mandatory true;
      description
        "Number of remote error symbol period log entries.";
    }
    leaf remote-error-frame-log-entries {
      if-feature "link-monitoring-remote";
      type yang:counter64;
      mandatory true;
      description
```

```
          "Number of remote error frame log entries.";
      }
      leaf remote-error-frame-period-log-entries {
        if-feature "link-monitoring-remote";
        type yang:counter64;
        mandatory true;
        description
          "Number of remote error frame period log entries.";
      }
      leaf remote-error-frame-second-log-entries {
        if-feature "link-monitoring-remote";
        type yang:counter64;
        mandatory true;
        description
          "Number of remote error frame second log entries.";
      }
    }
  }
 }
}
module ieee802-ethernet-lldp {
  yang-version 1.1;
  namespace "urn:ieee:std:802.3:yang:ieee802-ethernet-lldp";
  prefix ieee802-eth-lldp;

  import ieee802-dot1ab-lldp {
    prefix lldp;
    reference
      "IEEE Std 802.1ABcu-2021";
  }
  import ieee802-ethernet-phy-type {
    prefix ieee802-phy;
    reference
      "IEEE Std 802.3-2022";
  }

  organization
    "IEEE Std 802.3 Ethernet Working Group
     Web URL: http://www.ieee802.org/3/";
  contact
    "Web URL: http://www.ieee802.org/3/";
  description
    "This module contains YANG definitions for configuring LLDP for
     802.3 Ethernet Interfaces.
     In this YANG module, 'Ethernet interface' can be interpreted
     as referring to 'IEEE Std 802.3 compliant Ethernet
     interfaces'.";

  revision 2024-07-1712-23 {
    description
      "Updates under IEEE Std 802.3.2-202x, Draft D3.01";
    reference
      "IEEE Std 802.3-2022, unless dated explicitly";
  }

  typedef port-class-type {
    type enumeration {
      enum p-class-pse {
```

```
        value 0;
        description
          "Power Sourcing Equipment";
      }
      enum p-class-pd {
        value 1;
        description
          "Powered Device";
      }
    }
    description
      "Enumeration for the power port class";
    reference
      "IEEE Std 802.3, 30.12.2.1.5";
  }

  typedef pse-pinout-type {
    type enumeration {
      enum signal {
        value 0;
        description
          "PSE Pinout Alternative A";
      }
      enum spare {
        value 1;
        description
          "PSE Pinout Alternative B";
      }
    }
    description
      "Enumeration for the pinout alternatives used for PD
       detection and power ";
    reference
      "IEEE Std 802.3, 30.12.2.1.9";
  }

  typedef pse-power-class-type {
    type enumeration {
      enum class0 {
        value 0;
        description
          "Class 0 PD";
      }
      enum class1 {
        value 1;
        description
          "Class 1 PD";
      }
      enum class2 {
        value 2;
        description
          "Class 2 PD";
      }
      enum class3 {
        value 3;
        description
          "Class 3 PD";
      }
```

```
      enum class4 {
        value 4;
        description
          "Class 4 PD";
      }
    }
    description
      "Enumeration for the PD class";
    reference
      "IEEE Std 802.3, 30.12.2.1.10";
  }

  typedef power-class-ext-AB-type {
    type enumeration {
      enum singlesig {
        value 0;
        description
          "Single-signature PD or 2-pair only PSE";
      }
      enum class1 {
        value 1;
        description
          "Class 1";
      }
      enum class2 {
        value 2;
        description
          "Class 2";
      }
      enum class3 {
        value 3;
        description
          "Class 3";
      }
      enum class4 {
        value 4;
        description
          "Class 4";
      }
      enum class5 {
        value 5;
        description
          "Class 5";
      }
    }
    description
      "Enumeration for the assigned power class ";
    reference
      "IEEE Std 802.3, 30.12.3.1.26";
  }

  typedef power-class-ext-type {
    type enumeration {
      enum dualsig {
        value 0;
        description
          "Dual-signature PD";
      }
```

```
    enum class1 {
      value 1;
      description
        "Class 1";
    }
    enum class2 {
      value 2;
      description
        "Class 2";
    }
    enum class3 {
      value 3;
      description
        "Class 3";
    }
    enum class4 {
      value 4;
      description
        "Class 4";
    }
    enum class5 {
      value 5;
      description
        "Class 5";
    }
    enum class6 {
      value 6;
      description
        "Class 6";
    }
    enum class7 {
      value 7;
      description
        "Class 7";
    }
    enum class8 {
      value 8;
      description
        "Class 8";
    }
  }
  description
    "Enumeration for the assigned power class ";
  reference
    "IEEE Std 802.3, 30.12.3.1.28";
}

typedef power-type {
  type enumeration {
    enum type4dualsigPD {
      value 0;
      description
        "Type 4 dual-signature PD";
    }
    enum type4singlesigPD {
      value 1;
      description
        "Type 4 single-signature PD";
```

```
    }
    enum type3dualsigPD {
      value 2;
      description
        "Type 3 dual-signature PD";
    }
    enum type3singlesigPD {
      value 3;
      description
        "Type 3 single-signature PD";
    }
    enum type4PSE {
      value 4;
      description
        "Type 4 PSE";
    }
    enum type3PSE {
      value 5;
      description
        "Type 3 PSE";
    }
  }
  description
    "Enumeration for the PD class";
  reference
    "IEEE Std 802.3, 30.12.2.1.29";
}

typedef power-priority-type {
  type enumeration {
    enum low {
      value 0;
      description
        "low priority PD";
    }
    enum high {
      value 1;
      description
        "high priority PD";
    }
    enum critical {
      value 2;
      description
        "critical priority PD";
    }
    enum unknown {
      value 3;
      description
        "priority unknown";
    }
  }
  description
    "Enumeration for possible priorities of a PD system";
  reference
    "IEEE Std 802.3, 30.12.2.1.16";
}

typedef power-source-type {
```

```
    type enumeration {
      enum pse-primary {
        value 0;
        description
          "PSE powered by a primary power source";
      }
      enum pse-backup {
        value 1;
        description
          "PSE powered by a backup power source";
      }
      enum pse-unknwon {
        value 2;
        description
          "PSE powered by an unkown power source";
      }
      enum pd-pse-and-local {
        value 3;
        description
          "PD powered by a PSE and locally";
      }
      enum pd-local-only {
        value 4;
        description
          "PD powered only locally";
      }
      enum pd-pse-only {
        value 5;
        description
          "PD powered by PD only";
      }
      enum pd-unknown {
        value 6;
        description
          "PD powered by an uknown source";
      }
    }
    description
      "Enumeration for the power sources of the
       remote system. When the remote system is a PSE, it
       indicates whether it is being powered by a primary
       power source; a backup power source; or unknown.
       When the remote system is a PD, it indicates whether
       it is being powered by a PSE and locally;
       locally only; by a PSE only; or unknown.";
    reference
      "IEEE Std 802.3, 30.12.2.1.15";
  }

  typedef powering-status-type {
    type enumeration {
      enum 4PdualsigPD {
        value 0;
        description
          "4-pair powering a dual-signature PD";
      }
      enum 4PsinglesigPD {
        value 1;
```

```
      description
        "4-pair powering a single-signature PD";
    }
    enum 2P {
      value 2;
      description
        "2-pair powering";
    }
  }
  description
    "Enumeration for the power status of the PSE";
  reference
    "IEEE Std 802.3, 30.12.2.1.23";
}

typedef powered-status-type {
  type enumeration {
    enum 4PdualsigPD {
      value 0;
      description
        "4-pair powered dual-signature PD";
    }
    enum 2PdualsigPD {
      value 1;
      description
        "2-pair powered dual-signature PD";
    }
    enum singlesigPD {
      value 2;
      description
        "powered single-signature PD";
    }
  }
  description
    "Enumeration for the power status of the PSE";
  reference
    "IEEE Std 802.3, 30.12.2.1.24";
}

typedef power-pairs-type {
  type enumeration {
    enum altA {
      value 0;
      description
        "Alternative A";
    }
    enum altB {
      value 1;
      description
        "Alternative B";
    }
    enum both {
      value 2;
      description
        "both";
    }
  }
  description
```

"Enumeration for the PSE Pinout Alternative";
reference
"IEEE Std 802.3, 30.12.2.1.25";
}

~~augment "/lldp:lldp/lldp:port" {~~
~~description~~
~~"Augments port with 802.3 port config tlvs";~~
~~leaf tlvs-port-config-enable {~~
typedef measurement-source-type ~~bits~~ {
~~bit mac-phy-config-status {~~
~~position 0;~~
type enumeration {
enum noRequest {
description
"No Request";
}
enum altA {
description
~~"IEEE Std 802.3, 30.12.1.1.1~~"Pairset Alternative A / Mode A";
}
~~bit power-via-mdi~~enum altB {
~~position 1;~~
description
~~"IEEE Std 802.3, 30.12.1.1.1~~"Pairset Alternative B / Mode B";
}
~~bit unused~~enum total {
~~position 2;~~
description
~~"IEEE Std 802.3, 30.12.1.1.1~~"Port total";
}
~~bit max-frame-size {~~}
~~position 3;~~
description
~~"IEEE Std 802.3, 30.12.1.1.1~~"PoE measurement source";
~~}~~
~~bit eee-tlv {~~
~~position 4;~~
~~description~~
~~"IEEE Std 802.3, 30.12.1.1.1";~~
~~}~~
~~bit eee-fast-wake-tlv {~~
~~position 5;~~
~~description~~
~~"IEEE Std 802.3, 30.12.1.1.1";~~
~~}~~
~~bit additional-ethernet-capabilities-tlv {~~
~~position 6;~~
~~description~~
~~"IEEE Std 802.3, 30.12.1.1.1";~~
~~}~~
~~}~~
~~description~~
~~"Bitmap that corresponds to an IEEE 802.3 subtype~~
~~associated with a specific IEEE 802.3 port config TLV";~~
reference
"IEEE Std 802.3, 30.12.2.1.42 and 30.12.3.1.~~1~~42";
~~}~~

```
    }

      augment "/lldp:lldp/lldp:port" {
        description
          "Augments port with 802.3 port read-only objects";
        leaf auto-negotiation-supported {
          type boolean;
          config false;
          description
            "True if the port supports Auto-negotiation";
          reference
            "IEEE Std 802.3, 30.12.2.1.1";
        }
        leaf auto-negotiation-enabled {
          type boolean;
          config false;
          description
            "True if Auto-negotiation is enabled";
          reference
            "IEEE Std 802.3, 30.12.2.1.2";
        }
        /* Deprecated MAU objects */
        leaf auto-negotiation-cap {
          type binary {
            length "2";
          }
          config false;
          status deprecated;
          description
            "A read-only 2-octet value that contains the value (bitmap)
             of the ifMauAutoNegCapAdvertisedBits object
             (defined in IETF RFC 4836) which is associated with the
             given port on the local system.";.
             Superseeded by auto-negotiation-cap-bits.";
          reference
            "IEEE Std802.3, 30.12.2.1.3";
        }
        leaf operational-mau-type {
          type int32;
          config false;
          status deprecated;
          description
            "32-bit integer value that indicates the operational MAU
             type of the given port";.
             Superseeded by operational-pmd-type.";
          reference
            "IEEE Std 802.3, 30.12.2.1.4";
        }
        leaf auto-negotiation-cap-bits {
          type ieee802-phy:auto-neg-ability-bits;
          description
            "The capabilities advertised by the local Auto-Negotiation
             entity.";
          reference
            "IEEE Std 802.3, 30.6.1.1.6,
             aAutoNegadvertisedTechnologyAbility.";
        }
        leaf operational-pmd-type {
```

```
        type identityref {
          base ieee802-phy:pmd-type;
        }
        description
          "Operational PMD type of the given port";
        reference
          "IEEE Std 802.3, Clause 30.5.1.1.2 aMAUType";
      }
      /* Clause 30 & 145 PoE related LLDP definitions */
      leaf power-port-class {
        type port-class-type;
        config false;
        description
          "A read-only value that identifies the port Class of the
           given port";
        reference
          "IEEE Std 802.3, 30.12.2.1.5";
      }
      leaf mdi-power-supported {
        type boolean;
        config false;
        description
          "True if MDI power is supported";
        reference
          "IEEE Std 802.3, 30.12.2.1.6";
      }
      leaf mdi-power-enabled {
        type boolean;
        config false;
        description
          "True if MDI power is enabled";
        reference
          "IEEE Std 802.3, 30.12.2.1.7";
      }
      leaf power-pair-controlable {
        type boolean;
        config false;
        description
          "True if the pair selection can be controlled";
        reference
          "IEEE Std 802.3, 30.12.2.1.8";
      }
      leaf power-pairs {
        type pse-pinout-type;
        config false;
        description
          "Indicates which pinout alernative is used for PD detection
           and power";
        reference
          "IEEE Std 802.3, 30.12.2.1.9";
      }
      leaf local-power-class {
        type pse-power-class-type;
        config false;
        description
          "PD Power Class";
        reference
          "IEEE Std 802.3, 30.12.2.1.10";
```

```
    }
    /* Obsolete Link Aggregation objects
        Link aggregation has not been part of 802.3 since IEEE 802.1AX-
2008
        */
    leaf link-aggregation-status {
      type bits {
        bit aggregation-capability {
          position 0;
          description
            "IEEE Std 802.3, 79.3.3.1";
        }
        bit aggregation-status {
          position 1;
          description
            "IEEE Std 802.3, 79.3.3.1";
        }
      }
      config false;
      status obsolete;
      description
        "The bitmap value which contains the link aggregation
         capabilities and the current aggregation
         status of the link";
      reference
        "IEEE Std 802.3, 30.12.2.1.11";
    }
    leaf aggregation-port-id {
      type int32;
      config false;
      status obsolete;
      description
        "The unique identifier allocated to this Aggregation Port
         by the local System.";
      reference
        "IEEE Std 802.3, 30.12.2.1.12";
    }
    leaf local-max-frame-size {
      type int32;
      config false;
      status obsolete;
      description
        "An integer value indicating the maximum supported frame
         size in octets on the given port of the local system.";
      reference
        "IEEE Std 802.3, 30.12.2.1.13";
    }
    leaf power-type {
      type bits {
        bit type1-or-greater {
          position 0;
          description
            "0-type1, 1-greater than type1";
        }
        bit pse-or-pd {
          position 1;
          description
            "0-pse, 1-pd";
```

```
          }
        }
        config false;
        description
          "A read-only attribute that returns a bit string indicating
           whether the local system is a PSE or a PD and whether it
           is Type 1 or greater than Type 1. The first bit indicates
           Type 1 or greater than Type 1. The second bit indicates
           PSE or PD. A PSE sets this bit to indicate a PSE. A PD sets
           this bit to indicate a PD. See also aLldpXdot3LocPowerTypeExt.
           aLldpXdot3LocPowerTypeExt..";Superseeded by power-support";
        reference
          "IEEE Std 802.3, 30.12.2.1.14";
      }
      leaf power-support {
        type bits {
          bit pse-pd {
            description
              "0-PD, 1-PSE";
          }
          bit pse-power-support {
            description
              "0-not supported, 1-supported";
          }
          bit pse-power-state {
            description
              "0-disabled 1-enabled";
          }
          bit pse-pairs-selection {
            description
              "0-pair selection not controllable
               1-pair selection  controllable";
          }
        }
        description
          "A read-only attribute that returns a bit string indicating
           whether local system PoE support.";
        reference
          "IEEE Std 802.3, 79.3.2.1 MDI power support";
      }
      leaf power-source {
        type power-source-type;
        config false;
        description
          "Indicates the power sources of the local system. A PSE
           indicates whether it is being powered by a primary power
           source; a backup power source; or unknown. A PD indicates
           whether it is being powered by a PSE and locally; by a PSE
           only; or unknown.;";
        reference
          "IEEE Std 802.3, 30.12.2.1.15";
      }
      leaf local-power-priority {
        type power-priority-type;
        description
          "Priority of a PD system. For a PSE, this is the priority
           that the PSE assigns to the PD. For a PD, this is the
           priority that the PD requests from the PSE";
```

```
      reference
        "IEEE Std 802.3, 30.12.2.1.16";
    }
    leaf pd-requested-power-value {
      type int32;
      config false;
      description
        "PD requested power value. For a PD, it is the power value
         that the PD has currently requested from the remote
         system. For a PSE, it is the power value that the PSE
         mirrors back to the remote system";
      reference
        "IEEE Std 802.3, 30.12.2.1.17";
    }
    leaf pd-requested-power-value-a {
      type int32;
      config false;
      description
        "A read-only attribute that returns the PD requested power
         value for the Mode A pairset in units of 0.1 W. For a PD,
         it isit's the power value that the PD has currently requested
         from the remote system for the Mode A pairset. For a PSE,
         it is the power value for the Alternative A pairset that
         the PSE echoes back to the remote system";
      reference
        "IEEE Std 802.3, 30.12.2.1.18";
    }
    leaf pd-requested-power-value-b {
      type int32;
      config false;
      description
        "A read-only attribute that returns the PD requested power
         value for the Mode B pairset in units of 0.1 W.
         For a PD, it is the power value that the PD has currently
         requested from the remote system for the Mode B pairset.
         For a PSE, it is the power value for the Alternative B
         pairset that the PSE echoes back to the remote system";
      reference
        "IEEE Std 802.3, 30.12.2.1.19";
    }
    leaf pse-allocated-power-value {
      type int32;
      config false;
      description
        "PSE allocated power value. For a PSE, it is the power
         value that the PSE has currently allocated to the remote
         system. For a PD, it is the power value that the PD mirrors
         back to the remote syste";
      reference
        "IEEE Std 802.3, 30.12.2.1.20";
    }
    leaf pse-allocated-power-value-a {
      type int32;
      config false;
      description
        " PSE allocated power value for the Alternative A pairset
          in units of 0.1 W.For a PSE, it is the power value for the
          Alternative A pairset that the PSE has currently allocated
```

```
        to the remote system. For a PD, it is the power value for
        the Mode A pairset that the PD echoes back to the remote
        system.";
    reference
      "IEEE Std 802.3, 30.12.2.1.21";
  }
  leaf pse-allocated-power-value-b {
    type int32;
    config false;
    description
      "PSE allocated power value for the Alternative B pairset
       in units of 0.1 W. For a PSE, it is the power value for the
       Alternative B pairset that the PSE has currently
       allocated to the remote system. For a PD, it is the power
       value for the Mode B pairset that the PD echoes back to the
       remote system.";
    reference
      "IEEE Std 802.3, 30.12.2.1.22";
  }
  leaf pse-powering-status {
    type powering-status-type;
    config false;
    description
      "A read only value that indicates the powering status of
       the PSE. For a PD, the contents of this
       attribute are undefined.";
    reference
      "IEEE Std 802.3, 30.12.2.1.23";
  }
  leaf pd-powered-status {
    type powered-status-type;
    config false;
    description
      "A read only value that indicates the powering status of
       the PD. For a PSE, the contents of this attribute are
       undefined";
    reference
      "IEEE Std 802.3, 30.12.2.1.24";
  }
  leaf power-pairs-ext {
    type power-pairs-type;
    config false;
    description
      "A read-only value that identifies the supported PSE
       Pinout Alternative specified in 145.2.4. For a PSE, this
       attribute contains the value of the aPSEPowerPairs
        attribute (see 30.9.1.1.4). For a PD, the contents of this
        attribute are undefined";
    reference
      "IEEE Std 802.3, 30.12.2.1.25";
  }
  leaf power-class-ext-A {
    type power-class-ext-AB-type;
    config false;
    description
      "For a dual-signature PD, a read-only value that indicates
       the requested Class for Mode A during Physical Layer
       Classification (see 145.3.6). For a single-signature PD, a
```

```
     read-only value set to ‘singlesig’.For a PSE connected to
     a dual-signature PD, a read-only value that indicates the
     currently assigned Class for Mode A (see 145.2.8). For a
     PSE connected to a single-signature PD or a PSE that
     operates only in 2-pair mode, a read-only value set to
     ‘singlesig’";
  reference
    "IEEE Std 802.3, 30.12.2.1.26";
}
leaf power-class-ext-B {
  type power-class-ext-AB-type;
  config false;
  description
    "For a dual-signature PD, a read-only value that indicates
     the requested Class for Mode B during Physical Layer
     Classification (see 145.3.6). For a single-signature PD,
     a read-only value set to ‘singlesig’.For a PSE connected to
     a dual-signature PD, a read-only value that indicates the
     currently assigned Class for Mode B (see 145.2.8). For a
     PSE connected to a single-signature PD or a PSE that
     operates only in 2-pair mode, a read-only value set to
     ‘singlesig’";
  reference
    "IEEE Std 802.3, 30.12.2.1.27";
}
leaf power-class-ext {
  type power-class-ext-type;
  config false;
  description
    "For a single-signature PD, a read-only value that
     indicates the requested Class during Physical
     Layer Classification (see 145.3.6). For a dual-signature
     PD, a read-only value set to ‘dualsig’.
     For a PSE connected to a single-signature PD or a PSE that
     operates only in 2-pair mode, a read-only value that
     indicates the currently assigned Class (see 145.2.8).
     For a PSE connected to a dual-signature PD,
     a read-only value set to ‘dualsig’.’";
  reference
    "IEEE Std 802.3, 30.12.2.1.28";
}
leaf power-type-ext {
  type power-type;
  config false;
  description
    "A read-only attribute that returns a value to indicate if
     the local system is a Type 3 or Type 4 PSE or PD and, in
     the case of a Type 3 or Type 4 PD, if it is a
     single-signature PD or a dual-signature
     PD";
  reference
    "IEEE Std 802.3, 30.12.2.1.29";
}
leaf pd-load {
  type boolean;
  config false;
  description
    "For a dual-signature PD, a read-only attribute that
```

```
       returns whether the load of a dual-signature PD is
       electrically isolated, as defined in 79.3.2.10.2. For a
       single-signature PD or a PSE, the value of this attribute
       is FALSE";
    reference
      "IEEE Std 802.3, 30.12.2.1.30";
  }
  leaf pd-4pid {
    type boolean;
    config false;
    description
      "A read-only Boolean attribute indicating whether the local
       PD system supports powering of both PD Modes.";
    reference
      "IEEE Std 802.3, 30.12.2.1.31";
  }
  leaf pse-max-avail-power {
    type int32;
    config false;
    description
      "A read-only attribute that returns the local PSE maximum
       available power value in units of 0.1 W";
    reference
      "IEEE Std 802.3, 30.12.2.1.32";
  }
  leaf pse-autoclass-support {
    type boolean;
    config false;
    description
      "Indicates whether the local PSE system supports Autoclass.";
    reference
      "IEEE Std 802.3, 30.12.2.1.33";
  }
  leaf autoclass-completed {
    type boolean;
    config false;
    description
      "Indicates whether the local PSE system has completed the
       Autoclass measurement.";
    reference
      "IEEE Std 802.3, 30.12.2.1.34";
  }
  leaf autoclass-request {
    type boolean;
    config false;
    description
      "A read-only Boolean attribute indicating whether the local
       PD system is requesting an Autoclass measurement;";
    reference
      "IEEE Std 802.3, 30.12.2.1.35";
  }
  leaf power-down-request {
    type int32;
    description
      "A read-write attribute that indicates the local PD system
       is requesting a power down when the value is  0x1D.;";
    reference
      "IEEE Std 802.3, 30.12.2.1.36";
```

```
    }
    leaf power-down-time {
      type int32;
      description
        "A read-write attribute that indicates the number of
         seconds the PD requests to stay powered off. A value of
         zero indicates an indefinite amount of time;";
      reference
        "IEEE Std 802.3, 30.12.2.1.37";
    }
    /* PoE measurement objects */
    leaf meas-voltage-support {
      type boolean;
      config false;
      description
        "A read-only attribute that indicates the local device is
         capable of providing a voltage measurement.;";
      reference
        "IEEE Std 802.3, 30.12.2.1.38";
    }
    leaf meas-current-support {
      type boolean;
      config false;
      description
        "A read-only attribute that indicates the local device is
         capable of providing a current measurement.;";
      reference
        "IEEE Std 802.3, 30.12.2.1.39";
    }
    leaf meas-power-support {
      type boolean;
      config false;
      description
        "A read-only attribute that indicates the local device is
         capable of providing a power measurement.;";
      reference
        "IEEE Std 802.3, 30.12.2.1.40";
    }
    leaf meas-energy-support {
      type boolean;
      config false;
      description
        "A read-only attribute that indicates the local device is
         capable of providing a energy measurement.;";
      reference
        "IEEE Std 802.3, 30.12.2.1.41";
    }
    /* NOTE - What do these values mean? */
    leaf measurement-source {
      type bits {
        bit bit1 {
          position 0;
          description
            "-";"????";
        }
        bit bit2 {
          position 1;
          description
```

```
            "-";
          }
        }
        status obsolete;
        description
          "A read-write attribute value that indicates to local
           device on which Alternative or Mode the measurement
           is to be taken";
        reference
          "IEEE Std 802.3, 30.12.2.1.42";
      }
      leaf meas-voltage-request {
        type boolean;
        config false;
        description
          "A read-only attribute that indicates the local device is
           requesting a voltage measurement from the remote device.";
        reference
          "IEEE Std 802.3, 30.12.2.1.43";
      }
      leaf meas-current-request {
        type boolean;
        config false;
        description
          "A read-only attribute that indicates the local device is
           requesting a current measurement from the remote device.";
        reference
          "IEEE Std 802.3, 30.12.2.1.44";
      }
      leaf meas-power-request {
        type boolean;
        config false;
        description
          "A read-only attribute that indicates the local device is
           requesting a power measurement from the remote device.";
        reference
          "IEEE Std 802.3, 30.12.2.1.45";
      }
      leaf meas-energy-request {
        type boolean;
        config false;
        description
          "A read-only attribute that indicates the local device is
           requesting an energy measurement from the remote device.";
        reference
          "IEEE Std 802.3, 30.12.2.1.46";
      }
      leaf meas-voltage-valid {
        type boolean;
        config false;
        description
          "A read-only attribute that indicates the local device's
           voltage measurement is valid.";
        reference
          "IEEE Std 802.3, 30.12.2.1.47";
      }
      leaf meas-current-valid {
        type boolean;
```

```
    config false;
    description
      "A read-only attribute that indicates the local device's
       current measurement is valid.";
    reference
      "IEEE Std 802.3, 30.12.2.1.48";
  }
  leaf meas-power-valid {
    type boolean;
    config false;
    description
      "A read-only attribute that indicates the local device's
       power measurement is valid.";
    reference
      "IEEE Std 802.3, 30.12.2.1.49";
  }
  leaf meas-energy-valid {
    type boolean;
    config false;
    description
      "A read-only attribute that indicates the local device's
       energy measurement is valid.";
    reference
      "IEEE Std 802.3, 30.12.2.1.50";
  }
  leaf meas-voltage-uncertainty {
    type int32;
    config false;
    description
      "A read-only attribute that indicates the expanded
       uncertainty (coverage factor k = 2) for the device's
       voltage measurement.";
    reference
      "IEEE Std 802.3, 30.12.2.1.51";
  }
  leaf meas-current-uncertainty {
    type int32;
    config false;
    description
      "A read-only attribute that indicates the expanded
       uncertainty (coverage factor k = 2) for the device's
       current measurement.";
    reference
      "IEEE Std 802.3, 30.12.2.1.52";
  }
  leaf meas-power-uncertainty {
    type int32;
    config false;
    description
      "A read-only attribute that indicates the expanded
       uncertainty (coverage factor k = 2) for the device's
       power measurement.";
    reference
      "IEEE Std 802.3, 30.12.2.1.53";
  }
  leaf meas-energy-uncertainty {
    type int32;
    config false;
```

```
    description
      "A read-only attribute that indicates the expanded
       uncertainty (coverage factor k = 2) for the device's
       energy measurement.";
    reference
      "IEEE Std 802.3, 30.12.2.1.54";
  }
  leaf voltage-measurement {
    type int32;
    config false;
    description
      "A read-only attribute that returns the measured device
       voltage.";
    reference
      "IEEE Std 802.3, 30.12.2.1.55";
  }
  leaf current-measurement {
    type int32;
    config false;
    description
      "A read-only attribute that returns the measured device
       current.";
    reference
      "IEEE Std 802.3, 30.12.2.1.56";
  }
  leaf power-measurement {
    type int32;
    config false;
    description
      "A read-only attribute that returns the measured device
       power.";
    reference
      "IEEE Std 802.3, 30.12.2.1.57";
  }
  leaf energy-measurement {
    type int32;
    config false;
    description
      "A read-only attribute that returns the measured device
       energy.";
    reference
      "IEEE Std 802.3, 30.12.2.1.58";
  }
  leaf pse-power-price-index {
    type int32;
    config false;
    description
      "A read-only attribute that returns an index of the price
       of power being sourced by the PSE. For a PD, this value
       is undefined";
    reference
      "IEEE Std 802.3, 30.12.2.1.59";
  }
  leaf local-response {
    type int32;
    config false;
    description
      "The maximum time required to update
```

```
     pse-allocated-power-value";
  reference
    "IEEE Std 802.3, 30.12.2.1.60";
}
leaf local-system-ready {
  type boolean;
  config false;
  description
    "Initialization status of the Data Link Layer
     classification engine on the local system";
  reference
    "IEEE Std 802.3, 30.12.2.1.61";
}
leaf tx-system-value {
  type int32;
  config false;
  description
    "Returns the value of Tw_sys_tx that the local system can
     support in the transmit direction.";
  reference
    "IEEE Std 802.3, 30.12.2.1.62";
}
leaf tx-system-value-echo {
  type int32;
  config false;
  description
    "Returns the value of Tw_sys_tx that the emote system is
     advertising that it can support in the transmit direction
     and is echoed by the local system under the control of the
     EEE DLL receiver state diagram.";
  reference
    "IEEE Std 802.3, 30.12.2.1.63";
}
leaf rx-system-value {
  type int32;
  config false;
  description
    "Returns the value of Tw_sys_tx that the local system is
     requesting in the receive direction.";
  reference
    "IEEE Std 802.3, 30.12.2.1.64";
}
leaf rx-system-value-echo {
  type int32;
  config false;
  description
    "Returns the value of Tw_sys_tx that the remote system is
     advertising that it is requesting in the receive direction
     and is echoed by the local system under the control of the
     EEE DLL transmitter state diagram.";
  reference
    "IEEE Std 802.3, 30.12.2.1.65";
}
leaf fallback-system-value {
  type int32;
  config false;
  description
    "Returns the value of the fallback Tw_sys_tx that the local
```

```
           system is advertising to the remote system.";
        reference
          "IEEE Std 802.3, 30.12.2.1.66";
      }
      /* LDP EEE objects */
      leaf tx-dll-ready {
        type boolean;
        config false;
        description
          "Returns the initialization status of the EEE transmit Data
           Link Layer management function on the local system.";
        reference
          "IEEE Std 802.3, 30.12.2.1.67";
      }
      leaf rx-dll-ready {
        type boolean;
        config false;
        description
          "Returns the initialization status of the EEE receive Data
           Link Layer management function on the local system.";
        reference
          "IEEE Std 802.3, 30.12.2.1.68";
      }
      leaf dll-enabled {
        type boolean;
        config false;
        description
          "Returns the status of the EEE capability negotiation on
           the local system.";
        reference
          "IEEE Std 802.3, 30.12.2.1.69";
      }
      leaf tx-system-fw {
        type boolean;
        config false;
        description
          "Returns the value of LPI_FW that the local system can
           support in the transmit direction.";
        reference
          "IEEE Std 802.3, 30.12.2.1.70";
      }
      leaf tx-system-fw-echo {
        type boolean;
        config false;
        description
          "Returns the value of LPI_FW that the remote system is
           advertising that it can support in the transmit direction
           and is echoed by the local system under the control of the
           EEE DLL receiver state diagram.";
        reference
          "IEEE Std 802.3, 30.12.2.1.71";
      }
      leaf rx-system-fw {
        type boolean;
        config false;
        description
          "Returns the value of LPI_FW that the local system is
           requesting in the receive direction.";
```

```
      reference
        "IEEE Std 802.3, 30.12.2.1.72";
    }
    leaf rx-system-fw-echo {
      type boolean;
      config false;
      description
        "Returns the value of LPI_FW that the remote system is
         advertising that it is requesting in the receive direction
         and is echoed by the local system under the control of the
         EEE DLL transmitter state diagram.";
      reference
        "IEEE Std 802.3, 30.12.2.1.73";
    }
    /* LLDP preemption objects */
    leaf preemption-supported {
      type boolean;
      config false;
      description
        "Indicates whether the given port (associated with the local
         System) supports the preemption capability.";
      reference
        "IEEE Std 802.3, 30.12.2.1.74";
    }
    leaf preemption-enabled {
      type boolean;
      config false;
      description
        "Indicates whether the preemption capability is enabled on
         the given port associated with the local System.";
      reference
        "IEEE Std 802.3, 30.12.2.1.75";
    }
    leaf preemption-active {
      type boolean;
      config false;
      description
        "Indicates whether the preemption capability is active on
         the given port associated with the local System.;";
      reference
        "IEEE Std 802.3, 30.12.2.1.76";
    }
    leaf additional-fragment-size {
      type int32;
      config false;
      description
        "Indicate the minimum size of non-final fragments supported
         by the receiver on the given port associated with the local
         System. This value is expressed in units of 64 octets of
         additional fragment length.";
      reference
        "IEEE Std 802.3, 30.12.2.1.77";
    }
  }

  augment "/lldp:lldp/lldp:port" {
    description
      "Augments port with 802.3 port read-write objects";
```

```
    leaf tlvs-port-config-enable {
      type bits {
        bit mac-phy-config-status {
          position 0;
          description
            "IEEE Std 802.3, 30.12.1.1.1";
        }
        bit power-via-mdi {
          position 1;
          description
            "IEEE Std 802.3, 30.12.1.1.1";
        }
        bit unused {
          position 2;
          description
            "IEEE Std 802.3, 30.12.1.1.1";
        }
        bit max-frame-size {
          position 3;
          description
            "IEEE Std 802.3, 30.12.1.1.1";
        }
        bit eee-tlv {
          position 4;
          description
            "IEEE Std 802.3, 30.12.1.1.1";
        }
        bit eee-fast-wake-tlv {
          position 5;
          description
            "IEEE Std 802.3, 30.12.1.1.1";
        }
        bit additional-ethernet-capabilities-tlv {
          position 6;
          description
            "IEEE Std 802.3, 30.12.1.1.1";
        }
      }
      description
        "Bitmap that corresponds to an IEEE 802.3 subtype
         associated with a specific IEEE 802.3 port config TLV";
      reference
        "IEEE Std 802.3, 30.12.1.1.1";
    }
    leaf local-power-priority {
      type power-priority-type;
      description
        "Priority of a PD system. For a PSE, this is the priority
         that the PSE assigns to the PD. For a PD, this is the
         priority that the PD requests from the PSE.";
      reference
        "IEEE Std 802.3, 30.12.2.1.16";
    }
    leaf local-measurement-source {
      type measurement-source-type;
      description
        "A read-write attribute value that indicates to local
         device on which Alternative or Mode the measurement
```

```
        is to be taken.";
      reference
        "IEEE Std 802.3, 30.12.2.1.42";
    }
  }

  /* NOTE - lldp:remote-systems-data container is "config false;" */

  augment "/lldp:lldp/lldp:port/lldp:remote-systems-data" {
    description
      "Augments port with 802.3 port config tlvs";
    leaf auto-negotiation-supported {
      type boolean;
      ~~config false;~~
      description
        "True if the port supports Auto-negotiation";
      reference
        "IEEE Std 802.3, 30.12.3.1.1";
    }
    leaf auto-negotiation-enabled {
      type boolean;
      ~~config false;~~
      description
        "True if Auto-negotiation is enabled";
      reference
        "IEEE Std 802.3, 30.12.3.1.2";
    }
    /* Obsolete */
    leaf auto-negotiation-cap {
      type binary {
        length "2";
      }
      ~~config false~~status obsolete;
      description
        "A read-only 2-octet value that contains the value (bitmap)
         of the ifMauAutoNegCapAdvertisedBits object (defined in
         IETF RFC 4836)which is associated with the given port on
         the local system~~.";~~.
         Supersceeded by auto-negotiation-cap-bits.";
      reference
        "IEEE Std 802.3, 30.12.3.1.3";
    }
    leaf operational-mau-type {
      type int32;
      ~~config false~~status obsolete;
      description
        "32-bit integer value that indicates the operational MAU
         type of the given port~~";~~.
         Supersceeded by operational-pmd-type.";
      reference
        "IEEE Std 802.3, 30.12.3.1.4";
    }
    leaf auto-negotiation-cap-bits {
      type ieee802-phy:auto-neg-ability-bits;
      description
        "The capabilities advertised by the remote Auto-Negotiation
         entity.";
      reference
```

```
          "IEEE Std 802.3, 30.12.3.1.3,
           aAutoNegadvertisedTechnologyAbility.";
    }
    leaf operational-pmd-type {
      type identityref {
        base ieee802-phy:pmd-type;
      }
      description
        "The operational PMD type of the given port";
      reference
        "IEEE Std 802.3, 30.12.3.1.4";
    }
    leaf power-port-class {
      type port-class-type;
      config false;
      description
        "A read-only value that identifies the port Class of the
         given port";
      reference
        "IEEE Std 802.3, 30.12.3.1.5";
    }
    leaf mdi-power-supported {
      type boolean;
      config false;
      description
        "True if MDI power is supported";
      reference
        "IEEE Std 802.3, 30.12.3.1.6";
    }
    leaf mdi-power-enabled {
      type boolean;
      config false;
      description
        "True if MDI power is enabled";
      reference
        "IEEE Std 802.3, 30.12.3.1.7";
    }
    leaf power-pair-controlable {
      type boolean;
      config false;
      description
        "True if the pair selection can be controlled";
      reference
        "IEEE Std 802.3, 30.12.3.1.8";
    }
    leaf power-pairs {
      type pse-pinout-type;
      config false;
      description
        "Indicates which pinout alernative is used for PD
         detection and power";
      reference
        "IEEE Std 802.3, 30.12.3.1.9";
    }
    leaf power-class {
      type pse-power-class-type;
      config false;
      description
```

```
        "PD Power Class";
      reference
        "IEEE Std 802.3, 30.12.3.1.10";
    }
    /* Obsolete Link Aggregation objects
        Link aggregation has not been part of 802.3 since IEEE 802.1AX-
2008
        */
    leaf link-aggregation-status {
      type bits {
        bit aggregation-capability {
          position 0;
          description
            "IEEE Std 802.3, 79.3.3.1";
        }
        bit aggregation-status {
          position 1;
          description
            "IEEE Std 802.3, 79.3.3.1";
        }
        bit bit2-reserved {
          position 2;
          description
            "IEEE Std 802.3, 79.3.3.1";
        }
        bit bit3-reserved {
          position 3;
          description
            "IEEE Std 802.3, 79.3.3.1";
        }
        bit bit4-reserved {
          position 4;
          description
            "IEEE Std 802.3, 79.3.3.1";
        }
        bit bit5-reserved {
          position 5;
          description
            "IEEE Std 802.3, 79.3.3.1";
        }
        bit bit6-reserved {
          position 6;
          description
            "IEEE Std 802.3, 79.3.3.1";
        }
        bit bit7-reserved {
          position 7;
          description
            "IEEE Std 802.3, 79.3.3.1";
        }
      }
      config falsestatus obsolete;
      description
        "The bitmap value which contains the link aggregation
         capabilities and the current aggregation status
         of the link";
      reference
        "IEEE Std 802.3, 30.12.3.1.11";
```

```
      }
      leaf aggregation-port-id {
        type int32;
        config falsestatus obsolete;
        description
          "The unique identifier allocated to this Aggregation Port
           by the local System.";
        reference
          "IEEE Std 802.3, 30.12.3.1.12";
      }
      leaf local-max-frame-size {
        type int32;
        config falsestatus obsolete;
        description
          "An integer value indicating the maximum supported frame
           size in octets on the given port of the local system.";
        reference
          "IEEE Std 802.3, 30.12.3.1.13";
      }
      /* POE objects
          Consistency errors between PoE definitions in 30.9 (PoE),
          30.12 (LLDP), 79.3.2 (PoE TLV), 79.3.8 (PoE measurement TLV).
          Need to get these clauses reviewed against clause 145 and each
other
          */
      leaf power-type {
        type bits {
          bit type1-or-greater {
            position 0;
            description
              "0-type1, 1-greater than type1";
          }
          bit pse-or-pd {
            position 1;
            description
              "0-pse, 1-pd";
          }
        }
        config falsestatus obsolete;
        description
          "A read-only attribute that returns a bit string indicating
           whether the local system is a PSE or a PD and whether it
           is Type 1 or greater than Type 1. The first bit indicates
           Type 1 or greater than Type 1. The second bit indicates
           PSE or PD. A PSE sets this bit to indicate a PSE. A PD
           sets this bit to indicate a PD. See also
           aLldpXdot3LocPowerTypeExt.";
        reference
          "IEEE Std 802.3, 30.12.3.1.14";
      }
      leaf power-source {
        type power-source-type;
        config false;
        description
          "Indicates the power sources of the remote system. A PSE
           indicates whether it is being powered by a primary power
           source; a backup power source; or unknown. A PD indicates
           whether it is being powered by a PSE and locally; by a PSE
```

```
            only; or unknown.;";
        reference
          "IEEE Std 802.3, 30.12.3.1.15";
      }
      leaf power-priority {
        type power-priority-type;
        description
          "The priority of the PD system received from the remote
           system";
        reference
          "IEEE Std 802.3, 30.12.3.1.16";
      }
      leaf pd-requested-power-value {
        type int32;
        config false;
        description
          "PD requested power value that was used by the remote
           system to compute the power value that is has currently
           allocated to the PD.";
        reference
          "IEEE Std 802.3, 30.12.3.1.17";
      }
      leaf pd-requested-power-value-a {
        type int32;
        config false;
        description
          "A read-only attribute that returns the PD requested power
           value for the Mode A pairset that was used by the remote
           system to compute the power value that it has currently
           allocated to the PD. For a PSE, it is the PD requested
           power value for the Alternative A pairset received from
           the remote system. For a PD, it is the PD requested power
           value for the Alternative A pairset that the PSE echoes
           back to the remote system. The definition and encoding of
           PD requested power value for the Mode A pairset is the same
           as described in aLldpXdot3LocPDRequestedPowerValueA";
        reference
          "IEEE Std 802.3, 30.12.3.1.18";
      }
      leaf pd-requested-power-value-b {
        type int32;
        config false;
        description
          "A read-only attribute that returns the PD requested power
           value for the Mode B pairset that was used by the remote
           system to compute the power value that it has currently
           allocated to the PD. For a PSE, it is the PD requested
           power value for the Alternative B pairset received from
           the remote system. For a PD, it is the PD requested power
           value for the Alternative B pairset that the PSE echoes
           back to the remote system. The definition and encoding of
           PD requested power value for the Mode B pairset is the
           same as described in aLldpXdot3LocPDRequestedPowerValueB";
        reference
          "IEEE Std 802.3, 30.12.3.1.19";
      }
      leaf pse-allocated-power-value {
        type int32;
```

```
        ~~config false;~~
        description
          "PSE allocated power value. For a PSE, it is the power value
           that the PSE has currently allocated to the remote system.
           For a PD, it is the power value that the PD mirrors back to
           the remote syste";
        reference
          "IEEE Std 802.3, 30.12.3.1.20";
      }
      leaf pse-allocated-power-value-a {
        type int32;
        ~~config false;~~
        description
          "A read-only attribute that returns the PSE allocated power
           value for the Alternative A pairset received from the
           remote system. For a PSE, it is the PSE allocated power
           value for the Alternative A pairset that was echoed back
           by the remote PD. For a PD, it is the PSE allocated power
           value for the Mode A pairset received from the remote
           system. The definition and encoding of PSE allocated power
           value for the Alternative A pairset is the same as
           described in aLldpXdot3LocPSEAllocatedPowerValueA";
        reference
          "IEEE Std 802.3, 30.12.3.1.21";
      }
      leaf pse-allocated-power-value-b {
        type int32;
        ~~config false;~~
        description
          "A read-only attribute that returns the PSE allocated power
           value for the Alternative B pairset received from the
           value for the Alternative B pairset that was echoed back
           by the remote PD. For a PD, it is the PSE allocated power
           value for the Mode B pairset received from the remote
           system. The definition and encoding of PSE allocated power
           value for the Alternative B pairset is the same as
           described in aLldpXdot3LocPSEAllocatedPowerValueB";
        reference
          "IEEE Std 802.3, 30.12.3.1.22";
      }
      leaf pse-powering-status {
        type powering-status-type;
        ~~config false;~~
        description
          "A read only value that indicates the powering status of
           the remote PSE. For a PD, the contents of this attribute
           are undefined.";
        reference
          "IEEE Std 802.3, 30.12.3.1.23";
      }
      leaf pd-powered-status {
        type powered-status-type;
        ~~config false;~~
        description
          "A read only value that indicates the powering status of
           the PD. For a PSE, the contents of this attribute are
           undefined";
        reference
```

```
        "IEEE Std 802.3, 30.12.3.1.24";
    }
    leaf power-pairs-ext {
      type power-pairs-type;
      ~~config false;~~
      description
        "A read-only value that identifies the supported PSE
         Pinout Alternative specified in 145.2.4. For a PD, this
         attribute contains the value of the aPSEPowerPairs
         attribute (see 30.9.1.1.4). For a PSE, the contents of
         this attribute are undefined";
      reference
        "IEEE Std 802.3, 30.12.3.1.25";
    }
    leaf power-class-ext-A {
      type power-class-ext-AB-type;
      ~~config false;~~
      description
        "For a dual-signature PD, a read-only value that indicates
         the currently assigned Class for Mode A by the remote
         4-pair PSE. For a single-signature PD or a dual-signature
         PD connected to a 2-pair only PSE, a read-only value set
         to 'singlesig' by the remote PSE. For a PSE connected to a
         dual-signature PD, a read-only value that indicates the
         requested Class for Mode A during Physical Layer
         classification (see 145.2.8) by the remote PD. For a PSE
         connected to a single-signature PD, a read-only value set
         to 'singlesig' by the remote PD";
      reference
        "IEEE Std 802.3, 30.12.3.1.26";
    }
    leaf power-class-ext-B {
      type power-class-ext-AB-type;
      ~~config false;~~
      description
        "For a dual-signature PD, a read-only value that indicates
         the currently assigned Class for Mode B by the remote
         4-pair PSE. For a single-signature PD or a dual-signature
         PD connected to a 2-pair only PSE, a read-only value set
         to 'singlesig' by the remote PSE. For a PSE connected to a
         dual-signature PD, a read-only value that indicates the
         requested Class for Mode B during Physical Layer
         classification (see 145.2.8) by the remote PD. For a PSE
         connected to a single-signature PD, a read-only value set
         to 'singlesig' by the remote PD";
      reference
        "IEEE Std 802.3, 30.12.3.1.27";
    }
    leaf power-class-ext {
      type power-class-ext-type;
      ~~config false;~~
      description
        "For a single-signature PD or a dual-signature PD connected
         to a 2-pair only PSE, a read-only value that indicates the
         currently assigned Class by the remote PSE. For a
         dual-signature PD connected to a 4-pair capable PSE, a
         read-only value set to 'dualsig' by the remote PSE. For a
         PSE connected to a single-signature PD, a read-only value
```

```
           that indicates the requested Class during Physical Layer
           classification (see 145.2.8) by the remote PD. For a PSE
           connected to a dual-signature PD, a read-only value set to
           'dualsig' by the remote PD.";
        reference
          "IEEE Std 802.3, 30.12.3.1.28";
      }
      leaf power-type-ext {
        type power-type;
        config false;
        description
          "A read-only attribute that returns a value to indicate if
           the remote system is a Type 3 or Type 4 PSE or PD and, in
           the case of a Type 3 or Type 4 PD, if it is a
           single-signature PD or dual-signature PD.";
        reference
          "IEEE Std 802.3, 30.12.3.1.29";
      }
      leaf pd-load {
        type boolean;
        config false;
        description
          "For a PSE, a read-only attribute that returns whether the
           load of the remote dual-signature PD is electrically
           isolated, as defined in 79.3.2.10.2. For a PD, this
           attribute is set to FALSE.";
        reference
          "IEEE Std 802.3, 30.12.3.1.30";
      }
      leaf pd-4pid {
        type boolean;
        config false;
        description
          "A read-only Boolean attribute indicating whether the
           remote PD system supports powering of both PD Modes.";
        reference
          "IEEE Std 802.3, 30.12.3.1.31";
      }
      leaf pse-max-avail-power {
        type int32;
        config false;
        description
          "A read-only attribute that returns the remote PSE maximum
           available power value in units of 0.1 W";
        reference
          "IEEE Std 802.3, 30.12.3.1.32";
      }
      leaf pse-autoclass-support {
        type boolean;
        config false;
        description
          "Indicates whether the remote PSE system supports
           Autoclass.";
        reference
          "IEEE Std 802.3, 30.12.3.1.33";
      }
      leaf autoclass-completed {
        type boolean;
```

```
        config false;
        description
          "Indicates whether the remote PSE system has completed the
           Autoclass measurement.";
        reference
          "IEEE Std 802.3, 30.12.3.1.34";
      }
      leaf autoclass-request {
        type boolean;
        config false;
        description
          "A read-only Boolean attribute indicating whether the
           remote PD system is requesting an Autoclass measurement.";
        reference
          "IEEE Std 802.3, 30.12.3.1.35";
      }
      leaf power-down-request {
        type int32;
        description
          "A read-write attribute that indicates the remote PD system
           is requesting a power down when the value is  0x1D.;";
        reference
          "IEEE Std 802.3, 30.12.3.1.36";
      }
      leaf power-down-time {
        type int32;
        description
          "A read-only attribute that indicates the number of seconds
           the remote PD requests to stay powered off. A value of zero
           indicates an indefinite amount of time";
        reference
          "IEEE Std 802.3, 30.12.3.1.37";
      }
      /* PoE measurement objects */
      leaf meas-voltage-support {
        type boolean;
        config false;
        description
          "A read-only attribute that indicates the remote device is
           capable of providing a voltage measurement.;";
        reference
          "IEEE Std 802.3, 30.12.3.1.38";
      }
      leaf meas-current-support {
        type boolean;
        config false;
        description
          "A read-only attribute that indicates the remote device is
           capable of providing a current measurement.;";
        reference
          "IEEE Std 802.3, 30.12.3.1.39";
      }
      leaf meas-power-support {
        type boolean;
        config false;
        description
          "A read-only attribute that indicates the remote device is
           capable of providing a power measurement.;";
```

```
        reference
          "IEEE Std 802.3, 30.12.3.1.40";
      }
      leaf meas-energy-support {
        type boolean;
        config false;
        description
          "A read-only attribute that indicates the remote device is
           capable of providing a energy measurement.;";
        reference
          "IEEE Std 802.3, 30.12.3.1.41 ";
      }
      leaf measurement-source {
        type bits {
          bit bit1 {
            position 0;
            description
              "-";
          }
          bit bit2 {
            position 1;
            description
              "-";
          }
        }
        status obsolete;
        description
          "A read-write attribute value that indicates on which
           Alternative or Mode the measurement was taken by the remote
           device.";.
           Superseded by rem-measurement-source.";
        reference
          "IEEE Std 802.3, 30.12.3.1.42";
      }
      leaf rem-measurement-source {
        type measurement-source-type;
        description
          "A read-only attribute that indicates which Alternative or
           Mode the measurement is to be taken";
        reference
          "IEEE Std 802.3, 30.12.3.1.42";
      }
      leaf meas-voltage-request {
        type boolean;
        config false;
        description
          "A read-only attribute that indicates the rmoteremote device is
           requesting a voltage measurement from the local device.";
        reference
          "IEEE Std 802.3, 30.12.3.1.43";
      }
      leaf meas-current-request {
        type boolean;
        config false;
        description
          "A read-only attribute that indicates the remote device is
           requesting a current measurement from the local device.";
        reference
```

```
        "IEEE Std 802.3, 30.12.3.1.44";
    }
    leaf meas-power-request {
      type boolean;
      ~~config false;~~
      description
        "A read-only attribute that indicates the remote device is
         requesting a power measurement from the local device.";
      reference
        "IEEE Std 802.3, 30.12.3.1.45";
    }
    leaf meas-energy-request {
      type boolean;
      ~~config false;~~
      description
        "A read-only attribute that indicates the remote device is
         requesting an energy measurement from the local device.";
      reference
        "IEEE Std 802.3, 30.12.3.1.46";
    }
    leaf meas-voltage-valid {
      type boolean;
      ~~config false;~~
      description
        "A read-only attribute that indicates the remote device’s
         voltage measurement is valid.";
      reference
        "IEEE Std 802.3, 30.12.3.1.47";
    }
    leaf meas-current-valid {
      type boolean;
      ~~config false;~~
      description
        "A read-only attribute that indicates the remote device’s
         current measurement is valid.";
      reference
        "IEEE Std 802.3, 30.12.3.1.48";
    }
    leaf meas-power-valid {
      type boolean;
      ~~config false;~~
      description
        "A read-only attribute that indicates the remote device’s
         power measurement is valid.";
      reference
        "IEEE Std 802.3, 30.12.3.1.49";
    }
    leaf meas-energy-valid {
      type boolean;
      ~~config false;~~
      description
        "A read-only attribute that indicates the remote device’s
         energy measurement is valid.";
      reference
        "IEEE Std 802.3, 30.12.3.1.50";
    }
    leaf meas-voltage-uncertainty {
      type int32;
```

```
        config false;
        description
          "A read-only attribute that indicates the expanded
           uncertainty (coverage factor k = 2) for the remote device's
           voltage measurement.";
        reference
          "IEEE Std 802.3, 30.12.3.1.51";
      }
      leaf meas-current-uncertainty {
        type int32;
        config false;
        description
          "A read-only attribute that indicates the expanded
           uncertainty (coverage factor k = 2) for the remote device's
           current measurement.";
        reference
          "IEEE Std 802.3, 30.12.3.1.52";
      }
      leaf meas-power-uncertainty {
        type int32;
        config false;
        description
          "A read-only attribute that indicates the expanded
           uncertainty (coverage factor k = 2) for the remote device's
           power measurement.";
        reference
          "IEEE Std 802.3, 30.12.3.1.53";
      }
      leaf meas-energy-uncertainty {
        type int32;
        config false;
        description
          "A read-only attribute that indicates the expanded
           uncertainty (coverage factor k = 2) for the remote device's
           energy measurement.";
        reference
          "IEEE Std 802.3, 30.12.3.1.54";
      }
      leaf voltage-measurement {
        type int32;
        config false;
        description
          "A read-only attribute that returns the measured remote
           device voltage.";
        reference
          "IEEE Std 802.3, 30.12.3.1.55";
      }
      leaf current-measurement {
        type int32;
        config false;
        description
          "A read-only attribute that returns the measured remote
           device current.";
        reference
          "IEEE Std 802.3, 30.12.3.1.56";
      }
      leaf power-measurement {
        type int32;
```

```
        ~~config false;~~
        description
          "A read-only attribute that returns the measured remote
           device power.";
        reference
          "IEEE Std 802.3, 30.12.3.1.57";
      }
      leaf energy-measurement {
        type int32;
        ~~config false;~~
        description
          "A read-only attribute that returns the measured remote
           device energy.";
        reference
          "IEEE Std 802.3, 30.12.3.1.58";
      }
      leaf pse-power-price-index {
        type int32;
        ~~config false;~~
        description
          "A read-only attribute that returns an index of the price
           of power being sourced by the remote PSE. For a PSE, this
           value is undefined.";
        reference
          "IEEE Std 802.3, 30.12.3.1.59";
      }
      leaf tx-system-value {
        type int32;
        ~~config false;~~
        description
          "Returns the value of Tw_sys_tx that the remote system can
           support in the transmit direction.";
        reference
          "IEEE Std 802.3, 30.12.3.1.60";
      }
      leaf tx-system-value-echo {
        type int32;
        ~~config false;~~
        description
          "Returns the value of Tw_sys_tx that the local system is
           advertising that it can support in the transmit direction
           and is echoed by the local system under the control of the
           EEE DLL receiver state diagram.";
        reference
          "IEEE Std 802.3, 30.12.3.1.61";
      }
      leaf rx-system-value {
        type int32;
        ~~config false;~~
        description
          "Returns the value of Tw_sys_tx that the remote system is
           requesting in the receive direction.";
        reference
          "IEEE Std 802.3, 30.12.3.1.62";
      }
      leaf rx-system-value-echo {
        type int32;
        ~~config false;~~
```

```
        description
          "Returns the value of Tw_sys_tx that the local system is
           advertising that it is requesting in the receive direction
           and is echoed by the local system under the control of the
           EEE DLL transmitter state diagram.";
        reference
          "IEEE Std 802.3, 30.12.3.1.63 ";
      }
      leaf fallback-system-value {
        type int32;
        config false;
        description
          "Returns the value of the fallback Tw_sys_tx that the
           remote system is advertising to the remote system.";
        reference
          "IEEE Std 802.3, 30.12.3.1.64";
      }
      /* LDP EEE objects */
      leaf tx-system-fw {
        type boolean;
        config false;
        description
          "Returns the value of LPI_FW that the remote system can
           support in the transmit direction.";
        reference
          "IEEE Std 802.3, 30.12.3.1.65";
      }
      leaf tx-system-fw-echo {
        type boolean;
        config false;
        description
          "Returns the value of LPI_FW that the local system is
           advertising that it can support in the transmit direction
           and is echoed by the local system under the control of the
           EEE DLL receiver state diagram.";
        reference
          "IEEE Std 802.3, 30.12.3.1.66";
      }
      leaf rx-system-fw {
        type boolean;
        config false;
        description
          "Returns the value of LPI_FW that the remote system is
           requesting in the receive direction.";
        reference
          "IEEE Std 802.3, 30.12.3.1.67";
      }
      leaf rx-system-fw-echo {
        type boolean;
        config false;
        description
          "Returns the value of LPI_FW that the local system is
           advertising that it is requesting in the receive direction
           and is echoed by the local system under the control of the
           EEE DLL transmitter state diagram.";
        reference
          "IEEE Std 802.3, 30.12.3.1.68";
      }
```

```
    /* LLDP preemption objects */
    leaf preemption-supported {
      type boolean;
      ~~config false;~~
      description
        "Indicates whether the given port (associated with the
         remote System) supports the preemption capability.";
      reference
        "IEEE Std 802.3, 30.12.3.1.69";
    }
    leaf preemption-enabled {
      type boolean;
      ~~config false;~~
      description
        "Indicates whether the preemption capability is enabled on
         the given port associated with the remote System.";
      reference
        "IEEE Std 802.3, 30.12.3.1.70";
    }
    leaf preemption-active {
      type boolean;
      ~~config false;~~
      description
        "Indicates whether the preemption capability is active on
         the given port associated with the remote System.;";
      reference
        "IEEE Std 802.3, 30.12.3.1.72";
    }
    leaf additional-fragement-size {
      type int32;
      ~~config false;~~
      description
        "Indicate the minimum size of non-final fragments supported
         by the receiver on the given port associated with the
         remote System. This value is expressed in units of 64
         octets of additional fragment length.";
      reference
        "IEEE Std 802.3, 30.12.3.1.72 ";
    }
  }
}
module ieee802-ethernet-mac-merge {
  yang-version 1.1;
  namespace "urn:ieee:std:802.3:yang:ieee802-ethernet-mac-merge";
  prefix mac-merge;

  import ietf-yang-types {
    prefix yang;
    reference
      "IETF RFC 6991";
  }
  import ietf-interfaces {
    prefix if;
    reference
      "IETF RFC 8343";
  }
  import ieee802-ethernet-interface {
    prefix ieee802-eth-if;
```

```
    reference
      "IEEE Std 802.3.2-2019";
  }

  organization
    "IEEE Std 802.3 Ethernet Working Group
     Web URL: http://www.ieee802.org/3/";
  contact
    "Web URL: http://www.ieee802.org/3/";
  description
    "The Yang model for managing devices that support the MAC ~~merge~~Merge
     (aka preemption) sublayer as defined in Clause 99. Unless
     otherwise indicated, the references in this model module are to
         IEEE Std 802.3-2022.";

  revision 2024-~~07-17~~12-23 {
    description
      "Updates under IEEE Std 802.3.2-202x, Draft D3.~~0~~1";
    reference
      "IEEE Std 802.3-2022, unless dated explicitly";
  }

  feature mac-merge {
    description
      "~~Each Port supports the ~~MAC ~~merge sublayer~~Merge is supported.";
    reference
      "IEEE Std 802.3-2022";
  }

  augment "/if:interfaces/if:interface/ieee802-eth-if:ethernet" {
    if-feature "mac-merge";
    description
      "~~Missing description D1.2 2024-04-04";~~Interfaces supporting MAC
Merge.";
    container mac-merge {
      description
        "~~Missing description D1.2 2024-04-04";~~Container for MAC Merge
objects.";
      container admin-control {
        description
          "~~Missing description D1.2 2024-04-04";~~Container for MAC Merge
control objects.";
        leaf merge-enable-tx {
          type enumeration {
            enum Disabled {
              description
                "Transmit preemption is disabled";
            }
            enum Enabled {
              description
                "Transmit preemption is enabled";
            }
          }
          default "Disabled";
          description
            "~~This attribute indicates (when accessed via a
              READ operation) the status of the~~Is MAC Merge enabled when
transmitting?
```

```
                ~~sublayer on the given device in the~~
                ~~transmit direction. The status of the~~
                ~~MAC Merge sublayer~~It may be modified ~~to the~~using an edit
operation.";
                ~~indicated value via a WRITE operation.~~
                ~~This attribute maps to the variable pEnable~~
                ~~(see 99.4.7.3).";~~
          reference
            "IEEE Std 802.3, 30.14.1.3";
        }
        leaf verify-disable-tx {
          type enumeration {
            enum Disabled {
              description
                "Verify is disabled";
            }
            enum Enabled {
              description
                "Verify is enabled";
            }
          }
          default "Disabled";
          description
            "~~This attribute indicates (when accessed~~
             ~~via a READ operation) the status of the~~
             ~~Verify function of~~Does MAC Merge ~~sublayer on~~check to make
sure the link partner
             ~~the given device in the transmit~~supports MAC Merge?
             ~~direction.~~ The ~~status of the Verify~~check is peformed by
sending a verify request and
             ~~function~~getting a verify response.
             It may be modified ~~to the indicated~~using an edit
operation.;";
             ~~value  via a read-write operation.~~
             ~~This attribute maps to the variable~~
             ~~disableVerify (see 99.4.7.3).;";~~
          reference
            "IEEE Std 802.3, 30.14.1.4";
        }
        leaf verify-time {
          type uint16 {
            range "1..128";
          }
          units "milliseconds";
          default "10";
          description
            "The ~~value of this attribute defines the~~
             ~~nominal wait time between~~retry timer for verification
attempts~~ in milliseconds~~.
             ~~Valid~~The valid range is 1 to 128 inclusive.
             The default value is 10~~. This attribute~~.";
             ~~maps to the variable verifyTime~~
             ~~(see 99.4.7.3).;";~~
          reference
            "IEEE Std 802.3, 30.14.1.6";
        }
        leaf frag-size {
          type uint8 {
```

```
            range "0..3";
          }
          default "0";
          description
            "~~A 2-bit integer value used to indicate~~This controls minimum
frame size for packet fragments.
             It's specified in units of 64 bytes above the ~~value of
addFragSize variable used by~~minimum
             ~~the Transmit Processing State Diagram~~of 64 bytes. The
equation is:
             ~~(see Figure IEEE Std 802.3, 99-5).";~~minimum-size = 64 +
(frag-size*64)-4";
          reference
            "IEEE Std 802.3, 30.14.1.7";
        }
      }
      container admin-status {
        config false;
        description
          "~~Missing description D1.2 2024-04-04";~~Container for MAC Merge
status objects.";
        leaf merge-support {
          type enumeration {
            enum Supported {
              description
                "MAC Merge ~~sublayer ~~is supported on the device";
            }
            enum NotSupported {
              description
                "MAC Merge ~~sublayer ~~is not supported on the device";
            }
          }
          description
            "~~This attribute indicates (when accessed~~
~~             via a READ operation) whether the given~~
~~             device supports a~~Is MAC Merge ~~sublayer.~~is supported?";
~~             The WRITE operation shall have no effect~~
~~             on a device.";~~
          reference
            "IEEE Std 802.3, 30.14.1.1";
        }
        leaf verify-status {
          type enumeration {
            enum unknown {
              description
                "~~Verification status is ~~unknown";
            }
            enum initial {
              description
                "~~The Verify State diagram~~
~~                 Figure 99-8 IEEE Std 802.3 is in~~
~~                 the state INIT_VERIFICATION~~initial";
            }
            enum verifying {
              description
                "~~The Verify State diagram is in~~
~~                 the state VERIFICATION_IDLE,~~
~~                 SEND_VERIFY or WAIT_FOR_RESPONSE~~verifying";
```

```
          }
          enum succeeded {
            description
              "~~Indicates that the Verify State
              diagram is in the state VERIFIED~~succeeded";
          }
          enum failed {
            description
              "~~The Verify State diagram is in~~failed";
              ~~the state VERIFY_FAIL";~~
          }
          enum disabled {
            description
              "~~Verification of
              preemption operation is ~~disabled";
          }
        }
        description
          "~~This attribute indicates (when accessed
          via a READ operation) the~~The status of ~~the~~
          MAC Merge ~~sublayer verification on the
          given device. The WRITE operation shall
          have no effect on a device~~Verification.";
        reference
          "IEEE Std 802.3, 30.14.1.2";
      }
      leaf status-tx {
        type enumeration {
          enum unknown {
            description
              "~~transmit preemption status is
              ~~unknown";
          }
          enum inactive {
            description
              "~~transmit preemption is ~~inactive";
          }
          enum active {
            description
              "~~transmit preemption is ~~active";
          }
        }
        description
          "~~This attribute indicates (when accessed
          via a READ operation) the ~~The status of ~~the~~
          MAC Merge ~~sublayer on the given device ~~in the transmit
direction~~. The SET operation~~.";
          ~~shall have no effect on a device. This
          attribute maps to the variable preempt
          (see 99.4.7.3).";~~
        reference
          "IEEE Std 802.3, 30.14.1.5";
      }
    }
    container statistics {
      config false;
      description
```

```
          "~~Missing description D1.2 2024-04-04";~~Container for MAC Merge
counters.
           Discontinuities in the values of counters in
           this container can occur at re-initialization of
           the management system, and at other times as
           indicated by the value of the 'discontinuity-time'
           leaf defined in the ietf-interfaces YANG module
           (IETF RFC 8343).";
        leaf assembly-error-count {
          type yang:counter64;
          description
            "A count of MAC frames with reassembly errors.";
             ~~errors. The counter is incremented by one~~
             ~~every time the ASSEMBLY_ERROR state in~~
             ~~the Receive Processing State Diagram is~~
             ~~entered";~~
          reference
            "IEEE Std 802.3, 30.14.1.8";
        }
        leaf smd-error-count {
          type yang:counter64;
          description
            "A count of received MAC frames / MAC frame fragments
             ~~frame fragments~~ rejected ~~due to unknown~~
             ~~SMD value or arriving~~ with an ~~SMD-C when~~unexpected
preemption header.";
             ~~no frame is in progress. The counter is~~
             ~~incremented by one every time the~~
             ~~BAD_FRAG state in the Receive Processing~~
             ~~State Diagram is entered and every time~~
             ~~the WAIT_FOR_DV_FALSE state is entered~~
             ~~due to the invocation of the SMD_DECODE~~
             ~~function returning the value ERR";~~
          reference
            "IEEE Std 802.3, 30.14.1.9";
        }
        leaf assembly-ok-count {
          type yang:counter64;
          description
            "A count of preemptable MAC frames that were successfully
             reassembled and delivered to MAC~~. The counter is incremented~~
~~by one~~.";
             ~~every time the FRAME_COMPLETE state in the~~
             ~~Receive Processing state diagram~~
             ~~(see Figure 99-6) is entered if the state~~
             ~~CHECK_FOR_RESUME was previously entered~~
             ~~while processing the packet";~~
          reference
            "IEEE Std 802.3, 30.14.1.10";
        }
        leaf fragment-count-rx {
          type yang:counter64;
          description
            "~~A count of the~~ The number of ~~additional~~preemptable frame
fragments recieved.";
             ~~mPackets received due to preemption.~~
             ~~The counter is incremented by one every~~
             ~~time the state CHECK_FRAG_CNT in the~~
```

```
              Receive Processing State Diagram
              (see Figure 99-6) is entered";
          reference
            "IEEE Std 802.3, 30.14.1.11";
        }
        leaf fragment-count-tx {
          type yang:counter64;
          description
            "A count of theThe number of additional
             mPacketspreemptable frame fragments transmitted due to
preemption.
             This counter is incremented by one every
             time the SEND_SMD_C state in the Transmit
             Processing State Diagram
             (see Figure 99-5) is entered.";
          reference
            "IEEE Std 802.3, 30.14.1.12";
        }
        leaf hold-count {
          type yang:counter64;
          description
            "A count of the The number of times the
             variableMAC Merge hold (see 99.4.7.3) transitionsrequests
(i.e., to stop
             transmissions of preemptable traffic) received from FALSE to
TRUE.";
             the MAC";
          reference
            "IEEE Std 802.3, 30.14.1.13";
        }
      }
    }
  }
}
module ieee802-ethernet-phy-type {
  yang-version 1.1;
  namespace "urn:ieee:std:802.3:yang:ieee802-ethernet-phy-type";
  prefix ieee802-eth-phy-type;

  organization
    "IEEE Std 802.3 Ethernet Working Group
     Web URL: http://www.ieee802.org/3/";
  contact
    "Web URL: http://www.ieee802.org/3/";
  description
    "This module contains YANG definitions for Physical Layer device
        (PHY) and Physical Medium Dependent(PMD) indentities and related
        types.";

  revision 2024-12-23 {
    description
      "Initial revision derived from IANA-MAU-MIB REVISION
       202309070000Z using smidump.";
    reference
      "IEEE Std 802.3-2022, unless dated explicitly";
  }

  /* PHY type identities */
```

```
  identity phy-type {
    description
      "Base type for PHY types.";
    reference
      "IEEE Std 802.3, Clause 30.3.2.1.2 aPhyType";
  }

  identity phy-type-other {
    base phy-type;
    description
      "Undefined";
  }

  identity phy-type-unknown {
    base phy-type;
    description
      "Initializing, true state or type not yet known";
  }

  identity phy-type-none {
    base phy-type;
    description
      "MII present and nothing connected";
  }

  identity phy-type-10BASE-T {
    base phy-type;
    description
      "Clause 14 10 Mb/s PAM3";
  }

  identity phy-type-10BASE-T1L {
    base phy-type;
    description
      "Clause 146 10 Mb/s PAM3";
  }

  identity phy-type-10BASE-T1S {
    base phy-type;
    description
      "Clause 147 10 Mb/s DME";
  }

  identity phy-type-100BASE-T1 {
    base phy-type;
    description
      "Clause 96 100 Mb/s PAM3";
  }

  identity phy-type-100BASE-X {
    base phy-type;
    description
      "Clause 24 or subclause 66.1 100 Mb/s 4B/5B";
  }

  identity phy-type-1000BASE-H {
    base phy-type;
```

```
    description
      "Clause 115 1000 Mb/s PAM16-THP";
  }

  identity phy-type-1000BASE-T {
    base phy-type;
    description
      "Clause 40 1000 Mb/s 4D-PAM5";
  }

  identity phy-type-1000BASE-T1 {
    base phy-type;
    description
      "Clause 97 1000 Mb/s PAM3";
  }

  identity phy-type-1000BASE-X {
    base phy-type;
    description
      "Clause 36 or subclause 66.2 1000 Mb/s 8B/10B";
  }

  identity phy-type-2.5GBASE-T {
    base phy-type;
    description
      "Clause 126 2.5 Gb/s PAM16";
  }

  identity phy-type-2.5GBASE-T1 {
    base phy-type;
    description
      "Clause 149 2.5 Gb/s PAM4";
  }

  identity phy-type-2.5GBASE-X {
    base phy-type;
    description
      "Clause 127 2.5 Gb/s 8B/10B";
  }

  identity phy-type-5GBASE-R {
    base phy-type;
    description
      "Clause 129 5 Gb/s 64/66B";
  }

  identity phy-type-5GBASE-T {
    base phy-type;
    description
      "Clause 126 5 Gb/s PAM16";
  }

  identity phy-type-5GBASE-T1 {
    base phy-type;
    description
      "Clause 149 5 Gb/s PAM4";
  }
```

```
  identity phy-type-10G-1GBASE-PRX {
    base phy-type;
    description
      "Clause 76 10/1G-EPON 10 Gb/s 64B/66B downstream and
       1 Gb/s 8B/10B upstream";
  }

  identity phy-type-10GBASE-PR {
    base phy-type;
    description
      "Clause 76 10/10G-EPON 10 Gb/s 64B/66B";
  }

  identity phy-type-10GBASE-R {
    base phy-type;
    description
      "Clause 49 10 Gb/s 64B/66B";
  }

  identity phy-type-10GBASE-T {
    base phy-type;
    description
      "Clause 55 10 Gb/s DSQ128";
  }

  identity phy-type-10GBASE-T1 {
    base phy-type;
    description
      "Clause 149 10 Gb/s PAM4";
  }

  identity phy-type-10GBASE-X {
    base phy-type;
    description
      "Clause 48 10 Gb/s 4 lane 8B/10B";
  }

  identity phy-type-10GPASS-XR {
    base phy-type;
    description
      "Clause 101 PCS up to 10 Gb/s 64B/66B OFDM downstream
       and up to 1.6 Gb/s 64B/66B OFDMA upstream";
  }

  identity phy-type-10G-2p5GGBASE-SP {
    base phy-type;
    description
      "Clause 164.3 10 Gb/s downstream and 2.5 Gb/s upstream
       256B/257B";
  }

  identity phy-type-10GBASE-SP {
    base phy-type;
    description
      "Clause 164.3 10 Gb/s 256B/257B";
  }

  identity phy-type-25G-10GBASE-PQ {
```

```
    base phy-type;
    description
      "Clause 142 25/10G-EPON 256B/257B";
  }

  identity phy-type-25GBASE-PQ {
    base phy-type;
    description
      "Clause 142 25/25G-EPON 256B/257B";
  }

  identity phy-type-25GBASE-R {
    base phy-type;
    description
      "Clause 107 25 Gb/s 64B/66B";
  }

  identity phy-type-25GBASE-T {
    base phy-type;
    description
      "Clause 113 25 Gb/s DSQ128";
  }

  identity phy-type-40GBASE-R {
    base phy-type;
    description
      "Clause 82 40 Gb/s multi-PCS lane 64B/66B";
  }

  identity phy-type-40GBASE-T {
    base phy-type;
    description
      "Clause 113 40 Gb/s DSQ128";
  }

  identity phy-type-50G-10GBASE-PQ {
    base phy-type;
    description
      "Clause 142 50/10G-EPON 256B/257B";
  }

  identity phy-type-50G-25GBASE-PQ {
    base phy-type;
    description
      "Clause 142 50/25G-EPON 256B/257B";
  }

  identity phy-type-50GBASE-PQ {
    base phy-type;
    description
      "Clause 142 50/50G-EPON 256B/257B";
  }

  identity phy-type-50GBASE-R {
    base phy-type;
    description
      "Clause 133 50 Gb/s multi-PCS lane 64B/66B";
  }
```

```
  identity phy-type-100GBASE-P {
    base phy-type;
    description
      "Clause 82 100 Gb/s multi-PCS lane using >2-level PAM";
  }

  identity phy-type-100GBASE-R {
    base phy-type;
    description
      "Clause 82 100 Gb/s multi-PCS lane using 2-level PAM";
  }

  identity phy-type-200GBASE-R {
    base phy-type;
    description
      "Clause 119 200 Gb/s multi-PCS lane 64B/66B";
  }

  identity phy-type-400GBASE-R {
    base phy-type;
    description
      "Clause 119 400 Gb/s multi-PCS lane 64B/66B";
  }

  identity phy-type-800GBASE-R {
    base phy-type;
    description
      "Clause 172 800 Gb/s multi-PCS lane 64B/66B";
  }

  /* PMD  identities */

  identity pmd-type {
    description
      "Base type for Physical Medium Dependent(PMD)sublayer types.";
    reference
      "IEEE Std 802.3, Clause 30.5.1.1.2 aMAUType";
  }

  identity pmd-type-global {
    base pmd-type;
    description
      "Undefined";
  }

  identity pmd-type-other {
    base pmd-type;
    description
      "Nonconformant PMD.";
  }

  identity pmd-type-unknown {
    base pmd-type;
    description
      "Initializing, true state or type not yet known.";
  }
```

```
  identity pmd-type-10BASE-T {
    base pmd-type;
    description
      "Twisted-pair cabling MAU as specified in Clause 14";
  }

  identity pmd-type-10BASE-T1L {
    base pmd-type;
    description
      "Single balanced pair PHY as specified in Clause 146";
  }

  identity pmd-type-10BASE-T1S {
    base pmd-type;
    description
      "Single balanced pair PHY as specified in Clause 147, full
       duplex mode";
  }

  identity pmd-type-10BASE-T1SMD {
    base pmd-type;
    description
      "Single balanced pair PHY as specified in Clause 147,
       multidrop mode";
  }

  identity pmd-type-100BASE-BX10D {
    base pmd-type;
    description
      "One single-mode fiber OLT PHY as specified in Clause 58";
  }

  identity pmd-type-100BASE-BX10U {
    base pmd-type;
    description
      "One single-mode fiber ONU PHY as specified in Clause 58";
  }

  identity pmd-type-100BASE-FX {
    base pmd-type;
    description
      "100BASE-FX PMD as specified in Clause 26";
  }

  identity pmd-type-100BASE-LX10 {
    base pmd-type;
    description
      "Two fiber PHY as specified in Clause 58";
  }

  identity pmd-type-100BASE-T1 {
    base pmd-type;
    description
      "Single balanced twisted-pair copper cabling PHY as specified
       in Clause 96";
  }

  identity pmd-type-100BASE-TX {
```

```
    base pmd-type;
    description
      "Two-pair Category 5 twisted-pair cabling as specified in
       Clause 25";
  }

  identity pmd-type-1000BASE-BX10D {
    base pmd-type;
    description
      "One single-mode fiber OLT PHY as specified in Clause 59";
  }

  identity pmd-type-1000BASE-BX10U {
    base pmd-type;
    description
      "One single-mode fiber ONU PHY as specified in Clause 59";
  }

  identity pmd-type-1000BASE-KX {
    base pmd-type;
    description
      "1000BASE-KX PCS/PMA over an electrical backplane PMD as
       specified in Clause 70";
  }

  identity pmd-type-1000BASE-LX {
    base pmd-type;
    description
      "1000BASE-LX fiber over long-wavelength laser PMD as specified
       in Clause 38";
  }

  identity pmd-type-1000BASE-LX10 {
    base pmd-type;
    description
      "Two fiber 10 km PHY as specified in Clause 59";
  }

  identity pmd-type-1000BASE-PX10-D {
    base pmd-type;
    description
      "One single-mode fiber OMP OLT PHY, as specified in Clause 60,
       supporting a distance of at least 10 km, and a split of at
       least 1:16";
  }

  identity pmd-type-1000BASE-PX10-U {
    base pmd-type;
    description
      "One single-mode fiber OMP ONU PHY, as specified in Clause 60,
       supporting a distance of at least 10 km, and a split of at
       least 1:16";
  }

  identity pmd-type-1000BASE-PX20-D {
    base pmd-type;
    description
      "One single-mode fiber OMP OLT PHY, as specified in Clause 60,
```

```
         supporting a distance of at least 20 km, and a split of at
         least 1:16";
  }

  identity pmd-type-1000BASE-PX20-U {
    base pmd-type;
    description
      "One single-mode fiber OMP ONU PHY, as specified in Clause 60,
       supporting a distance of at least 20 km, and a split of at
       least 1:16";
  }

  identity pmd-type-1000BASE-PX30-D {
    base pmd-type;
    description
      "One single-mode fiber OMP OLT PHY, as specified in Clause 60,
       supporting a distance of at least 20 km, and a split of at
       least 1:32";
  }

  identity pmd-type-1000BASE-PX30-U {
    base pmd-type;
    description
      "One single-mode fiber OMP ONU PHY, as specified in Clause 60,
       supporting a distance of at least 20 km, and a split of at
       least 1:32";
  }

  identity pmd-type-1000BASE-PX40-D {
    base pmd-type;
    description
      "One single-mode fiber OMP OLT PHY as specified in Clause 60,
       supporting a distance of at least 20 km, and a split of at
       least 1:64";
  }

  identity pmd-type-1000BASE-PX40-U {
    base pmd-type;
    description
      "One single-mode fiber OMP ONU PHY as specified in Clause 60,
       supporting a distance of at least 20 km, and a split of at
       least 1:64";
  }

  identity pmd-type-1000BASE-RHA {
    base pmd-type;
    description
      "Plastic optical fiber PHY as specified in Clause 115.";
  }

  identity pmd-type-1000BASE-RHB {
    base pmd-type;
    description
      "Plastic optical fiber PHY as specified in Clause 115.";
  }

  identity pmd-type-1000BASE-RHC {
    base pmd-type;
```

```
    description
      "Plastic optical fiber PHY as specified in Clause 115.";
  }

  identity pmd-type-1000BASE-SX {
    base pmd-type;
    description
      "1000BASE-SX fiber over short-wavelength laser PMD as
       specified in Clause 38";
  }

  identity pmd-type-1000BASE-T {
    base pmd-type;
    description
      "Four-pair Category 5 twisted-pair cabling PHY as specified in
       Clause 40";
  }

  identity pmd-type-1000BASE-T1 {
    base pmd-type;
    description
      "Single twisted-pair copper cable PHY as specified in Clause
       97";
  }

  identity pmd-type-1000BASE-X {
    base pmd-type;
    description
      "1000BASE-X PCS/PMA as specified in Clause 36 over undefined
       PMD";
  }

  identity pmd-type-2.5GBASE-KX {
    base pmd-type;
    description
      "2.5GBASE-X PMD as specified in Clause 128 over an electrical
       backplane as specified in Clause 128";
  }

  identity pmd-type-2.5GBASE-T {
    base pmd-type;
    description
      "Four-pair twisted-pair balanced copper cabling PHY as
       specified in Clause 126";
  }

  identity pmd-type-2.5GBASE-T1 {
    base pmd-type;
    description
      "Single balanced pair of conductors PHY as specified in Clause
       149";
  }

  identity pmd-type-2.5GBASE-X {
    base pmd-type;
    description
      "2.5GBASE-X PCS/PMA as specified in Clause 127 over undefined
       PMD";
```

```
  }

  identity pmd-type-5GBASE-KR {
    base pmd-type;
    description
      "5GBASE-KR PMD as specified in Clause 130 over an electrical
       backplane as specified in Clause 130";
  }

  identity pmd-type-5GBASE-R {
    base pmd-type;
    description
      "5GBASE-R PCS/PMA as specified in Clause 129 over undefined
       PMD";
  }

  identity pmd-type-5GBASE-T {
    base pmd-type;
    description
      "Four-pair twisted-pair balanced copper cabling PHY as
       specified in Clause 126";
  }

  identity pmd-type-5GBASE-T1 {
    base pmd-type;
    description
      "Single balanced pair of conductors PHY as specified in Clause
       149";
  }

  identity pmd-type-10G-1GBASE-PRX-D1 {
    base pmd-type;
    description
      "One single-mode fiber 10.3125 GBd continuous downstream /
       1.25 GBd burst mode upstream OLT PHY as specified in Clause 75";
  }

  identity pmd-type-10G-1GBASE-PRX-D2 {
    base pmd-type;
    description
      "One single-mode fiber 10.3125 GBd continuous downstream /
       1.25 GBd burst mode upstream OLT PHY as specified in Clause 75";
  }

  identity pmd-type-10G-1GBASE-PRX-D3 {
    base pmd-type;
    description
      "One single-mode fiber 10.3125 GBd continuous downstream /
       1.25 GBd burst mode upstream OLT PHY as specified in Clause 75";
  }

  identity pmd-type-10G-1GBASE-PRX-D4 {
    base pmd-type;
    description
      "One single-mode fiber 10.3125 GBd continuous downstream /
       1.25 GBd burst mode upstream OLT PHY as specified in Clause 75";
  }
```

```
  identity pmd-type-10G-1GBASE-PRX-U1 {
    base pmd-type;
    description
      "One single-mode fiber 10.3125 GBd continuous downstream /
       1.25 GBd burst mode upstream ONU PHY as specified in Clause 75";
  }

  identity pmd-type-10G-1GBASE-PRX-U2 {
    base pmd-type;
    description
      "One single-mode fiber 10.3125 GBd continuous downstream /
       1.25 GBd burst mode upstream ONU PHY as specified in Clause 75";
  }

  identity pmd-type-10G-1GBASE-PRX-U3 {
    base pmd-type;
    description
      "One single-mode fiber 10.3125 GBd continuous downstream /
       1.25 GBd burst mode upstream ONU PHY as specified in Clause 75";
  }

  identity pmd-type-10G-1GBASE-PRX-U4 {
    base pmd-type;
    description
      "One single-mode fiber 10.3125 GBd continuous downstream /
       1.25 GBd burst mode upstream ONU PHY as specified in Clause 75";
  }

  identity pmd-type-10GBASE-BR10-D {
    base pmd-type;
    description
      "One single-mode fiber OLT PHY supporting a distance of at
       least 10 km as specified in Clause 158";
  }

  identity pmd-type-10GBASE-BR10-U {
    base pmd-type;
    description
      "One single-mode fiber ONU PHY supporting a distance of at
       least 10 km as specified in Clause 158";
  }

  identity pmd-type-10GBASE-BR20-D {
    base pmd-type;
    description
      "One single-mode fiber OLT PHY supporting a distance of at
       least 20 km as specified in Clause 158";
  }

  identity pmd-type-10GBASE-BR20-U {
    base pmd-type;
    description
      "One single-mode fiber ONU PHY supporting a distance of at
       least 20 km as specified in Clause 158";
  }

  identity pmd-type-10GBASE-BR40-D {
    base pmd-type;
```

```
    description
      "One single-mode fiber OLT PHY upporting a distance of at
       least 40 km as specified in Clause 158";
  }

  identity pmd-type-10GBASE-BR40-U {
    base pmd-type;
    description
      "One single-mode fiber ONU PHY supporting a distance of at
       least 40 km as specified in Clause 158";
  }

  identity pmd-type-10GBASE-CX4 {
    base pmd-type;
    description
      "10GBASE-CX4 copper over 8 pair 100-Ohm balanced cable as
       specified in Clause 54";
  }

  identity pmd-type-10GBASE-ER {
    base pmd-type;
    description
      "10GBASE-ER fiber over 1550nm optics as specified in Clause 52";
  }

  identity pmd-type-10GBASE-EW {
    base pmd-type;
    description
      "10GBASE-EW fiber over 1550nm optics as specified in Clause 52";
  }

  identity pmd-type-10GBASE-KR {
    base pmd-type;
    description
      "10GBASE-KR PCS/PMA over an electrical backplane PMD as
       specified in Clause 72";
  }

  identity pmd-type-10GBASE-KX4 {
    base pmd-type;
    description
      "10GBASE-KX4 PCS/PMA over an electrical backplane PMD as
       specified in Clause 71";
  }

  identity pmd-type-10GBASE-LR {
    base pmd-type;
    description
      "10GBASE-LR fiber over 1310nm optics as specified in Clause 52";
  }

  identity pmd-type-10GBASE-LRM {
    base pmd-type;
    description
      "10GBASE-LRM fiber over 1310 nm optics as specified in Clause
       68";
  }
```

```
  identity pmd-type-10GBASE-LW {
    base pmd-type;
    description
      "10GBASE-LW fiber over 1310nm optics as specified in Clause 52";
  }

  identity pmd-type-10GBASE-LX4 {
    base pmd-type;
    description
      "10GBASE-LX4 fiber over 4 lane 1310nm optics as specified in
       Clause 53";
  }

  identity pmd-type-10GBASE-PR-D1 {
    base pmd-type;
    description
      "One single-mode fiber 10.3125 GBd continuous downstream /
       1.25 GBd burst mode upstream OLT PHY as specified in Clause 75";
  }

  identity pmd-type-10GBASE-PR-D2 {
    base pmd-type;
    description
      "One single-mode fiber 10.3125 GBd continuous downstream /
       1.25 GBd burst mode upstream OLT PHY as specified in Clause 75";
  }

  identity pmd-type-10GBASE-PR-D3 {
    base pmd-type;
    description
      "One single-mode fiber 10.3125 GBd continuous downstream /
       1.25 GBd burst mode upstream OLT PHY as specified in Clause 75";
  }

  identity pmd-type-10GBASE-PR-D4 {
    base pmd-type;
    description
      "One single-mode fiber 10.3125 GBd continuous downstream /
       1.25 GBd burst mode upstream OLT PHY as specified in Clause 75";
  }

  identity pmd-type-10GBASE-PR-U1 {
    base pmd-type;
    description
      "One single-mode fiber 10.3125 GBd continuous downstream /
       1.25 GBd burst mode upstream ONU PHY as specified in Clause 75";
  }

  identity pmd-type-10GBASE-PR-U3 {
    base pmd-type;
    description
      "One single-mode fiber 10.3125 GBd continuous downstream /
       1.25 GBd burst mode upstream ONU PHY as specified in Clause 75";
  }

  identity pmd-type-10GBASE-PR-U4 {
    base pmd-type;
    description
```

```
      "One single-mode fiber 10.3125 GBd continuous downstream /
       1.25 GBd burst mode upstream ONU PHY as specified in Clause 75";
  }

  identity pmd-type-10GBASE-R {
    base pmd-type;
    description
      "10GBASE-R PCS/PMA as specified in Clause 49 over undefined
       PMD";
  }

  identity pmd-type-10GBASE-SR {
    base pmd-type;
    description
      "10GBASE-SR fiber over 850nm optics as specified in Clause 52";
  }

  identity pmd-type-10GBASE-SW {
    base pmd-type;
    description
      "10GBASE-SW fiber over 850nm optics as specified in Clause 52";
  }

  identity pmd-type-10GBASE-T {
    base pmd-type;
    description
      "Four-pair twisted-pair balanced copper cabling PHY as
       specified in Clause 55";
  }

  identity pmd-type-10GBASE-T1 {
    base pmd-type;
    description
      "Single balanced pair of conductors PHY as specified in Clause
       149";
  }

  identity pmd-type-10GBASE-X {
    base pmd-type;
    description
      "10GBASE-X PCS/PMA as specified in Clause 48 over undefined
       PMD";
  }

  identity pmd-type-10GPASS-XR {
    base pmd-type;
    description
      "Coax cable distribution network PHY continuous
       downstream/burst mode upstream PHY as specified in Clause 100
       and Clause 101";
  }

  identity pmd-type-10G-2p5GGBASE-SP1-Dx {
    base pmd-type;
    description
      "One single-mode fiber 10.3125 GBd continuous downstream /
       2.578125 GBd burst mode upstream OLT PHY supporting
       channel x of FSR set 1, as specified in 164.2,";
```

```
  }

  identity pmd-type-10G-2p5GGBASE-SP1-Uxy {
    base pmd-type;
    description
      "One single-mode fiber 10.3125 GBd continuous downstream /
       2.578125 GBd burst mode upstream ONU PHY supporting
       channels x to y of FSR set 1, as specified in 164.2";
  }

  identity pmd-type-10GBASE-SP {
    base pmd-type;
    description
      "Clause 164.3 10 Gb/s 256B/257B";
  }

  identity pmd-type-25G-10GBASE-PQG-D2 {
    base pmd-type;
    description
      "One single mode fiber, 1 × 25.78125 GBd continuous
       transmission / 1 × 10.3125 GBd burst mode reception, medium
       power class, as specified in Clause 141";
  }

  identity pmd-type-25G-10GBASE-PQG-D3 {
    base pmd-type;
    description
      "One single mode fiber, 1 × 25.78125 GBd continuous
       transmission / 1 × 10.3125 GBd burst mode reception, high
       power class, as specified in Clause 141";
  }

  identity pmd-type-25G-10GBASE-PQG-U2 {
    base pmd-type;
    description
      "One single mode fiber, 1 × 25.78125 GBd continuous reception
       / 1 × 10.3125 GBd burst mode transmission, medium power class,
       as specified in Clause 141";
  }

  identity pmd-type-25G-10GBASE-PQG-U3 {
    base pmd-type;
    description
      "One single mode fiber, 1 × 25.78125 GBd continuous reception
       / 1 × 10.3125 GBd burst mode transmission, high power class,
       as specified in Clause 141";
  }

  identity pmd-type-25G-10GBASE-PQX-D2 {
    base pmd-type;
    description
      "One single mode fiber, 1 × 25.78125 GBd continuous
       transmission / 1 × 10.3125 GBd burst mode reception, medium
       power class, as specified in Clause 141";
  }

  identity pmd-type-25G-10GBASE-PQX-D3 {
    base pmd-type;
```

```
    description
      "One single mode fiber, 1 × 25.78125 GBd continuous
       transmission / 1 × 10.3125 GBd burst mode reception, high
       power class, as specified in Clause 141";
  }

  identity pmd-type-25G-10GBASE-PQX-U2 {
    base pmd-type;
    description
      "One single mode fiber, 1 × 25.78125 GBd continuous reception
       / 1 × 10.3125 GBd burst mode transmission, medium power class,
       as specified in Clause 141";
  }

  identity pmd-type-25G-10GBASE-PQX-U3 {
    base pmd-type;
    description
      "One single mode fiber, 1 × 25.78125 GBd continuous reception
       / 1 × 10.3125 GBd burst mode transmission, high power class,
       as specified in Clause 141";
  }

  identity pmd-type-25GBASE-BR10-D {
    base pmd-type;
    description
      "One single-mode fiber OLT PHY supporting a distance of at
       least 10 km as specified in Clause 159";
  }

  identity pmd-type-25GBASE-BR10-U {
    base pmd-type;
    description
      "One single-mode fiber ONU PHY supporting a distance of at
       least 10 km as specified in Clause 159";
  }

  identity pmd-type-25GBASE-BR20-D {
    base pmd-type;
    description
      "One single-mode fiber OLT PHY supporting a distance of at
       least 20 km as specified in Clause 159";
  }

  identity pmd-type-25GBASE-BR20-U {
    base pmd-type;
    description
      "One single-mode fiber ONU PHY supporting a distance of at
       least 20 km as specified in Clause 159";
  }

  identity pmd-type-25GBASE-BR40-D {
    base pmd-type;
    description
      "One single-mode fiber OLT PHY supporting a distance of at
       least 40 km as specified in Clause 159";
  }

  identity pmd-type-25GBASE-BR40-U {
```

```
    base pmd-type;
    description
      "One single-mode fiber ONU PHY supporting a distance of at
       least 40 km as specified in Clause 159";
  }

  identity pmd-type-25GBASE-CR {
    base pmd-type;
    description
      "25GBASE-R PCS/PMA over shielded balanced copper cable PMD as
       specified in Clause 110";
  }

  identity pmd-type-25GBASE-CR-S {
    base pmd-type;
    description
      "25GBASE-R PCS/PMA over shielded balanced copper cable PMD as
       specified in Clause 110 without support for RS-FEC ";
  }

  identity pmd-type-25GBASE-ER {
    base pmd-type;
    description
      "25GBASE-R PCS/PMA over single-mode fiber PMD, with extended
       reach, as specified in Clause 114";
  }

  identity pmd-type-25GBASE-KR {
    base pmd-type;
    description
      "25GBASE-R PCS/PMA over an electrical backplane PMD as
       specified in Clause 111";
  }

  identity pmd-type-25GBASE-KR-S {
    base pmd-type;
    description
      "25GBASE-R PCS/PMA over an electrical backplane PMD as
       specified in Clause 111 without support for RS-FEC";
  }

  identity pmd-type-25GBASE-LR {
    base pmd-type;
    description
      "25GBASE-R PCS/PMA over single-mode fiber PMD, with long
       reach, as specified in Clause 114";
  }

  identity pmd-type-25GBASE-PQG-D2 {
    base pmd-type;
    description
      "One single mode fiber, 1 × 25.78125 GBd continuous
       transmission / 1 × 25.78125 GBd burst mode reception, medium
       power class,  as specified in Clause 141";
  }

  identity pmd-type-25GBASE-PQG-D3 {
    base pmd-type;
```

```
    description
      "One single mode fiber, 1 × 25.78125 GBd continuous
       transmission / 1 × 25.78125 GBd burst mode reception, high
       power class, as specified in Clause 141";
  }

  identity pmd-type-25GBASE-PQG-U2 {
    base pmd-type;
    description
      "One single mode fiber, 1 × 25.78125 GBd continuous reception
       / 1 × 25.78125 GBd burst mode transmission, medium power
       class, as specified in Clause 141";
  }

  identity pmd-type-25GBASE-PQG-U3 {
    base pmd-type;
    description
      "One single mode fiber, 1 × 25.78125 GBd continuous reception
       / 1 × 25.78125 GBd burst mode transmission, high power class,
       as specified in Clause 141";
  }

  identity pmd-type-25GBASE-PQX-D2 {
    base pmd-type;
    description
      "One single mode fiber, 1 × 25.78125 GBd continuous
       transmission / 1 × 25.78125 GBd burst mode reception, medium
       power class, as specified in Clause 141";
  }

  identity pmd-type-25GBASE-PQX-D3 {
    base pmd-type;
    description
      "One single mode fiber, 1 × 25.78125 GBd continuous
       transmission / 1 × 25.78125 GBd burst mode reception, high
       power class, as specified in Clause 141";
  }

  identity pmd-type-25GBASE-PQX-U2 {
    base pmd-type;
    description
      "One single mode fiber, 1 × 25.78125 GBd continuous reception
       / 1 × 25.78125 GBd burst mode transmission, medium power
       class, as specified in Clause 141";
  }

  identity pmd-type-25GBASE-PQX-U3 {
    base pmd-type;
    description
      "One single mode fiber, 1 × 25.78125 GBd continuous reception
       / 1 × 25.78125 GBd burst mode transmission, high power class,
       as specified in Clause 141";
  }

  identity pmd-type-25GBASE-R {
    base pmd-type;
    description
      "PCS as specified in Clause 107 with PMA as specified in
```

```
       Clause 109 over undefined PMD";
  }

  identity pmd-type-25GBASE-SR {
    base pmd-type;
    description
      "25GBASE-R PCS/PMA over multimode fiber PMD as specified in
       Clause 112";
  }

  identity pmd-type-25GBASE-T {
    base pmd-type;
    description
      "Four-pair twisted-pair balanced copper cabling PHY as
       specified in Clause 113";
  }

  identity pmd-type-40GBASE-CR4 {
    base pmd-type;
    description
      "40GBASE-R PCS/PMA over 4 lane shielded copper balanced cable
       PMD as specified in Clause 85";
  }

  identity pmd-type-40GBASE-ER4 {
    base pmd-type;
    description
      "40GBASE-R PCS/PMA over 4 WDM lane single mode fiber PMD, with
       extended reach, as specified in Clause 87";
  }

  identity pmd-type-40GBASE-FR {
    base pmd-type;
    description
      "40GBASE-R PCS/PMA over single mode fiber PMD as specified in
       Clause 89";
  }

  identity pmd-type-40GBASE-KR4 {
    base pmd-type;
    description
      "40GBASE-R PCS/PMA over an electrical backplane PMD as
       specified in Clause 84";
  }

  identity pmd-type-40GBASE-LR4 {
    base pmd-type;
    description
      "40GBASE-R PCS/PMA over 4 WDM lane single mode fiber PMD, with
       long reach, as specified in Clause 87";
  }

  identity pmd-type-40GBASE-R {
    base pmd-type;
    description
      "Multi-lane PCS as specified in Clause 82 over undefined
       PMA/PMD";
  }
```

```
  identity pmd-type-40GBASE-SR4 {
    base pmd-type;
    description
      "40GBASE-R PCS/PMA over 4 lane multimode fiber PMD as
       specified in Clause 86";
  }

  identity pmd-type-40GBASE-T {
    base pmd-type;
    description
      "Four-pair twisted-pair balanced copper cabling PHY as
       specified in Clause 113";
  }

  identity pmd-type-50G-10GBASE-PQG-D2 {
    base pmd-type;
    description
      "One single mode fiber, 2 × 25.78125 GBd continuous
       transmission / 1 × 10.3125 GBd burst mode reception, medium
       power class, as specified in Clause 141";
  }

  identity pmd-type-50G-10GBASE-PQG-D3 {
    base pmd-type;
    description
      "One single mode fiber, 2 × 25.78125 GBd continuous
       transmission / 1 × 10.3125 GBd burst mode reception, high
       power class, as specified in Clause 141";
  }

  identity pmd-type-50G-10GBASE-PQG-U2 {
    base pmd-type;
    description
      "One single mode fiber, 2 × 25.78125 GBd continuous reception
       / 1 × 10.3125 GBd burst mode transmission, medium power class,
       as specified in Clause 141";
  }

  identity pmd-type-50G-10GBASE-PQG-U3 {
    base pmd-type;
    description
      "One single mode fiber, 2 × 25.78125 GBd continuous reception
       / 1 × 10.3125 GBd burst mode transmission, high power class,
       as specified in Clause 141";
  }

  identity pmd-type-50G-10GBASE-PQX-D2 {
    base pmd-type;
    description
      "One single mode fiber, 2 × 25.78125 GBd continuous
       transmission / 1 × 10.3125 GBd burst mode reception, medium
       power class, as specified in Clause 141";
  }

  identity pmd-type-50G-10GBASE-PQX-D3 {
    base pmd-type;
    description
```

```
      "One single mode fiber, 2 × 25.78125 GBd continuous
       transmission / 1 × 10.3125 GBd burst mode reception, high
       power class, as specified in Clause 141";
  }

  identity pmd-type-50G-10GBASE-PQX-U2 {
    base pmd-type;
    description
      "One single mode fiber, 2 × 25.78125 GBd continuous reception
       / 1 × 10.3125 GBd burst mode transmission, medium power class,
       as specified in Clause 141";
  }

  identity pmd-type-50G-10GBASE-PQX-U3 {
    base pmd-type;
    description
      "One single mode fiber, 2 × 25.78125 GBd continuous reception
       / 1 × 10.3125 GBd burst mode transmission, high power class,
       as specified in Clause 141";
  }

  identity pmd-type-50G-25GBASE-PQG-D2 {
    base pmd-type;
    description
      "One single mode fiber, 2 × 25.78125 GBd continuous
       transmission / 1 × 25.78125 GBd burst mode reception, medium
       power class, as specified in Clause 141";
  }

  identity pmd-type-50G-25GBASE-PQG-D3 {
    base pmd-type;
    description
      "One single mode fiber, 2 × 25.78125 GBd continuous
       transmission / 1 × 25.78125 GBd burst mode reception, high
       power class, as specified in Clause 141";
  }

  identity pmd-type-50G-25GBASE-PQG-U2 {
    base pmd-type;
    description
      "One single mode fiber, 2 × 25.78125 GBd continuous reception
       / 1 × 25.78125 GBd burst mode transmission, medium power
       class, as specified in Clause 141";
  }

  identity pmd-type-50G-25GBASE-PQG-U3 {
    base pmd-type;
    description
      "One single mode fiber, 2 × 25.78125 GBd continuous reception
       / 1 × 25.78125 GBd burst mode transmission, high power class,
       as specified in Clause 141";
  }

  identity pmd-type-50G-25GBASE-PQX-D2 {
    base pmd-type;
    description
      "One single mode fiber, 2 × 25.78125 GBd continuous
       transmission / 1 × 25.78125 GBd burst mode reception, medium
```

```
        power class, as specified in Clause 141";
  }

  identity pmd-type-50G-25GBASE-PQX-D3 {
    base pmd-type;
    description
      "One single mode fiber, 2 × 25.78125 GBd continuous
       transmission / 1 × 25.78125 GBd burst mode reception, high
       power class, as specified in Clause 141";
  }

  identity pmd-type-50G-25GBASE-PQX-U2 {
    base pmd-type;
    description
      "One single mode fiber, 2 × 25.78125 GBd continuous reception
       / 1 × 25.78125 GBd burst mode transmission, medium power
       class, as specified in Clause 141";
  }

  identity pmd-type-50G-25GBASE-PQX-U3 {
    base pmd-type;
    description
      "One single mode fiber, 2 × 25.78125 GBd continuous reception
       / 1 × 25.78125 GBd burst mode transmission, high power class,
       as specified in Clause 141";
  }

  identity pmd-type-50GBASE-BR10-D {
    base pmd-type;
    description
      "One single-mode fiber OLT PHY supporting a distance of at
       least 10 km as specified in Clause 160";
  }

  identity pmd-type-50GBASE-BR10-U {
    base pmd-type;
    description
      "One single-mode fiber ONU PHY supporting a distance of at
       least 10 km as specified in Clause 160";
  }

  identity pmd-type-50GBASE-BR20-D {
    base pmd-type;
    description
      "One single-mode fiber OLT PHY supporting a distance of at
       least 20 km as specified in Clause 160";
  }

  identity pmd-type-50GBASE-BR20-U {
    base pmd-type;
    description
      "One single-mode fiber ONU PHY supporting a distance of at
       least 20 km as specified in Clause 160";
  }

  identity pmd-type-50GBASE-BR40-D {
    base pmd-type;
    description
```

```
      "One single-mode fiber OLT PHY supporting a distance of at
       least 40 km as specified in Clause 160";
  }

  identity pmd-type-50GBASE-BR40-U {
    base pmd-type;
    description
      "One single-mode fiber ONU PHY supporting a distance of at
       least 40 km as specified in Clause 160";
  }

  identity pmd-type-50GBASE-CR {
    base pmd-type;
    description
      "50GBASE-R PCS/PMA over shielded copper balanced cable PMD as
       specified in Clause 136 ";
  }

  identity pmd-type-50GBASE-ER {
    base pmd-type;
    description
      "50GBASE-R PCS/PMA over single-mode fiber PMD with reach up to
       at least 40 km as specified in Clause 139";
  }

  identity pmd-type-50GBASE-FR {
    base pmd-type;
    description
      "50GBASE-R PCS/PMA over single mode fiber PMD as specified in
       Clause 139";
  }

  identity pmd-type-50GBASE-KR {
    base pmd-type;
    description
      "50GBASE-R PCS/PMA over an electrical backplane PMD as
       specified in Clause 137";
  }

  identity pmd-type-50GBASE-LR {
    base pmd-type;
    description
      "50GBASE-R PCS/PMA over single mode fiber PMD as specified in
       Clause 139";
  }

  identity pmd-type-50GBASE-PQG-D2 {
    base pmd-type;
    description
      "One single mode fiber, 2 × 25.78125 GBd continuous
       transmission / 2 × 25.78125 GBd burst mode reception, medium
       power class, as specified in Clause 141";
  }

  identity pmd-type-50GBASE-PQG-D3 {
    base pmd-type;
    description
      "One single mode fiber, 2 × 25.78125 GBd continuous
```

```
        transmission / 2 × 25.78125 GBd burst mode reception, high
        power class, as specified in Clause 141";
  }

  identity pmd-type-50GBASE-PQG-U2 {
    base pmd-type;
    description
      "One single mode fiber, 2 × 25.78125 GBd continuous reception
       / 2 × 25.78125 GBd burst mode transmission, medium power
       class, as specified in Clause 141";
  }

  identity pmd-type-50GBASE-PQG-U3 {
    base pmd-type;
    description
      "One single mode fiber, 2 × 25.78125 GBd continuous reception
       / 2 × 25.78125 GBd burst mode transmission, high power class,
       as specified in Clause 141";
  }

  identity pmd-type-50GBASE-PQX-D2 {
    base pmd-type;
    description
      "One single mode fiber, 2 × 25.78125 GBd continuous
       transmission / 2 × 25.78125 GBd burst mode reception, medium
       power class, as specified in Clause 141";
  }

  identity pmd-type-50GBASE-PQX-D3 {
    base pmd-type;
    description
      "One single mode fiber, 2 × 25.78125 GBd continuous
       transmission / 2 × 25.78125 GBd burst mode reception, high
       power class, as specified in Clause 141";
  }

  identity pmd-type-50GBASE-PQX-U2 {
    base pmd-type;
    description
      "One single mode fiber, 2 × 25.78125 GBd continuous reception
       / 2 × 25.78125 GBd burst mode transmission, medium power
       class, as specified in Clause 141";
  }

  identity pmd-type-50GBASE-PQX-U3 {
    base pmd-type;
    description
      "One single mode fiber, 2 × 25.78125 GBd continuous reception
       / 2 × 25.78125 GBd burst mode transmission, high power class,
       as specified in Clause 141";
  }

  identity pmd-type-50GBASE-R {
    base pmd-type;
    description
      "Multi-lane PCS as specified in Clause 133 with PMA as
       specified in Clause 135 over undefined PMD";
  }
```

```
  identity pmd-type-50GBASE-SR {
    base pmd-type;
    description
      "50GBASE-R PCS/PMA over multimode fiber PMD as specified in
       Clause 138";
  }

  identity pmd-type-100CR1 {
    base pmd-type;
    description
      "100GBASE-CR1 as specified in Clause 162 or 100GBASE-KR1 as
       specified in Clause 163";
  }

  identity pmd-type-100GBASE-CR2 {
    base pmd-type;
    description
      "100GBASE-R PCS/PMA over 2 lane shielded copper balanced cable
       PMD as specified in Clause 136";
  }

  identity pmd-type-100GBASE-CR4 {
    base pmd-type;
    description
      "100GBASE-R PCS/PMA over 4 lane shielded copper balanced cable
       PMD as specified in Clause 92";
  }

  identity pmd-type-100GBASE-CR10 {
    base pmd-type;
    description
      "100GBASE-R PCS/PMA over 10 lane shielded copper balanced
       cable PMD as specified in Clause 85";
  }

  identity pmd-type-100GBASE-DR {
    base pmd-type;
    description
      "100GBASE-R PCS/PMA over single mode fiber PMD as specified in
       Clause 140";
  }

  identity pmd-type-100GBASE-ER4 {
    base pmd-type;
    description
      "100GBASE-R PCS/PMA over 4 WDM lane single mode fiber PMD,
       with extended reach, as specified in Clause 88";
  }

  identity pmd-type-100GBASE-FR1 {
    base pmd-type;
    description
      "100GBASE-R PCS/PMA over single-mode fiber PMD with reach up
       to at least 2 km as specified in Clause 140";
  }

  identity pmd-type-100GBASE-KP4 {
```

```
    base pmd-type;
    description
      "100GBASE-P PCS/PMA over an electrical backplane PMD as
       specified in Clause 94";
  }

  identity pmd-type-100GBASE-KR2 {
    base pmd-type;
    description
      "100GBASE-R PCS/PMA over an electrical backplane PMD as
       specified in Clause 137";
  }

  identity pmd-type-100GBASE-KR4 {
    base pmd-type;
    description
      "100GBASE-R PCS/PMA over an electrical backplane PMD as
       specified in Clause 93";
  }

  identity pmd-type-100GBASE-LR1 {
    base pmd-type;
    description
      "100GBASE-R PCS/PMA over single-mode fiber PMD with reach up
       to at least 10 km as specified in Clause 140";
  }

  identity pmd-type-100GBASE-LR4 {
    base pmd-type;
    description
      "100GBASE-R PCS/PMA over 4 WDM lane single mode fiber PMD,
       with long reach, as specified in Clause 88";
  }

  identity pmd-type-100GBASE-R {
    base pmd-type;
    description
      "Multi-lane PCS as specified in Clause 82 over undefined
       100GBASE-R or 100GBASE-P PMA/PMD";
  }

  identity pmd-type-100GBASE-SR1 {
    base pmd-type;
    description
      "100GBASE-R PCS/PMA over multimode fiber PMD with
       reach up to at least 100 m as specified in Clause 167";
  }

  identity pmd-type-100GBASE-SR2 {
    base pmd-type;
    description
      "100GBASE-R PCS/PMA over 2 lane multimode fiber PMD as
       specified in Clause 138";
  }

  identity pmd-type-100GBASE-SR4 {
    base pmd-type;
    description
```

```
      "100GBASE-R PCS/PMA over 4 lane multimode fiber PMD as
       specified in Clause 95";
  }

  identity pmd-type-100GBASE-SR10 {
    base pmd-type;
    description
      "100GBASE-R PCS/PMA over 10 lane multimode fiber PMD as
       specified in Clause 86";
  }

  identity pmd-type-100GBASE-VR1 {
    base pmd-type;
    description
      "100GBASE-R PCS/PMA over multimode fiber PMD with
       reach up to at least 50 m as specified in Clause 167";
  }

  identity pmd-type-100GBASE-ZR {
    base pmd-type;
    description
      "100GBASE-R PCS/100GBASE-ZR PMA over a PMD with reach up to at
       least 80 km as specified in Clause 154";
  }

  identity pmd-type-200GBASE-CR2 {
    base pmd-type;
    description
      "200GBASE-CR2 as specified in Clause 162 or 200GBASE-KR2 as
       specified in Clause 163";
  }

  identity pmd-type-200GBASE-CR4 {
    base pmd-type;
    description
      "200GBASE-R PCS/PMA over 4 lane shielded copper balanced cable
       PMD as specified in Clause 136 ";
  }

  identity pmd-type-200GBASE-DR4 {
    base pmd-type;
    description
      "200GBASE-R PCS/PMA over 4-lane single-mode fiber PMD as
       specified in Clause 121";
  }

  identity pmd-type-200GBASE-ER4 {
    base pmd-type;
    description
      "200GBASE-R PCS/PMA over 4 WDM lane single-mode fiber PMD with
       reach up to at least 40 km as specified in Clause 122";
  }

  identity pmd-type-200GBASE-FR4 {
    base pmd-type;
    description
      "200GBASE-R PCS/PMA over 4 WDM lane single-mode fiber PMD with
       reach up to at least 2 km as specified in Clause 122";
```

```
  }

  identity pmd-type-200GBASE-KR2 {
    base pmd-type;
    description
      "200GBASE-R PCS/PMA over an electrical backplane PMD
       as specified in Clause 163";
  }

  identity pmd-type-200GBASE-KR4 {
    base pmd-type;
    description
      "200GBASE-R PCS/PMA over an electrical backplane PMD as
       specified in Clause 137";
  }

  identity pmd-type-200GBASE-LR4 {
    base pmd-type;
    description
      "200GBASE-R PCS/PMA over 4 WDM lane single-mode fiber PMD with
       reach up to at least 10 km as specified in Clause 122";
  }

  identity pmd-type-200GBASE-R {
    base pmd-type;
    description
      "Multi-lane PCS as specified in Clause 119 over undefined
       PMA/PMD";
  }

  identity pmd-type-200GBASE-SR2 {
    base pmd-type;
    description
      "200GBASE-R PCS/PMA over 2 lane multimode fiber PMD with
       reach up to at least 100 m as specified in Clause 167";
  }

  identity pmd-type-200GBASE-SR4 {
    base pmd-type;
    description
      "200GBASE-R PCS/PMA over 4 lane multimode fiber PMD as
       specified in Clause 138";
  }

  identity pmd-type-200GBASE-VR2 {
    base pmd-type;
    description
      "200GBASE-R PCS/PMA over 2 lane multimode fiber PMD with
       reach up to at least 50 m as specified in Clause 167";
  }

  identity pmd-type-400GBASE-DR4 {
    base pmd-type;
    description
      "400GBASE-R PCS/PMA over 4-lane single-mode fiber PMD as
       specified in Clause 124";
  }
```

```
  identity pmd-type-400GBASE-DR4-2 {
    base pmd-type;
    description
      "400GBASE-R PCS/PMA over 4 single-mode fibers in each
       direction PMD with reach up to at least 2 km as specified
       in Clause 124";
  }

  identity pmd-type-400GBASE-ER8 {
    base pmd-type;
    description
      "400GBASE-R PCS/PMA over 8 WDM lane single-mode fiber PMD with
       reach up to at least 40 km as specified in Clause 122";
  }

  identity pmd-type-400GBASE-FR4 {
    base pmd-type;
    description
      "400GBASE-R PCS/PMA over 4 WDM lane single-mode fiber PMD with
       reach up to at least 2 km as specified in Clause 151";
  }

  identity pmd-type-400GBASE-FR8 {
    base pmd-type;
    description
      "400GBASE-R PCS/PMA over 8 WDM lane single-mode fiber PMD with
       reach up to at least 2 km as specified in Clause 122";
  }

  identity pmd-type-400GBASE-LR4-6 {
    base pmd-type;
    description
      "400GBASE-R PCS/PMA over 4 WDM lane single-mode fiber PMD with
       reach up to at least 6 km as specified in Clause 151";
  }

  identity pmd-type-400GBASE-LR8 {
    base pmd-type;
    description
      "400GBASE-R PCS/PMA over 8 WDM lane single-mode fiber PMD with
       reach up to at least 10 km as specified in Clause 122";
  }

  identity pmd-type-400GBASE-R {
    base pmd-type;
    description
      "Multi-lane PCS as specified in Clause 119 over undefined
       PMA/PMD";
  }

  identity pmd-type-400GBASE-SR4 {
    base pmd-type;
    description
      "400GBASE-R PCS/PMA over 4 lane multimode fiber PMD with
       reach up to at least 100 m as specified in Clause 167";
  }

  identity pmd-type-400GBASE-SR4d2 {
```

```
    base pmd-type;
    description
      "400GBASE-R PCS/PMA over 8-lane multimode fiber PMD as
       specified in Clause 150";
  }

  identity pmd-type-400GBASE-SR8 {
    base pmd-type;
    description
      "400GBASE-R PCS/PMA over 8-lane multimode fiber PMD as
       specified in Clause 138";
  }

  identity pmd-type-400GBASE-SR16 {
    base pmd-type;
    description
      "400GBASE-R PCS/PMA over 16-lane multimode fiber PMD as
       specified in Clause 123";
  }

  identity pmd-type-400GBASE-VR4 {
    base pmd-type;
    description
      "400GBASE-R PCS/PMA over 4 lane multimode fiber PMD with
       reach up to at least 50 m as specified in Clause 167";
  }

  identity pmd-type-800GBASE-CR8 {
    base pmd-type;
    description
      "800GBASE-R PCS/PMA over 8-lane shielded balanced copper cable
       PMD as specified in Clause 162";
  }

  identity pmd-type-800GBASE-DR8 {
    base pmd-type;
    description
      "800GBASE-R PCS/PMA over 8 single-mode fibers in each
          direction PMD with reach up to at least 500 m as specified
          in Clause 124";
  }

  identity pmd-type-800GBASE-DR8-2 {
    base pmd-type;
    description
      "800GBASE-R PCS/PMA over 8 single-mode fibers in each
       direction PMD with reach up to at least 2 km as specified
       in Clause 124";
  }

  identity pmd-type-800GBASE-KR8 {
    base pmd-type;
    description
      "800GBASE-R PCS/PMA over an electrical backplane PMD as
       specified in Clause 163";
  }

  identity pmd-type-800GBASE-R {
```

```
    base pmd-type;
    description
      "Multi-lane PCS as specified in Clause 172 over
       undefined PMA/PMD";
  }

  identity pmd-type-800GBASE-SR8 {
    base pmd-type;
    description
      "800GBASE-R PCS/PMA over 8 multimode fibers in each
       direction PMD with reach up to at least 100 m as specified
       in Clause 167";
  }

  identity pmd-type-800GBASE-VR8 {
    base pmd-type;
    description
      "800GBASE-R PCS/PMA over 8 multimode fibers in each
       direction PMD with reach up to at least 50 m as specified
       in Clause 167";
  }

  typedef pmd-type-list {
    type bits {
      /* 802.3-2022 */
      bit bOther {
        description
          "Other";
      }
      bit b10baseT {
        description
          "10BASE-T";
      }
      bit b10baseFL {
        description
          "10BASE-FL";
      }
      bit b100baseTX {
        description
          "100BASE-TX";
      }
      bit b100baseFX {
        description
          "100BASE-FX";
      }
      bit b1000baseX {
        description
          "1000BASE-X";
      }
      bit b1000baseLX {
        description
          "1000BASE-LX";
      }
      bit b1000baseSX {
        description
          "1000BASE-SX";
      }
      bit b1000baseCX {
```

```
        description
          "1000BASE-CX";
      }
      bit b1000baseT {
        description
          "1000BASE-T";
      }
      bit b10GbaseX {
        description
          "10GBASE-X";
      }
      bit b10GbaseLX4 {
        description
          "10GBASE-LX4";
      }
      bit b10GbaseR {
        description
          "10GBASE-R";
      }
      bit b10GbaseER {
        description
          "10GBASE-ER";
      }
      bit b10GbaseLR {
        description
          "10GBASE-LR";
      }
      bit b10GbaseSR {
        description
          "10GBASE-SR";
      }
      bit b10GbaseSW {
        description
          "10GBASE-SW";
      }
      bit b10GbaseCX4 {
        description
          "10GBASE-CX4";
      }
      bit b100BaseBX10D {
        description
          "100BASE-BX10D";
      }
      bit b100BaseBX10U {
        description
          "100BASE-BX10U";
      }
      bit b100BaseLX10 {
        description
          "100BASE-LX10";
      }
      bit b1000BaseBX10D {
        description
          "1000BASE-BX10D";
      }
      bit b1000BaseBX10U {
        description
          "1000BASE-BX10U";
```

```
      }
      bit b1000BaseLX10 {
        description
          "1000BASE-LX10";
      }
      bit b1000BasePX10D {
        description
          "1000BASE-PX10D";
      }
      bit b1000BasePX10U {
        description
          "1000BASE-PX10U";
      }
      bit b1000BasePX20D {
        description
          "1000BASE-PX20D";
      }
      bit b1000BasePX20U {
        description
          "1000BASE-PX20U";
      }
      bit b10GbaseT {
        description
          "10GBASE-T";
      }
      bit b10GbaseLRM {
        description
          "10GBASE-LRM";
      }
      bit b1000baseKX {
        description
          "1000BASE-KX";
      }
      bit b10GbaseKX4 {
        description
          "10GBASE-KX4";
      }
      bit b10GbaseKR {
        description
          "10GBASE-KR";
      }
      bit b10G1GbasePRXD1 {
        description
          "10/1GBASE-PRXD1";
      }
      bit b10G1GbasePRXD2 {
        description
          "10/1GBASE-PRXD2";
      }
      bit b10G1GbasePRXD3 {
        description
          "10/1GBASE-PRXD3";
      }
      bit b10G1GbasePRXU1 {
        description
          "10/1GBASE-PRXU1";
      }
      bit b10G1GbasePRXU2 {
```

```
        description
          "10/1GBASE-PRXU2";
      }
      bit b10G1GbasePRXU3 {
        description
          "10/1GBASE-PRXU3";
      }
      bit b10GbasePRD1 {
        description
          "10GBASE-PRD1";
      }
      bit b10GbasePRD2 {
        description
          "10GBASE-PRD2";
      }
      bit b10GbasePRD3 {
        description
          "10GBASE-PRD3";
      }
      bit b10GbasePRU1 {
        description
          "10GBASE-PRU1";
      }
      bit b10GbasePRU3 {
        description
          "10GBASE-PRU3";
      }
      bit b40GbaseKR4 {
        description
          "40GBASE-KR4";
      }
      bit b40GbaseCR4 {
        description
          "40GBASE-CR4";
      }
      bit b40GbaseSR4 {
        description
          "40GBASE-SR4";
      }
      bit b40GbaseFR {
        description
          "40GBASE-FR";
      }
      bit b40GbaseLR4 {
        description
          "40GBASE-LR4";
      }
      bit b100GbaseCR10 {
        description
          "100GBASE-CR10";
      }
      bit b100GbaseSR10 {
        description
          "100GBASE-SR10";
      }
      bit b100GbaseLR4 {
        description
          "100GBASE-LR4";
```

```
      }
      bit b100GbaseER4 {
        description
          "100GBASE-ER4";
      }
      bit b1000baseT1 {
        description
          "1000BASE-T1";
      }
      bit b1000basePX30D {
        description
          "1000BASE-PX30D";
      }
      bit b1000basePX30U {
        description
          "1000BASE-PX30U";
      }
      bit b1000basePX40D {
        description
          "1000BASE-PX40D";
      }
      bit b1000basePX40U {
        description
          "1000BASE-PX40U";
      }
      bit b10G1GbasePRXD4 {
        description
          "10/1GBASE-PRXD4";
      }
      bit b10G1GbasePRXU4 {
        description
          "10/1GBASE-PRXU4";
      }
      bit b10GbasePRD4 {
        description
          "10GBASE-PRD4";
      }
      bit b10GbasePRU4 {
        description
          "10GBASE-PRU4";
      }
      bit b25GbaseCR {
        description
          "25GBASE-CR";
      }
      bit b25GbaseCRS {
        description
          "25GBASE-CRS";
      }
      bit b25GbaseKR {
        description
          "25GBASE-KR";
      }
      bit b25GbaseKRS {
        description
          "25GBASE-KRS";
      }
      bit b25GbaseR {
```

```
          description
            "25GBASE-R";
        }
        bit b25GbaseSR {
          description
            "25GBASE-SR";
        }
        bit b25GbaseT {
          description
            "25GBASE-T";
        }
        bit b40GbaseER4 {
          description
            "40GBASE-ER4";
        }
        bit b40GbaseR {
          description
            "40GBASE-R";
        }
        bit b40GbaseT {
          description
            "40GBASE-T";
        }
        bit b100GbaseCR4 {
          description
            "100GBASE-CR4";
        }
        bit b100GbaseKR4 {
          description
            "100GBASE-KR4";
        }
        bit b100GbaseKP4 {
          description
            "100GBASE-KP4";
        }
        bit b100GbaseR {
          description
            "100GBASE-R";
        }
        bit b100GbaseSR4 {
          description
            "100GBASE-SR4";
        }
        bit b2p5GbaseT {
          description
            "2.5GBASE-T";
        }
        bit b5GbaseT {
          description
            "5GBASE-T";
        }
        bit b100baseT1 {
          description
            "100BASE-T1";
        }
        bit b1000baseRHA {
          description
            "1000BASE-RHA";
```

```
        }
        bit b1000baseRHB {
          description
            "1000BASE-RHB";
        }
        bit b1000baseRHC {
          description
            "1000BASE-RHC";
        }
        bit b2p5GbaseKX {
          description
            "2.5GBASE-KX";
        }
        bit b2p5GbaseX {
          description
            "2.5GBASE-X";
        }
        bit b5GbaseKR {
          description
            "5GBASE-KR";
        }
        bit b5GbaseR {
          description
            "5GBASE-R";
        }
        bit b10GpassXR {
          description
            "10GPASS-XR";
        }
        bit b25GbaseLR {
          description
            "25GBASE-LR";
        }
        bit b25GbaseER {
          description
            "25GBASE-ER";
        }
        bit b50GbaseR {
          description
            "50GBASE-R";
        }
        bit b50GbaseCR {
          description
            "50GBASE-CR";
        }
        bit b50GbaseKR {
          description
            "50GBASE-KR";
        }
        bit b50GbaseSR {
          description
            "50GBASE-SR";
        }
        bit b50GbaseFR {
          description
            "50GBASE-FR";
        }
        bit b50GbaseLR {
```

```
          description
            "50GBASE-LR";
        }
        bit b50GbaseER {
          description
            "50GBASE-ER";
        }
        bit b100GbaseCR2 {
          description
            "100GBASE-CR2";
        }
        bit b100GbaseKR2 {
          description
            "100GBASE-KR2";
        }
        bit b100GbaseSR2 {
          description
            "100GBASE-SR2";
        }
        bit b100GbaseDR {
          description
            "100GBASE-DR";
        }
        bit b200GbaseR {
          description
            "200GBASE-R";
        }
        bit b200GbaseDR4 {
          description
            "200GBASE-DR4";
        }
        bit b200GbaseFR4 {
          description
            "200GBASE-FR4";
        }
        bit b200GbaseLR4 {
          description
            "200GBASE-LR4";
        }
        bit b200GbaseCR4 {
          description
            "200GBASE-CR4";
        }
        bit b200GbaseKR4 {
          description
            "200GBASE-KR4";
        }
        bit b200GbaseSR4 {
          description
            "200GBASE-SR4";
        }
        bit b200GbaseER4 {
          description
            "200GBASE-ER4";
        }
        bit b400GbaseR {
          description
            "400GBASE-R";
```

```
        }
        bit b400GbaseSR16 {
          description
            "400GBASE-SR16";
        }
        bit b400GbaseDR4 {
          description
            "400GBASE-DR4";
        }
        bit b400GbaseFR8 {
          description
            "400GBASE-FR8";
        }
        bit b400GbaseLR8 {
          description
            "400GBASE-LR8";
        }
        bit b400GbaseER8 {
          description
            "400GBASE-ER8";
        }
        bit b10baseT1L {
          description
            "10BASE-T1L";
        }
        bit b10baseT1S {
          description
            "10BASE-T1S";
        }
        bit b10baseT1SMD {
          description
            "10BASE-T1SMD";
        }
        bit b100GbaseFR1 {
          description
            "100GBASE-FR1";
        }
        bit b100GbaseLR1 {
          description
            "100GBASE-LR1";
        }
        bit b100GBASE-CR1 {
          description
            "100GBASE-LR1";
        }
        bit b400GbaseFR4 {
          description
            "400GBASE-FR4";
        }
        bit b400GbaseLR46 {
          description
            "400GBASE-LR46";
        }
        bit b400GbaseSR8 {
          description
            "400GBASE-SR8";
        }
        bit b400GbaseSR4p2 {
```

```
          description
            "400GBASE-SR4p2";
        }
        bit b2p5GbaseT1 {
          description
            "2.5GBASE-T1";
        }
        bit b5GbaseT1 {
          description
            "5GBASE-T1";
        }
        bit b10GbaseT1 {
          description
            "10GBASE-T1";
        }
        bit b25G10GbasePQGD2 {
          description
            "25/10GBASE-PQGD2";
        }
        bit b25G10GbasePQGD3 {
          description
            "25/10GBASE-PQGD3";
        }
        bit b25G10GbasePQGU2 {
          description
            "25/10GBASE-PQGU2";
        }
        bit b25G10GbasePQGU3 {
          description
            "25/10GBASE-PQGU3";
        }
        bit b25G10GbasePQXD2 {
          description
            "25/10GBASE-PQXD2";
        }
        bit b25G10GbasePQXD3 {
          description
            "25/10GBASE-PQXD3";
        }
        bit b25G10GbasePQXU2 {
          description
            "25/10GBASE-PQXU2";
        }
        bit b25G10GbasePQXU3 {
          description
            "25/10GBASE-PQXU3";
        }
        bit b25GbasePQGD2 {
          description
            "25GBASE-PQGD2";
        }
        bit b25GbasePQGD3 {
          description
            "25GBASE-PQGD3";
        }
        bit b25GbasePQGU2 {
          description
            "25GBASE-PQGU2";
```

```
      }
      bit b25GbasePQGU3 {
        description
          "25GBASE-PQGU3";
      }
      bit b25GbasePQXD2 {
        description
          "25GBASE-PQXD2";
      }
      bit b25GbasePQXD3 {
        description
          "25GBASE-PQXD3";
      }
      bit b25GbasePQXU2 {
        description
          "25GBASE-PQXU2";
      }
      bit b25GbasePQXU3 {
        description
          "25GBASE-PQXU3";
      }
      bit b50G10GbasePQGD2 {
        description
          "50/10GBASE-PQGD2";
      }
      bit b50G10GbasePQGD3 {
        description
          "50/10GBASE-PQGD3";
      }
      bit b50G10GbasePQGU2 {
        description
          "50/10GBASE-PQGU2";
      }
      bit b50G10GbasePQGU3 {
        description
          "50/10GBASE-PQGU3";
      }
      bit b50G10GbasePQXD2 {
        description
          "50/10GBASE-PQXD2";
      }
      bit b50G10GbasePQXD3 {
        description
          "50/10GBASE-PQXD3";
      }
      bit b50G10GbasePQXU2 {
        description
          "50/10GBASE-PQXU2";
      }
      bit b50G10GbasePQXU3 {
        description
          "50/10GBASE-PQXU3";
      }
      bit b50G25GbasePQGD2 {
        description
          "50/25GBASE-PQGD2";
      }
      bit b50G25GbasePQGD3 {
```

```
          description
            "50/25GBASE-PQGD3";
        }
        bit b50G25GbasePQGU2 {
          description
            "50/25GBASE-PQGU2";
        }
        bit b50G25GbasePQGU3 {
          description
            "50/25GBASE-PQGU3";
        }
        bit b50G25GbasePQXD2 {
          description
            "50/25GBASE-PQXD2";
        }
        bit b50G25GbasePQXD3 {
          description
            "50/25GBASE-PQXD3";
        }
        bit b50G25GbasePQXU2 {
          description
            "50/25GBASE-PQXU2";
        }
        bit b50G25GbasePQXU3 {
          description
            "50/25GBASE-PQXU3";
        }
        bit b50GbasePQGD2 {
          description
            "50GBASE-PQGD2";
        }
        bit b50GbasePQGD3 {
          description
            "50GBASE-PQGD3";
        }
        bit b50GbasePQGU2 {
          description
            "50GBASE-PQGU2";
        }
        bit b50GbasePQGU3 {
          description
            "50GBASE-PQGU3";
        }
        bit b50GbasePQXD2 {
          description
            "50GBASE-PQXD2";
        }
        bit b50GbasePQXD3 {
          description
            "50GBASE-PQXD3";
        }
        bit b50GbasePQXU2 {
          description
            "50GBASE-PQXU2";
        }
        bit b50GbasePQXU3 {
          description
            "50GBASE-PQXU3";
```

```
        }
        bit b100GbaseZR {
          description
            "100GBASE-ZR";
        }
        bit b10GbaseBR10D {
          description
            "10GBASE-BR10D";
        }
        bit b10GbaseBR10U {
          description
            "10GBASE-BR10U";
        }
        bit b10GbaseBR20D {
          description
            "10GBASE-BR20D";
        }
        bit b10GbaseBR20U {
          description
            "10GBASE-BR20U";
        }
        bit b10GbaseBR40D {
          description
            "10GBASE-BR40D";
        }
        bit b10GbaseBR40U {
          description
            "10GBASE-BR40U";
        }
        bit b25GbaseBR10D {
          description
            "25GBASE-BR10D";
        }
        bit b25GbaseBR10U {
          description
            "25GBASE-BR10U";
        }
        bit b25GbaseBR20D {
          description
            "25GBASE-BR20D";
        }
        bit b25GbaseBR20U {
          description
            "25GBASE-BR20U";
        }
        bit b25GbaseBR40D {
          description
            "25GBASE-BR40D";
        }
        bit b25GbaseBR40U {
          description
            "25GBASE-BR40U";
        }
        bit b50GbaseBR10D {
          description
            "50GBASE-BR10D";
        }
        bit b50GbaseBR10U {
```

```
          description
            "50GBASE-BR10U";
        }
        bit b50GbaseBR20D {
          description
            "50GBASE-BR20D";
        }
        bit b50GbaseBR20U {
          description
            "50GBASE-BR20U";
        }
        bit b50GbaseBR40D {
          description
            "50GBASE-BR40D";
        }
        bit b50GbaseBR40U {
          description
            "50GBASE-BR40U";
        }
        /* 802.3ck-2022 */
        bit b100GBaseCR1 {
          description
            "100GBASE-CR1";
        }
        bit b100GBaseKR1 {
          description
            "100GBASE-KR1";
        }
        bit b200GBaseCR2 {
          description
            "200GBASE-CR2";
        }
        bit b200GBaseKR2 {
          description
            "200GBASE-KR2";
        }
        bit b400GBaseCR4 {
          description
            "400GBASE-CR4";
        }
        bit b400GBaseKR4 {
          description
            "400GBASE-KR4";
        }
        /* 802.3cs-2022 */
        bit b10G2p5GBaseSP1Dx {
          description
            "10/2.5GBASE-SP1-Dx";
        }
        bit b10G2p5GBaseSP1Uxy {
          description
            "10/2.5GBASE-SP1-Uxy";
        }
        bit b10GBaseSP1Dx {
          description
            "10GBASE-SP1-Dx";
        }
        bit b10GBaseSP1Uxy {
```

```
          description
            "10GBASE-SP1-Uxy";
        }
        /* 802.3db-2022 */
        bit b100GBaseSR1 {
          description
            "100GBASE-SR1";
        }
        bit b100GBaseVR1 {
          description
            "100GBASE-VR1";
        }
        bit bit200GBaseSR2 {
          description
            "200GBASE-SR2";
        }
        bit bit200GBaseVR2 {
          description
            "200GBASE-VR2";
        }
        bit bit400GBaseSR4 {
          description
            "400GBASE-SR4";
        }
        bit bit400GBaseVR4 {
          description
            "400GBASE-VR4";
        }
        /* 802.3cz-2023 */
        bit b2p5GbaseAU {
          description
            "2.5GBASE-AU";
        }
        bit b5GbaseAU {
          description
            "5GBASE-AU";
        }
        bit b10GbaseAU {
          description
            "10GBASE-AU";
        }
        bit b25GbaseAU {
          description
            "25GBASE-AU";
        }
        bit b50GbaseAU {
          description
            "50GBASE-AU";
        }
        /* 802.3cy-2023 */
        bit b25GbaseT1 {
          description
            "25GBASE-T1";
        }
        /* 802.3ca-2023 */
        bit bit400GBaseDR4d2 {
          description
            "400GBASE-DR4-2";
```

```
        }
        bit bit800GBaseCR8 {
          description
            "800GBASE-CR8";
        }
        bit bit800GBaseDR8 {
          description
            "800GBASE-DR8";
        }
        bit bit800GBaseDR8d2 {
          description
            "800GBASE-DR8-2";
        }
        bit bit800GBaseKR8 {
          description
            "800GBASE-KR8";
        }
        bit bit800GBaseR {
          description
            "800GBASE-R";
        }
        bit bit800GBaseSR8 {
          description
            "800GBASE-SR8";
        }
        bit bit800GBaseVR8 {
          description
            "800GBASE-VR8";
        }
      }
      description
        "This data type is used to report the set of pmd-type values
         an interface can implement.";
      reference
        "IEEE Std 802.3, Clause 30.5.1.1.3 aMAUTypeList";
    }

    typedef media-available-type {
      type enumeration {
        /* 802.3-2022 */
        enum other {
          description
            "Undefined";
        }
        enum unknown {
          description
            "initializing, true state not yet known";
        }
        enum available {
          description
            "link or light normal, loopback normal";
        }
        enum notAvailable {
          description
            "link loss or low light, no loopback";
        }
        enum remoteFault {
          description
```

```
            "remote fault with no detail";
        }
        enum remoteJabber {
          description
            "remote fault, reason known to be jabber";
        }
        enum remoteLinkLoss {
          description
            "remote fault, reason known to be far-end link loss";
        }
        enum remoteTest {
          description
            "remote fault, reason known to be test";
        }
        enum offline {
          description
            "offline, applies only to Clause 37 Auto-Negotiation";
        }
        enum autoNegError {
          description
            "Auto-Negotiation Error, applies only to Clause 37
                  auto-Negotiation";
        }
        enum pmdLinkFault {
          description
            "PMD/PMA receive link fault";
        }
        enum pcsLinkFault {
          description
            "PCS receive link fault";
        }
        enum excessiveBER {
          description
            "PCS Bit Error Ratio monitor reporting excessive error
             ratio";
        }
        enum dxsLinkFault {
          description
            "DTE XGXS receive link fault, applies only to XAUI ";
        }
        enum pxsLinkFault {
          description
            "PHY XGXS transmit link fault, applies only to XAUI";
        }
      }
      description
        "This data type is used to report media availability.";
      reference
        "IEEE Std 802.3, Clause 30.5.1.1.4 aMediaAvailable";
    }

  typedef auto-neg-ability-bits {
    type bits {
      /* Not PHY specific */
      bit bOther {
        description
          "Other abilities beyond the scope of this module.";
      }
```

```
      bit bHalfDuplex {
        description
          "Half duplex";
      }
      bit bFullDuplex {
        description
          "Full duplex";
      }
      bit bPause {
        description
          "PAUSE operation as defined in Annex 31B";
      }
      bit bAPause {
        description
          "Asymmetric PAUSE operation as defined in
           Clause 37, Annex 28B, and Annex 31B";
      }
      bit bSPause {
        description
          "Symmetric PAUSE operation as defined in
           Clause 37, Annex 28B, and Annex 31B";
      }
      bit bBPause {
        description
          "Asymmetric and Symmetric PAUSE operation
           as defined in Clause 37, Annex 28B, and Annex 31B";
      }
      bit bRemFault {
        description
          "Remote fault bit (RF) as specified in Clause 73
           and Clause 98";
      }
      bit bRemFault1 {
        description
          "Remote fault bit 1 (RF1) as specified in Clause 37";
      }
      bit bRemFault2 {
        description
          "Remote fault bit 2 (RF2) as specified in Clause 37";
      }
      bit bFECCapable {
        description
          "FEC ability as specified in Clause 73 (see 73.6.5)
           and Clause 74";
      }
      bit bFECRequested {
        description
          "FEC requested as specified in Clause 73 (see 73.6.5)
           and Clause 74";
      }
      bit bRSFEC25Greq {
        description
          "25G RS-FEC requested as specified in Clause 73 (see 73.6.5)
           and Clause 108";
      }
      bit bBaseFEC25Greq {
        description
          "25Gb/s BASE-R FEC ";
```

```
        }
        bit bForceMS {
          description
            "Force MASTER-SLAVE as specified in Clause 98 (see
             98.2.1.2.5)";
        }
        /* PHY/PMD  */
        bit b10baseT {
          description
            "10BASE-T as defined in Clause 14";
        }
        bit b10baseT1L {
          description
            "10BASE-T1L as specified in Clause 146";
        }
        bit b10baseT1S {
          description
            "10BASE-T1S as specified in Clause 147";
        }
        bit b100baseTX {
          description
            "100BASE-TX as defined in Clause 25";
        }
        bit b1000baseX {
          description
            "1000BASE-X half duplex as specified in Clause 36";
        }
        bit b1000baseT {
          description
            "1000BASE-T PHY as specified in Clause 40";
        }
        bit b1000baseT1 {
          description
            "1000BASE-T1 as specified in Clause 97";
        }
        bit b2p5GbaseT {
          description
            "2.5GBASE-T PHY as specified in Clause 126";
        }
        bit b2p5GbaseT1 {
          description
            "2.5GBASE-T1 as specified in Clause 149";
        }
        bit b2p5GbaseKX {
          description
            "2.5GBASE-KX as specified in Clause 128";
        }
        bit b5GBaseT {
          description
            "5GBASE-T PHY as specified in Clause 126";
        }
        bit b5GbaseT1 {
          description
            "5GBASE-T1 as specified in Clause 149";
        }
        bit b5GbaseKR {
          description
            "5GBASE-KR as specified in Clause 130";
```

```
      }
      bit b10GbaseT {
        description
          "10GBASE-T PHY as specified in Clause 55";
      }
      bit b1000baseKX {
        description
          "1000BASE-KX as specified in Clause 70";
      }
      bit b10GbaseKX4 {
        description
          "10GBASE-KX4 as specified in Clause 71";
      }
      bit b10GbaseKR {
        description
          "10GBASE-KR as specified in Clause 72";
      }
      bit b25GbaseT {
        description
          "25GBASE-T as specified in Clause 113";
      }
      bit b25GbaseRS {
        description
          "25GBASE-CR-S as specified in Clause 110 or
           25GBASE-KR-S as specified in Clause 111";
      }
      bit b25GbaseR {
        description
          "25GBASE-CR as specified in Clause 110 or
           25GBASE-KR as specified in Clause 111";
      }
      bit b25GbaseT1 {
        description
          "25GBASE-T1 as specified in Clause 165";
      }
      bit b40GbaseKR4 {
        description
          "40GBASE-KR4 as specified in Clause 84";
      }
      bit b40GbaseCR4 {
        description
          "40GBASE-CR4 as specified in Clause 85";
      }
      bit b40GbaseT {
        description
          "40GBASE-T as specified in Clause 113";
      }
      bit b50GbaseR {
        description
          "50GBASE-CR as specified in Clause 136 or
           50GBASE-KR as specified in Clause 137";
      }
      bit b100GbaseCR10 {
        description
          "100GBASE-CR10 as specified in Clause 85";
      }
      bit b100GbaseCR4 {
        description
```

```
          "100GBASE-CR4 as specified in Clause 92";
      }
      bit b100GbaseKR4 {
        description
          "100GBASE-KR4 as specified in Clause 93";
      }
      bit b100GbaseKP4 {
        description
          "100GBASE-KP4 as specified in Clause 94";
      }
      bit b100GbaseR2 {
        description
          "100GBASE-CR2 as specified in Clause 136 or
           100GBASE-KR2 as specified in Clause 137";
      }
      bit b200GbaseR4 {
        description
          "200GBASE-CR4 as specified in Clause 136
           or 200GBASE-KR4 as specified in Clause 137";
      }
    }
    description
      "This data type is used as the syntax of the
       ieee802-eth-phy-auto-neg-capability-bits,
       ieee802-eth-phy-auto-neg-cap-advertised-bits,
       and ieee802-eth-phy-auto-neg-cap-received-bits
       objects in ieee802-eth-phy-auto-neg-table.";
    reference
      "IEEE Std 802.3, Clause 30.6.1.1.5";
  }
}
module ieee802-ethernet-phy {
  yang-version 1.1;
  namespace "urn:ieee:std:802.3:yang:ieee802-ethernet-phy";
  prefix ieee802-eth-phy;

  import ietf-yang-types {
    prefix yang;
    reference
      "IETF RFC 6991 YANG Types";
  }
  import iana-if-type {
    prefix ianaift;
    reference
      "IANA Interface Types";
  }
  import ietf-interfaces {
    prefix if;
    reference
      "IETF RFC 8343";
  }
  import ieee802-ethernet-phy-type {
    prefix ieee802-phy;
    reference
      "IEEE Std 802.3-2022";
  }

  organization
```

```
    "IEEE Std 802.3 Ethernet Working Group";
  contact
    "Web URL: http://www.ieee802.org/3/";
  description
    "This module contains YANG definitions for  802.3 PHYs and PMDs.";

  revision 2024-12-23 {
    description
      "Initial revision derived from IEEE8023-MAU-MIB
       using smidump.";
    reference
      "IEEE Std 802.3-2022, unless dated explicitly";
  }

  feature fec-support {
    description
      "This object indicates support for one or more FEC modes.";
  }

  feature baset-snr-support {
    description
      "This object indicates support for SNR measurement for
       BASE-T interfaces.";
  }

  feature time-sync-support {
    description
      "This object indicates support for time syncronisation.";
  }

  feature pcs-lane-statistics-support {
    description
      "This object indicates support for PCS Lane statistics.";
  }

  feature auto-negotiation-support {
    description
      "This object indicates support for Auto-Negotiation.";
  }

  typedef snr-value {
    type int32 {
      range "-127..127";
    }
    units "0.1dB";
    description
      "SNR units of 0.1 dB to an accuracy of 0.5 dB within
       the range of -12.7 dB to 12.7 dB.";
  }

  typedef counter-1000-second {
    type yang:counter64;
    description
      "This counter increments no more than 1000 times per second.";
  }

  typedef counter-5000-second {
    type yang:counter64;
```

```
    description
      "This counter increments no more than 5000 times per second.";
  }

  typedef counter-pcs-lane {
    type yang:counter64;
    description
      "This counter has a maximum per second increment rate of:
       1.2 million for 1000 Mb/s
       5 million for 10 Gb/s and 40 Gb/s
       2.5 million for 100 Gb/s";
  }

  typedef enabled-state {
    type enumeration {
      enum unknown {
        description
          "Initializing.";
      }
      enum disabled {
        description
          "Disabled";
      }
      enum enabled {
        description
          "Enabled";
      }
    }
    description
      "Enabled, disabled or unknown.";
  }

  augment "/if:interfaces/if:interface" {
    when "derived-from-or-self(if:type,'ianaift:ethernetCsmacd')" {
      description
        "Applies to all Ethernet interfaces.";
    }
    description
      "Augment interface model with Ethernet interface
       information";
    container phy-table {
      description
        "Table of descriptive and status information
         about the Ethernet interface physical layer.";
      container control {
        description
          "Control information for the Ethernet interface
           physical layer.";
        leaf phy-reset {
          type boolean;
          description
            "Setting this to 'True' resets the interface in the
             same manner as would a power-off, power-on cycle of
             at least 0.5 s duration.";
          reference
            "IEEE Std 802.3, 30.5.1.2.1 acResetMAU";
        }
        leaf phy-admin-control {
```

```
          type enumeration {
            enum operational {
              description
                "Operational";
            }
            enum standby {
              description
                "Standby";
            }
            enum shutdown {
              description
                "Shutdown";
            }
          }
          description
            "The configured state of the interface. This object
             MAY be implemented as a read-only object by devices
             that do not implement software control of the interface
             state. Some devices may not support setting the value
             tof this object to some of the enumerated values.

             This can be set to “standby”, “operational”, and
             “shutdown” state, except that a “standby” action
             to a multidrop interface will cause the interface
             to enter the “shutdown” management state.";
          reference
            "IEEE Std 802.3, 30.5.1.2.2 acMAUAdminControl";
        }
        container fec-control {
          if-feature "fec-support";
          presence "The port supports one or more FEC modes.";
          description
            "FEC Control.";
          leaf fec-mode-control {
            type enumeration {
              enum disabled {
                description
                  "FEC disabled";
              }
              enum enabled {
                description
                  "FEC enabled";
              }
              enum BASE-R-enabled {
                description
                  "BASE-R FEC enabled";
              }
              enum RS-FEC-enabled {
                description
                  "RS-FEC enabled";
              }
              enum RS-FEC-Int-enabled {
                description
                  "RS-FEC-Int enabled";
              }
            }
            description
              "The FEC mode across the MDI of the port.
```

```
               BASE-R-enabled, RS-FEC-enabled and
               RS-FEC-Int-enabled are only used for PHYs which
               support more than one type of FEC operation.
               When Clause 73 Auto-Negotiation is enabled, this
               leaf becomes read-only.";
            reference
              "IEEE Std 802.3, 30.5.1.1.16";
          }
          leaf rs-fec-bypass-control {
            type enabled-state;
            description
              "Is RS-FEC bypass enabled?";
            reference
              "IEEE Std 802.3, 30.5.1.1.31";
          }
        }
      }
      container status {
        config false;
        description
          "Status for the port.";
        leaf phy-type {
          type identityref {
            base ieee802-phy:phy-type;
          }
          description
            "This object identifies the PHY type.";
          reference
            "IEEE Std 802.3, 30.5.1.1.2, aPHYType.";
        }
        leaf pmd-type {
          type identityref {
            base ieee802-phy:pmd-type;
          }
          description
            "This object identifies the PMD type.";
          reference
            "IEEE Std 802.3, 30.5.1.1.2, aMAUType.";
        }
        leaf pmd-type-list {
          type ieee802-phy:pmd-type-list;
          description
            "PMD types this interface supports.";
        }
        leaf phy-status {
          type enumeration {
            enum unknown {
              description
                "Initializing, true state
                 not yet known";
            }
            enum operational {
              description
                "Powered and connected";
            }
            enum standby {
              description
                "Inactive but on";
```

```
            }
            enum shutdown {
              description
                "Inactive and powered down";
            }
          }
          description
            "The current state of the interface.";
          reference
            "IEEE Std 802.3, 30.5.1.1.7, aMAUAdminState";
        }
        leaf phy-media-available {
          type ieee802-phy:media-available-type;
          description
            "This object reports if the media is availeable.";
          reference
            "IEEE Std 802.3, 30.5.1.1.4, aMediaAvailable.";
        }
        /* Jabber */
        leaf jabber-state {
          type enumeration {
            enum other {
              description
                "undefined";
            }
            enum unknown {
              description
                "initializing, true state
                 not yet known";
            }
            enum normal {
              description
                "state is true or normal";
            }
            enum fault {
              description
                "state is false, fault, or
                 abnormal";
            }
          }
          description
            "The JABBER state of the multidrop interface.";
          reference
            "IEEE Std 802.3, 30.5.1.1.6, aJabber.jabberFlag.";
        }
        leaf jabbering-state-enters {
          type yang:counter64;
          description
            "The number of times  this interface detected JABBER.";
          reference
            "IEEE Std 802.3,
             30.5.1.1.6, aJabber.jabberCounter.";
        }
        /* FEC */
        container fec-status {
          if-feature "fec-support";
          presence "The port supports one or more FEC modes.";
          description
```

```
            "FEC Status.";
          leaf fec-ability {
            type bits {
              bit other {
                description
                  "Other";
              }
              bit base-r {
                description
                  "BASE-R FEC supported";
              }
              bit rs-fec {
                description
                  "RS-FEC supported";
              }
              bit rs-fec-bypass {
                description
                  "RS-FEC bypass supported";
              }
              bit rs-fec-bypass-indication {
                description
                  "RS-FEC error indication bypass supported";
              }
              bit pcs-fec-bypass {
                description
                  "PCS FEC error bypass supported";
              }
              bit pcs-fec-bypass-indication {
                description
                  "PCS FEC error indication bypass supported";
              }
            }
            description
              "The FEC modes this port supports.";
            reference
              "IEEE Std 802.3, 30.5.1.1.15, 30.5.1.1.16
               30.5.1.1.29, 30.5.1.1.30, 30.5.1.1.33,
               30.5.1.1.34";
          }
          leaf fec-mode-status {
            type enumeration {
              enum unknown {
                description
                  "Initializing, true state not yet known.";
              }
              enum disabled {
                description
                  "FEC disabled";
              }
              enum enabled {
                description
                  "FEC enabled";
              }
              enum BASE-R-enabled {
                description
                  "BASE-R FEC enabled";
              }
              enum RS-FEC-enabled {
```

```
                description
                  "RS-FEC enabled";
              }
              enum RS-FEC-Int-enabled {
                description
                  "RS-FEC-Int enabled";
              }
            }
            description
              "The FEC mode of the PHY.
               BASE-R-enabled, RS-FEC-enabled and
               RS-FEC-Int-enabled are only used by PHYs which
               support more than one type of FEC operation.";
            reference
              "IEEE Std 802.3, 30.5.1.1.16";
          }
          leaf rs-fec-bypass-enabled-status {
            type enabled-state;
            description
              "Is RS-FEC bypass indication enabled?";
            reference
              "IEEE Std 802.3, 30.5.1.1.32";
          }
        }
        /* MultiGBASE-T SNR */
        container baset-snr {
          if-feature "baset-snr-support";
          presence "The port supports BASE-T SNR measurements.";
          description
            "BASE-T SNR Status.";
          leaf snr-op-margin-chnl-a {
            type snr-value;
            description
              "The current SNR operating margin on channel A
               on this port for MultiGBASE-T PMAs.";
            reference
              "IEEE Std 802.3, 30.5.1.1.19";
          }
          leaf snr-op-margin-chnl-b {
            type snr-value;
            description
              "The current SNR operating margin on channel B
               on this port for MultiGBASE-T PMAs.";
            reference
              "IEEE Std 802.3, 30.5.1.1.20";
          }
          leaf snr-op-margin-chnl-c {
            type snr-value;
            description
              "The current SNR operating margin on channel C
               on this port for MultiGBASE-T PMAs.";
            reference
              "IEEE Std 802.3, 30.5.1.1.21";
          }
          leaf snr-op-margin-chnl-d {
            type snr-value;
            description
              "The current SNR operating margin on channel CD
```

```
               on this port for MultiGBASE-T PMAs.";
            reference
              "IEEE Std 802.3, 30.5.1.1.22";
          }
        }
        /* TimeSync */
        container time-sync {
          if-feature "time-sync-support";
          presence "The port supports Time Sync.";
          description
            "Time Sync Status.";
          leaf time-sync-ability {
            type bits {
              bit transmit {
                description
                  "Transmit";
              }
              bit receive {
                description
                  "Receive";
              }
            }
            description
              "The TimeSync capabilities of this port.";
            reference
              "IEEE Std 802.3, 30.13.1.1, 30.13.1.2";
          }
          leaf time-sync-delay-tx-max {
            type uint32;
            units "nano-seconds";
            description
              "The maximum transmit data delay between
               the xMII input to the MDI output.";
            reference
              "IEEE Std 802.3, 30.13.1.3";
          }
          leaf time-sync-delay-tx-min {
            type uint32;
            units "nano-seconds";
            description
              "The minimum transmit data delay between
               the xMII input to the MDI output.";
            reference
              "IEEE Std 802.3, 30.13.1.4";
          }
          leaf time-sync-delay-rx-max {
            type int32;
            units "nano-seconds";
            description
              "The maximum receive data delay between
               the MDI input to the xMII output.";
            reference
              "IEEE Std 802.3, 30.13.5";
          }
          leaf time-sync-delay-rx-min {
            type int32;
            units "nano-seconds";
            description
```

```
              "The minimum receive data delay between
               the MDI input to the xMII output.";
            reference
              "IEEE Std 802.3, 30.13.5";
          }
        }
      }
      container statistics {
        config false;
        description
          "Counters for the port.";
        leaf phy-media-losts {
          type yang:counter64;
          description
            "The number of times phy-media-available changed
             from 'available'.";
          reference
            "IEEE Std 802.3, 30.5.1.1.5 aLoseMediaCounter.";
        }
        leaf jabbering-state-enters {
          type yang:counter64;
          description
            "The number of times this interface detected JABBER.";
          reference
            "IEEE Std 802.3, 30.5.1.1.6, aJabber.jabberCounter.";
        }
        leaf false-carriers {
          type yang:counter64;
          description
            "The number of false carrier
             events during IDLE in 100BASE-X and
             1000BASE-X links.

             For all other interface types, this counter
             will always indicate zero.

             It can increment at a maximum rate of once
             per 100 ms for 100BASE-X and once per 10us
             for 1000BASE-X.";
          reference
            "IEEE Std 802.3, 30.5.1.1.10, aFalseCarriers.";
        }
        leaf pcs-coding-violations {
          type yang:counter64;
          description
            "The number of PCS coding violations for 100Mb/s and
             1000Mb/s ports.

             This counter increments at a maximum rate of 25 thousand
             per second for 100 Mb/s implementations and
             125 thousand per second for 1000 Mb/s
             implementations.

             The contents of this
             attribute is undefined when FEC is operating.";
          reference
            "IEEE Std 802.3, 30.5.1.1.14";
        }
```

```
        leaf fast-retrain-count-local {
          type counter-1000-second;
          description
            "The number of fast retrains initiated
             by the local device.";
          reference
            "IEEE Std 802.3, 30.5.1.1.24";
        }
        leaf fast-retrain-count-peer {
          type counter-1000-second;
          description
            "The number of fast retrains initiated
             by the link partner.";
          reference
            "IEEE Std 802.3, 30.5.1.1.25";
        }
      }
      container pcs-lane-statistics {
        if-feature "pcs-lane-statistics-support";
        config false;
        description
          "Per PCS lane counters for the port.
           These counters apply to the following:
           1000BASE-PX,
           10/25/40/50/100/200/400GBASE-R
           100GBASE-P
           10GBASE-PR
           10/1GBASE-PRX";
        list pcs-lane {
          key "pcs-lane-index";
          description
            "Per PCS lane counters for the port.";
          leaf pcs-lane-index {
            type uint32 {
              range "0..255";
            }
            description
              "PCS lane index.";
          }
          leaf fec-corrected-blocks {
            type counter-pcs-lane;
            description
              "The number blocks that FEC corrected.";
            reference
              "IEEE Std 802.3, 30.5.1.1.17";
          }
          leaf fec-uncorrected-blocks {
            type counter-pcs-lane;
            description
              "The number blocks that FEC could not correct.";
            reference
              "IEEE Std 802.3, 30.5.1.1.18";
          }
        }
      }
    }
    container auto-neg-table {
      if-feature "auto-negotiation-support";
```

```
      description
        "Configuration and status objects for Auto-Negotiation
         on Ethernet interfaces.";
      container control {
        description
          "Auto-Negotiation control.";
        leaf auto-neg-control {
          type boolean;
          default "true";
          description
            "Enable/disable Auto-Negotiation for this interface.";
          reference
            "IEEE Std 802.3, 30.6.1.1.2, acAutoNegAdminControl.";
        }
        leaf auto-neg-restart {
          type boolean;
          description
            "Setting this to true restarts Auto-Negotiation if enabled.";
          reference
            "IEEE Std 802.3,
             30.6.1.2.1, acAutoNegRestartAutoConfig.";
        }
      }
      container status {
        config false;
        description
          "Auto-Negotiation status.";
        leaf auto-neg-enabled {
          type boolean;
          description
            "Is Auto-Negotiation enabled for this interface.";
          reference
            "IEEE Std 802.3,
             30.6.1.1.2 aAutoNegAdminState.";
        }
        leaf auto-neg-remote-signaling {
          type boolean;
          description
            "Is the link partner using Auto-Negotiation?";
          reference
            "IEEE Std 802.3, 30.6.1.1.3, aAutoNegRemoteSignaling.";
        }
        leaf auto-neg-status {
          type enumeration {
            enum other {
              description
                "Unknown state.";
            }
            enum configuring {
              description
                "Auto-Negotiation is active.";
            }
            enum complete {
              description
                "Auto-Negotiation is complete.";
            }
            enum disabled {
              description
```

```
                "Auto-Negotiation s disabled.";
            }
            enum parallelDetectFail {
              description
                "Auto-Negotiation failed because the
                 link partner does not suport, or has
                 disabled, Auto-Negotiation.";
            }
          }
          description
            "The current status of Auto-Negotiation.";
          reference
            "IEEE Std 802.3, 30.6.1.1.4, aAutoNegAutoConfig.";
        }
        leaf auto-neg-capability-bits {
          type ieee802-phy:auto-neg-ability-bits;
          description
            "The capabilities of the local Auto-Negotiation entity.";
          reference
            "IEEE Std 802.3, 30.6.1.1.5,
             aAutoNegLocalTechnologyAbility.";
        }
        leaf auto-neg-cap-advertised-bits {
          type ieee802-phy:auto-neg-ability-bits;
          description
            "The capabilities advertised by the local
             Auto-Negotiation entity.";
          reference
            "IEEE Std 802.3, 30.6.1.1.6,
             aAutoNegadvertisedTechnologyAbility.";
        }
        leaf auto-neg-cap-received-bits {
          type ieee802-phy:auto-neg-ability-bits;
          description
            "Auto-Negotiation capabilities received from
             the link partner.";
          reference
            "IEEE Std 802.3, 30.6.1.1.7,
             aAutoNegreceivedTechnologyAbility.";
        }
      }
    }
  }
}
module ieee802-ethernet-pon {
  yang-version 1.1;
  namespace "urn:ieee:std:802.3:yang:ieee802-ethernet-pon";
  prefix ieee802-eth-pon;

  import ieee802-types {
    prefix ieee;
    reference
      "IEEE 802 types";
  }
  import ietf-yang-types {
    prefix yang;
    reference
      "IETF RFC 6991";
```

```
}
import ietf-interfaces {
  prefix if;
  reference
    "IETF RFC 8343";
}
import ieee802-ethernet-interface {
  prefix ieee802-eth-if;
}

organization
  "IEEE 802.3 Ethernet Working Group
   Web URL: http://www.ieee802.org/3/";
contact
  "Web URL: http://www.ieee802.org/3/";
description
  "This module contains a collection of YANG definitions for
   managing the Multi Point Control Protocol for
   Ethernet PON (EPON),as defined in IEEE Std 802.3, Clause 64
   and Clause 77.

   This YANG module augments the 'ethernet' module.";

revision 2024-~~07-17~~12-23 {
  description
    "Updates under IEEE Std 802.3.2-202x, Draft D3.~~0~~1";
  reference
    "IEEE Std 802.3-2022, unless dated explicitly";
}

feature trx-power-level-reporting-supported {
  description
    "This object indicates the support for optical transceiver
     power level monitoring and reporting capability.
     When 'true', the given interface supports the optical power
     level monitoring and reporting function. Otherwise,
     the value is 'false'.";
}

feature fec-supported {
  description
    "This object indicates the support of operation of the
     optional FEC sublayer of the 1G-EPON PHY specified in
     IEEE Std 802.3, 65.2. The value of 'unknown' is reported in
     the initialization, for non FEC support state or type not
     yet known. The value of 'not supported' is reported when the
     sublayer is not supported. The value of 'supported' is
     reported when the sublayer is supported. This object is
     applicable for an OLT, with the same value for all logical
     links, and for an ONU.";
  reference
    "IEEE Std 802.3, 30.5.1.1.15";
}

identity state-change-action-type {
  description
    "Type of interface state change requested.";
}
```

```
identity power-down {
  base state-change-action-type;
  description
    "Power down the EPON logical interface.
     Power-down actions are applicable for the OLT and ONU.  A
     power down of a specific logical interface affects only
     the logical interface (and not the physical interface).
     the logical interface will be unavailable while the
     power-down occurs and data may be lost. Other logical
     interface are unaffected by power-down.

     This action is relevant when the admin state is active.";
}

identity power-up {
  base state-change-action-type;
  description
    "Exit EPON logical interface power-down state.";
}

identity reset-action-type {
  description
    "Type of reset action requested.";
}

identity reset-interface {
  base reset-action-type;
  description
    "Reset the EPON logical interface. Resetting an interface
     can lead an interruption of service for the users connected
     to the respective EPON interface.

     This object is applicable for an OLT and an ONU. At the
     OLT, it has a distinct value for each logical interface.
     A reset for a specific logical interface resets only
     this logical interface and not the physical interface.

     Thus, a logical link that is malfunctioning can be
     reset without affecting the operation of other logical
     interfaces.

     The reset can cause Discontinuities in the values of the
     counters of the interface, similar to re-initialization
     of the management system.";
}

identity register-type {
  description
    "Type of registration requested.";
}

identity register {
  base register-type;
  description
    "Register indicates a request to register an LLID.
     This action applies to an OLT or ONU logical interface.";
}
```

```
identity reregister {
  base register-type;
  description
    "Re-register indicates an request to re-register an LLID.
     This action applies to an OLT or ONU logical interface.";
}

identity deregister {
  base register-type;
  description
    "De-register indicates an request to de-register an LLID.
     This action applies to an OLT or ONU logical interface.
     Deregister may result in an interruption of service to
     users connected to the respective EPON interface.";
}

typedef mpcp-supported {
  type boolean;
  description
    "This object indicates that the given interface supports
     MPCP, i.e., it is an Ethernet PON (EPON) interface.";
}

typedef mpcp-llid {
  type uint64 {
    range "0 .. 32767";
  }
  description
    "Logical Link Identifiers (LLIDs) are used to identify a
     single MAC from a number of MACs which may be present in the
     EPON OLT or ONU. LLIDs between the value of 0x07FFE
     and 0x7FFF are assigned for ONU discovery and registration.
     Other LLIDs are dynamically assigned by the OLT during the
     registration process. For a complete description of how the
     LLID is used in an EPON device, see IEEE Std 802.3,
     Clause 65 for 1G-EPON and Clause 76 for 10G-EPON.";
  reference
    "IEEE Std 802.3, 65.1.3.3 for 1G-EPON and
     76.2.6.1.3 for 10G-EPON";
}

typedef mpcp-maximum-queue-count-per-report {
  type uint8 {
    range "0..7";
  }
  default "0";
  description
    "Defines the maximum number of queues (0-7) in the REPORT
     MPCPDU as defined in IEEE Std 802.3, Clause 64 and
     Clause 77.";
}

typedef mpcp-llid-count {
  type uint64 {
    range "0 .. 32767";
  }
  description
```

```
    "Indicates the number of registered LLIDs. The initialization
     value is 0. This is applicable for an OLT with the same
     value for all logical interfaces and for an ONU.";
  reference
    "IEEE Std 802.3, 65.1.3.3 for 1G-EPON and
        76.2.6.1.3 for 10G-EPON";
}

typedef mpcp-admin-state {
  type enumeration {
    enum enabled {
      description
        "When selecting the value of 'enabled', the MultiPoint
         Control Protocol sublayer on the OLT / ONU is enabled.";
    }
    enum disabled {
      description
        "When selecting the value of 'disabled', the MultiPoint
         Control Protocol sublayer on the OLT / ONU is
         disabled.";
    }
  }
  description
    "Enumeration of valid administrative states for a
     MultiPoint MAC Control sublayer on the OLT or ONU.";
  reference
    "IEEE Std 802.3, 30.3.5.2.1";
}

typedef mpcp-mode {
  type enumeration {
    enum olt {
      description
        "MPCP mode: olt";
    }
    enum onu {
      description
        "MPCP mode: onu";
    }
  }
  description
    "Enumeration of valid MPCP modes for EPON interfaces.";
  reference
    "IEEE Std 802.3, 30.3.5.1.3";
}

typedef mpcp-logical-link-state {
  type enumeration {
    enum unregistered {
      description
        "MPCP registration state: logical link is
         NOT registered.";
    }
    enum registering {
      description
        "MPCP registration state: logical link is currently in
         the process of registering.";
    }
```

```
    enum registered {
      description
        "MPCP registration state: logical link is currently
         registered.";
    }
  }
  description
    "Enumeration of valid MPCP registration states for EPON
     interfaces.";
  reference
    "IEEE Std 802.3, 30.3.5.1.6";
}

typedef mpcp-logical-link-admin-state {
  type enumeration {
    enum reset {
      description
        "When read, the value of 'reset' indicates that the given
         logical link on the OLT / ONU has been reset.";
    }
    enum operate {
      description
        "When read, the value of 'operate' indicates that the
         given logical link on the OLT / ONU has moved into
         operating mode.";
    }
    enum unknown {
      description
        "When read, the value of 'unknown' indicates that the
         status of the given logical link on the OLT / ONU is
         currently not known.";
    }
    enum registered {
      description
        "When read, the value of 'registered' indicates that the
         given logical link on the OLT / ONU has been
         registered.";
    }
    enum deregistered {
      description
        "When read, the value of 'deregistered' indicates that the
         given logical link on the OLT / ONU has been
         deregistered.";
    }
    enum reregistered {
      description
        "When read, the value of 'reregistered' indicates that the
         given logical link on the OLT / ONU has been
         reregistered.";
    }
  }
  description
    "Enumeration of valid administrative states for a logical
     link on the OLT or ONU.";
}

typedef ompe-mode {
  type enumeration {
```

```
      enum unknown {
        description
          "omp-emulation mode: unknown = system is initializing";
      }
      enum olt {
        description
          "omp-emulation mode: olt";
      }
      enum onu {
        description
          "omp-emulation mode: onu";
      }
    }
    description
      "Enumeration of valid OMP-Emulation modes for EPON
       interfaces.";
    reference
      "IEEE Std 802.3, 30.3.7.1.2";
  }

  typedef fec-capability {
    type enumeration {
      enum unknown {
        description
          "FEC capability: unknown = system is initializing.";
      }
      enum supported {
        description
          "FEC capability: supported.";
      }
      enum NotSupported {
        description
          "FEC capability: not supported.";
      }
    }
    description
      "Enumeration of valid FEC capability values for EPON
       interfaces with enabled MPCP.";
    reference
      "IEEE Std 802.3, 30.5.1.1.15";
  }

  typedef fec-mode {
    type enumeration {
      enum unknown {
        description
          "FEC mode: unknown = system is initializing.";
      }
      enum disabled {
        description
          "FEC mode: disabled = FEC is disabled for the given
           logical link (both Tx and Rx directions).";
      }
      enum enabled-Tx-Rx {
        description
          "FEC mode: enabled-Tx-Rx = FEC is enabled for the given
           logical link in both Tx and Rx directions.";
      }
```

```
    enum enabled-Tx-only {
      description
        "FEC mode: enabled-Tx-only = FEC is enabled for
         the given logical link but only in Tx direction.";
    }
    enum enabled-Rx-only {
      description
        "FEC mode: enabled-Rx-only = FEC is enabled for
         the given logical link but only in Rx direction.";
    }
  }
  description
    "Enumeration of valid FEC modes for EPON interfaces.";
  reference
    "IEEE Std 802.3, 30.5.1.1.16";
}

typedef power-level {
  type int32;
  units "0.1 dBm";
  description
    "Power level reflects the value of power, as measured at the
     optical transceiver, expressed in units of 0.1 dBm.";
}

typedef trx-admin-state {
  type enumeration {
    enum enabled {
      description
        "When read as 'enabled', the transmitter is enabled and
         operating under the control of the logical control
         protocol. When set to 'enabled', the transmitter is
         enabled to operate under the control of the logical
         control protocol.";
    }
    enum disabled {
      description
        "When read as 'disabled', the transmitter is currently
         disabled (not transmitting). When set to 'disabled',
         the transmitter is expected to be disabled
         (to stop transmitting).";
    }
  }
  description
    "Enumeration of valid administrative states for an optical
     transceiver.";
  reference
    "IEEE Std 802.3.1, dot3ExtPkgOptIfTransmitEnable";
}

augment "/if:interfaces/if:interface/ieee802-eth-if:ethernet" {
  description
    "Augments the definition of Ethernet interface (
        /if:interfaces/if:interface/ieee802-eth-if:ethernet)
        module with nodes specific to Ethernet PON (EPON).";
  leaf fec-mode {
    if-feature "fec-supported";
    type fec-mode;
```

```
  description
    "This object reflects the current administrative state of
     the FEC function for the given logical link on an ONU or
     OLT.

     When reading the value of 'disabled', the FEC function on
     the given logical link is disabled.

     When reading the value of 'enabled-Tx-Rx', the FEC function
     on the given logical link is enabled in both Tx and Rx
     directions.

     When reading the value of 'enabled-Tx-only', the FEC
     function on the given logical link is enabled in Tx
     direction only.

     When reading the value of 'enabled-Rx-only', the FEC
     function on the given logical link is enabled in Rx
     direction only.

     When reading the value of 'unknown', the state of the FEC
     function on the given logical link is unknown or the FEC
     function is currently initializing.

     This object is applicable for an OLT and an ONU. This
     object has the same value for each logical link.";
  reference
    "IEEE Std 802.3, 30.5.1.1.16";
}
leaf mpcp-admin-state {
  type mpcp-admin-state;
  description
    "This object reflects the current administrative state of
     the MultiPoint MAC Control sublayer, as defined in
     IEEE Std 802.3, Clause 64 and Clause 77, for the
     OLT / ONU.

     When reading the value of 'enabled', the MultiPoint
     Control Protocol on the OLT / ONU is enabled.

     When reading the value of 'disabled', the MultiPoint
     Control Protocol on the OLT / ONU is disabled.

     This object is applicable for an OLT and an ONU. It has
     the same value for all logical links.";
  reference
    "IEEE Std 802.3, 30.3.5.1.2";
}
leaf mpcp-logical-link-admin-state {
  type mpcp-logical-link-admin-state;
  config false;
  description
    "This object reflects the current administrative state of a
     logical link on an ONU or OLT.

     When reading the value of 'reset', the given logical link
     is undergoing a reset.
```

```
        When reading the value of 'unknown', the current status of
        the given logical link is unknown and the link might be
        undergoing initialization.

        When reading the value of 'operate', the given logical
        link is operating  normally.

        When reading the value of 'registered', the given logical
        link was requested to perform registration.

        When reading the value of 'deregistered', the given
        logical link was requested to perform deregistration.

        When reading the value of 'reregistered', the given
        logical link was requested to perform reregistration.

        This object is applicable for an OLT and an ONU. It has a
        distinct value for each logical link.";
    reference
      "IEEE Std 802.3.1, dot3ExtPkgObjectRegisterAction";
  }
  leaf trx-transmit-admin-state {
    when "../../ieee802-eth-if:ethernet/
                 ieee802-eth-pon:mpcp-admin-state = 'enabled'";
    if-feature "trx-power-level-reporting-supported";
    type trx-admin-state;
    description
      "This object reflects the current status of the transmitter
       in the optical transceiver.

       When read as 'enabled', the optical transmitter is enabled
       and operating under the control of the logical control
       protocol.

       When read as 'disabled', the optical transmitter is
       disabled.

       This object is applicable for an OLT and an ONU.
       At the OLT, this object has a distinct value for each
       logical link.

       The value of this object is only reliable when
       /if:interfaces-state/if:interface/ieee802-eth-if:ethernet/
       'mpcp-admin-state' is equal to 'enabled'.";
    reference
      "IEEE Std 802.3.1, dot3ExtPkgOptIfTransmitEnable";
  }
  container capabilities {
    config false;
    description
      "This container includes all EPON interface-specific
       capabilities.";
    leaf mpcp-supported {
      type mpcp-supported;
      default "true";
      description
        "This object indicates that the given interface supports
         MPCP, i.e., it is an Ethernet PON (EPON) interface.";
```

```
    }
  }
  container statistics-mpcp {
    config false;
    description
      "This container defines a set of MPCP-related statistics
       counters of an EPON interface, as defined in
       IEEE Std 802.3, Clause 64 and Clause 77.

       Discontinuities in the values of counters in this
       container can occur at re-initialization of the management
       system, and at other times as indicated by the value of
       the 'discontinuity-time' leaf defined in the ietf-interfaces
       YANG module (IETF RFC 8343).";
    leaf out-mpcp-mac-ctrl-frames {
      type yang:counter64;
      units "frames";
      config false;
      description
        "A count of MPCP frames passed to the MAC sublayer for
         transmission.

         This counter is incremented when a MA_CONTROL.request
         service primitive is generated within the MAC control
         sublayer with an opcode indicating an MPCP frame.

         This object is applicable for an OLT and an ONU. It has
         a distinct value for each logical link.";
      reference
        "IEEE Std 802.3, 30.3.5.1.7";
    }
    leaf in-mpcp-mac-ctrl-frames {
      type yang:counter64;
      units "frames";
      config false;
      description
        "A count of MPCP frames passed by the MAC sublayer to the
         MAC Control sublayer.

         This counter is incremented when a frame is received at
         the interface which is an MPCP frame or has a
         Length/Type Ethernet header field value equal to the
         Type assigned for 802.3_MAC_Control as specified in
         IEEE Std 802.3, 31.4.1.3.

         This object is applicable for an OLT and an ONU. It has
         a distinct value for each logical link.";
      reference
        "IEEE Std 802.3, 30.3.5.1.8";
    }
    leaf mpcp-discovery-window-count {
      when "../../ompe-mode = 'olt'";
      type yang:counter64;
      units "discovery windows";
      config false;
      description
        "A count of discovery windows generated by the OLT.
```

```
           The counter is incremented by one for each generated
           discovery window.

           This object is applicable for an OLT and has the same
           value for each logical link.";
        reference
          "IEEE Std 802.3, 30.3.5.1.22";
      }
      leaf mpcp-discovery-timeout-count {
        when "../../ompe-mode = 'olt'";
        type yang:counter64;
        units "discovery timeouts";
        config false;
        description
          "A count of the number of times a discovery timeout
           occurs.

           This counter is incremented by one for each discovery
           processing state-machine reset resulting from timeout
           waiting for message arrival.

           This object is applicable for an OLT and has the same
           value for each logical link.";
        reference
          "IEEE Std 802.3, 30.3.5.1.23";
      }
      leaf out-mpcp-register-req {
        when "../../ompe-mode = 'onu'";
        type yang:counter64;
        units "frames";
        config false;
        description
          "A count of the number of times a REGISTER_REQ MPCP frame
           transmission occurs.

           This counter is incremented by one for each
           REGISTER_REQ MPCP frame transmitted as defined in
           IEEE Std 802.3, Clause 64 and Clause 77.

           This object is applicable for an ONU and has the same
           value for each logical link.";
        reference
          "IEEE Std 802.3, 30.3.5.1.12";
      }
      leaf in-mpcp-register-req {
        when "../../ompe-mode = 'olt'";
        type yang:counter64;
        units "frames";
        config false;
        description
          "A count of the number of times a REGISTER_REQ MPCP frame
           reception occurs.

           This counter is incremented by one for each
           REGISTER_REQ MPCP frame received as defined in
           IEEE Std 802.3, Clause 64 and Clause 77.

           This object is applicable for an OLT and has the same
```

```
      value for each logical link.";
  reference
    "IEEE Std 802.3, 30.3.5.1.17";
}
leaf out-mpcp-register-ack {
  when "../../ompe-mode = 'onu'";
  type yang:counter64;
  units "frames";
  config false;
  description
    "A count of the number of times a REGISTER_ACK MPCP frame
     transmission occurs.

     This counter is incremented by one for each
     REGISTER_ACK MPCP frame transmitted as defined in
     IEEE Std 802.3, Clause 64 and Clause 77.

     This object is applicable for an ONU and has a distinct
     value for each logical link.";
  reference
    "IEEE Std 802.3, 30.3.5.1.10";
}
leaf in-mpcp-register-ack {
  when "../../ompe-mode = 'olt'";
  type yang:counter64;
  units "frames";
  config false;
  description
    "A count of the number of times a REGISTER_ACK MPCP frame
     reception occurs.

     This counter is incremented by one for each
     REGISTER_ACK MPCP frame received as defined in
     IEEE Std 802.3, Clause 64 and Clause 77.

     This object is applicable for an OLT and has a distinct
     value for each logical link.";
  reference
    "IEEE Std 802.3, 30.3.5.1.15";
}
leaf out-mpcp-report {
  when "../../ompe-mode = 'onu'";
  type yang:counter64;
  units "frames";
  config false;
  description
    "A count of the number of times a REPORT MPCP frame
     transmission occurs.

     This counter is incremented by one for each REPORT MPCP
     frame transmitted as defined in IEEE Std 802.3,
     Clause 64 and Clause 77.

     This object is applicable for an ONU and has a distinct
     value for each logical link.";
  reference
    "IEEE Std 802.3, 30.3.5.1.13";
}
```

```
leaf in-mpcp-report {
  when "../../ompe-mode = 'olt'";
  type yang:counter64;
  units "frames";
  config false;
  description
    "A count of the number of times a REPORT MPCP frame
     reception occurs.

     This counter is incremented by one for each REPORT MPCP
     frame received as defined in IEEE Std 802.3,
     Clause 64 and Clause 77.

     This object is applicable for an OLT and has a distinct
     value for each logical link.";
  reference
    "IEEE Std 802.3, 30.3.5.1.18";
}
leaf out-mpcp-gate {
  when "../../ompe-mode = 'olt'";
  type yang:counter64;
  units "frames";
  config false;
  description
    "A count of the number of times a GATE MPCP frame
     transmission occurs.

     This counter is incremented by one for each GATE MPCP
     frame transmitted as defined in IEEE Std 802.3,
     Clause 64 and Clause 77.

     This object is applicable for an OLT and has a distinct
     value for each logical link.";
  reference
    "IEEE Std 802.3, 30.3.5.1.9";
}
leaf in-mpcp-gate {
  when "../../ompe-mode = 'onu'";
  type yang:counter64;
  units "frames";
  config false;
  description
    "A count of the number of times a GATE MPCP frame
     reception occurs.

     This counter is incremented by one for each GATE MPCP
     frame received as defined in IEEE Std 802.3,
     Clause 64 and Clause 77.

     This object is applicable for an ONU and has a distinct
     value for each logical link.";
  reference
    "IEEE Std 802.3, 30.3.5.1.14";
}
leaf out-mpcp-register {
  when "../../ompe-mode = 'olt'";
  type yang:counter64;
  units "frames";
```

```
      config false;
      description
        "A count of the number of times a REGISTER MPCP frame
         transmission occurs.

         This counter is incremented by one for each
         REGISTER MPCP frame transmitted as defined in
         IEEE Std 802.3, Clause 64 and Clause 77.

         This object is applicable for an OLT and has a distinct
         value for each logical link.";
      reference
        "IEEE Std 802.3, 30.3.5.1.11";
    }
    leaf in-mpcp-register {
      when "../../ompe-mode = 'onu'";
      type yang:counter64;
      units "frames";
      config false;
      description
        "A count of the number of times a REGISTER MPCP frame
         reception occurs.

         This counter is incremented by one for each
         REGISTER MPCP frame received as defined in
         IEEE Std 802.3, Clause 64 and Clause 77.

         This object is applicable for an ONU and has a distinct
         value for each logical link.";
      reference
        "IEEE Std 802.3, 30.3.5.1.16";
    }
  }
  container statistics-ompe {
    description
      "This container defines a set of OMP-Emulation-related
       statistics counters of an EPON interface, as defined in
       IEEE Std 802.3, Clause 65 and Clause 76.

       Discontinuities in the values of counters in this
       container can occur at re-initialization of the management
       system, and at other times as indicated by the value of
       the 'discontinuity-time' leaf defined in the
       ietf-interfaces YANG module (IETF RFC 8343).";
    reference
      "IEEE Std 802.3.1, dot3OmpEmulationStatEntry";
    leaf in-ompe-frames-errored-sld {
      type yang:counter64;
      units "frames";
      config false;
      description
        "A count of frames received that do not contain a valid
         SLD field as defined in IEEE Std 802.3, 65.1.3.3.1 or
         76.2.6.1.3.1, as appropriate.

         This object is applicable for an OLT and an ONU.
         It has a distinct value for each logical link.";
      reference
```

```
    "IEEE Std 802.3, 30.3.7.1.3";
}
leaf in-ompe-frames-errored-crc8 {
  type yang:counter64;
  units "frames";
  config false;
  description
    "A count of frames received that contain a valid SLD
     field, as defined in IEEE Std 802.3, 65.1.3.3.1 or
     76.2.6.1.3.1 as appropriate, but do not pass the CRC-8
     check as defined in IEEE Std 802.3, 65.1.3.3.3 or
     76.2.6.1.3.3 as appropriate.

     This object is applicable for an OLT and an ONU.
     It has a distinct value for each logical link.";
  reference
    "IEEE Std 802.3, 30.3.7.1.4";
}
leaf ompe-onu-frames-with-good-llid-good-crc8 {
  when "../../ompe-mode = 'onu'";
  type yang:counter64;
  units "frames";
  config false;
  description
    "A count of frames received that 1) contain a valid SLD
     field in an ONU, 2) meet the rule for frame acceptance,
     and 3) pass the CRC-8 check.

     The SLD is defined in IEEE Std 802.3, 65.1.3.3.1 or
     76.2.6.1.3.1, as appropriate.

     The rules for LLID acceptance are defined in
     IEEE Std 802.3, 65.1.3.3.2 or 76.2.6.1.3.2,
     as appropriate.

     The CRC-8 check is defined in IEEE Std 802.3,
     65.1.3.3.3 or 76.2.6.1.3.3, as appropriate.

     This object is applicable for an ONU and has a distinct
             value for each logical link.";
  reference
    "IEEE Std 802.3, 30.3.7.1.6";
}
leaf ompe-olt-frames-with-good-llid-good-crc8 {
  when "../../ompe-mode = 'olt'";
  type yang:counter64;
  units "frames";
  config false;
  description
    "A count of frames received that 1) contain a valid SLD
     field in an OLT, and 2) pass the CRC-8 check.

     The SLD is defined in IEEE Std 802.3, 65.1.3.3.1 or
     76.2.6.1.3.1, as appropriate.

     The frame acceptance are defined in IEEE Std 802.3,
     65.1.3.3.2 or 76.2.6.1.3.2, as appropriate.
```

```
      The CRC-8 check is defined in IEEE Std 802.3,
      65.1.3.3.3 or 76.2.6.1.3.3, as appropriate.

      This object is applicable for an OLT and has a distinct
      value for each logical link.";
  reference
    "IEEE Std 802.3, 30.3.7.1.6";
}
leaf in-ompe-frames-with-bad-llid {
  when "../../ompe-mode = 'olt'";
  type yang:counter64;
  units "frames";
  config false;
  description
    "A count of frames received that contain a valid SLD
     field, as defined in IEEE Std 802.3, 65.1.3.3.1 or
     76.2.6.1.3.1, as appropriate, and pass the CRC-8 check
     as defined in IEEE Std 802.3, 65.1.3.3.3 or
     76.2.6.1.3.3, as appropriate, but are discarded due to
     the LLID check.

     This object is applicable for an OLT and has a distinct
             value for each logical link.";
  reference
    "IEEE Std 802.3, 30.3.7.1.8";
}
leaf in-ompe-frames-with-good-llid {
  type yang:counter64;
  units "frames";
  config false;
  description
    "A count of frames received that contain a valid SLD
     field, as defined in IEEE Std 802.3, 65.1.3.3.1 or
     76.2.6.1.3.1 as appropriate, but do not pass the CRC-8
     check as defined in IEEE Std 802.3, 65.1.3.3.3 or
     76.2.6.1.3.3 as appropriate.

     This object is applicable for an OLT and an ONU. It has
     a distinct value for each logical link.";
  reference
    "IEEE Std 802.3, 30.3.7.1.4";
}
leaf in-ompe-frames {
  type yang:counter64;
  units "frames";
  config false;
  description
    "A count of frames received that contain a valid SLD
     field, as defined in IEEE Std 802.3, 65.1.3.3.1 or
     76.2.6.1.3.1, as appropriate, and pass the CRC-8
     check as defined in IEEE Std 802.3, 65.1.3.3.3 or
     76.2.6.1.3.3, as appropriate.

     This object is applicable for an OLT and an ONU. It has
     a distinct value for each logical link.";
  reference
    "IEEE Std 802.3, 30.3.7.1.6 (ONU) and 30.3.7.1.7 (OLT)";
}
```

```
leaf in-ompe-frames-not-match-onu-llid-broadcast {
  when "../../ompe-mode = 'onu'";
  type yang:counter64;
  units "frames";
  config false;
  description
    "A count of frames received that contain a valid SLD
     field, as defined in IEEE Std 802.3, 65.1.3.3.1 or
     76.2.6.1.3.1, as appropriate, pass the CRC-8 check, as
     defined in IEEE Std 802.3, 65.1.3.3.3 or 76.2.6.1.3.3,
     as appropriate, and contain the broadcast bit in the
     LLID and not the ONU's LLID (frame accepted) as defined
     in IEEE Std 802.3,Clause 65 and Clause 76,
     as appropriate.

     This object is applicable for an ONU only.";
  reference
    "IEEE Std 802.3.1,
              dot3OmpEmulationBroadcastBitNotOnuLlid";
}
leaf in-ompe-frames-match-onu-llid-not-broadcast {
  when "../../ompe-mode = 'onu'";
  type yang:counter64;
  units "frames";
  config false;
  description
    "A count of frames received that contain a valid SLD
     field, as defined in IEEE Std 802.3, 65.1.3.3.1 or
     76.2.6.1.3.1, as appropriate, pass the CRC-8 check, as
     defined in IEEE Std 802.3, 65.1.3.3.3 or 76.2.6.1.3.3,
     as appropriate, and contain the ONU's LLID
     (frame accepted) as defined in IEEE Std 802.3, Clause 65
     and Clause 76, as appropriate.

     This object is applicable for an ONU only.";
  reference
    "IEEE Std 802.3.1, dot3OmpEmulationOnuLLIDNotBroadcast";
}
leaf in-ompe-frames-match-onu-llid-broadcast {
  when "../../ompe-mode = 'onu'";
  type yang:counter64;
  units "frames";
  config false;
  description
    "A count of frames received that contain a valid SLD
     field, as defined in IEEE Std 802.3, 65.1.3.3.1 or
     76.2.6.1.3.1, as appropriate, pass the CRC-8 check, as
     defined in IEEE Std 802.3, 65.1.3.3.3 or 76.2.6.1.3.3,
     as appropriate, and contain the broadcast bit in the
     LLID and the ONU's LLID (frame accepted) as defined in
     IEEE Std 802.3, Clause 65 and Clause 76, as appropriate.

     This object is applicable for an ONU only.";
  reference
    "IEEE Std 802.3.1,
              dot3OmpEmulationBroadcastBitPlusOnuLlid";
}
leaf in-ompe-frames-not-match-onu-llid-not-broadcast {
```

```
        when "../../ompe-mode = 'onu'";
        type yang:counter64;
        units "frames";
        config false;
        description
          "A count of frames received that contain a valid SLD
           field, as defined in IEEE Std 802.3, 65.1.3.3.1 or
           76.2.6.1.3.1, as appropriate, pass the CRC-8 check,
           as defined in IEEE Std 802.3, 65.1.3.3.3 or
           76.2.6.1.3.3, as appropriate, do not contain the
           broadcast bit in the LLID and do not contain the ONU's
           LLID (frame is NOT accepted) as defined in
           IEEE Std 802.3, Clause 65 and Clause 76, as appropriate.

           This object is applicable for an ONU only.";
        reference
          "IEEE Std 802.3.1,
                   dot3OmpEmulationNotBroadcastBitNotOnuLlid";
      }
    }
    container thresholds-trx {
      if-feature "trx-power-level-reporting-supported";
      description
        "This container defines a set of optical transceiver
         thresholds of an EPON interface as defined in
         IEEE Std 802.3, Clause 60 and Clause 75.";
      reference
        "IEEE Std 802.3.1, dot3ExtPkgOptIfEntry";
      leaf in-trx-power-low-threshold {
        if-feature "trx-power-level-reporting-supported";
        type power-level;
        description
          "This object reflects the current setting of low alarm
           threshold for the input power into the optical receiver.
           If the value reported in 'in-trx-power' object drops
           below the value set in 'in-trx-power-low-threshold', a
           'in-trx-power-low-threshold-crossing' event is
           generated.

           This object is applicable for an OLT and an ONU. It has
           a distinct value for each logical link.";
        reference
          "IEEE Std 802.3.1,
           dot3ExtPkgOptIfLowerInputPowerThreshold";
      }
      leaf in-trx-power-high-threshold {
        if-feature "trx-power-level-reporting-supported";
        type power-level;
        description
          "This object reflects the current setting of high alarm
           threshold for the input power into the optical receiver.
           If the value reported in 'in-trx-power' object exceeds
           the value set in 'in-trx-power-high-threshold', a
           'in-trx-power-high-threshold-crossing' event is
           generated.

           This object is applicable for an OLT and an ONU. It has
           a distinct value for each logical link.";
```

```
      reference
        "IEEE Std 802.3.1,
         dot3ExtPkgOptIfUpperInputPowerThreshold";
    }
    leaf out-trx-power-low-threshold {
      if-feature "trx-power-level-reporting-supported";
      type power-level;
      description
        "This object reflects the current setting of low alarm
         threshold for the output power out of the optical
         transmitter. If the value reported in 'out-trx-power'
         object drops below the value set in
         'out-trx-power-low-threshold', a
         'out-trx-power-low-threshold-crossing' event is
         generated.

         This object is applicable for an OLT and an ONU. It has
         a distinct value for each logical link.";
      reference
        "IEEE Std 802.3.1,
         dot3ExtPkgOptIfLowerOutputPowerThreshold";
    }
    leaf out-trx-power-high-threshold {
      if-feature "trx-power-level-reporting-supported";
      type power-level;
      description
        "This object reflects the current setting of high alarm
         threshold for the output power out of the optical
         transmitter. If the value reported in 'out-trx-power'
         object exceeds the value set in
         'out-trx-power-high-threshold', a
         'out-trx-power-high-threshold-crossing' event is
         generated.

         This object is applicable for an OLT and an ONU.
                 It has a distinct value for each logical link.";
      reference
        "IEEE Std 802.3.1,
                 dot3ExtPkgOptIfUpperOutputPowerThreshold";
    }
  }
  container statistics-trx {
    if-feature "trx-power-level-reporting-supported";
    status deprecated;
    description
      "This container defines a set of optical transceiver
       statistics counters of an EPON interface as defined in
       IEEE Std 802.3, Clause 60 and Clause 75.";
    reference
      "IEEE Std 802.3.1, dot3ExtPkgOptIfEntry";
    leaf in-trx-power-signal-detect {
      type boolean;
      config false;
      description
        "This object indicates whether a valid optical signal was
         detected (when read as 'true') or not
         (when read as 'false') at the input to the
         optical transceiver.
```

```
     This object is applicable for an OLT and an ONU. It has
     a distinct value for each logical link.";
  reference
    "IEEE Std 802.3.1, dot3ExtPkgOptIfSignalDetect";
}
leaf in-trx-power {
  type power-level;
  config false;
  description
    "This object reflects the value of the input power, as
     measured at the optical transceiver, expressed in units
     of 0.1 dBm.

     At the ONU, the measurement is performed in a continuous
     manner.

     At the OLT, the measurement is performed in a burst-mode
     manner, for each incoming data burst.

     This object is applicable for an OLT and an ONU. It has
     a distinct value for each logical link.";
  reference
    "IEEE Std 802.3.1, dot3ExtPkgOptIfInputPower";
}
leaf in-trx-power-low-15-minutes-bin {
  type power-level;
  config false;
  description
    "This object reflects the lowest value of the input power
     during the period of the last 15 minutes, as measured at
             the optical transceiver, and expressed in units of
     0.1 dBm.

     At the ONU, the measurement is performed in a continuous
     manner and stored in a rolling 15-minutes' long
     observation bin.

     At the OLT, the measurement is the average power for
     each incoming data burst, and stored in a rolling
     15-minutes' long observation bin.

     This object is applicable for an OLT and an ONU. It has
     a distinct value for each logical link.";
  reference
    "IEEE Std 802.3.1, dot3ExtPkgOptIfLowInputPower";
}
leaf in-trx-power-high-15-minutes-bin {
  type power-level;
  config false;
  description
    "This object reflects the highest value of the input
     power during the period of the last 15 minutes, as
     measured at the optical transceiver, and expressed in
     units of 0.1 dBm.

     At the ONU, the measurement is performed in a continuous
     manner and stored in a rolling 15-minutes' long
```

```
        observation bin.

        At the OLT, the measurement is the average power for
        each incoming data burst, and stored in a rolling
        15-minutes' long observation bin.

        This object is applicable for an OLT and an ONU. It has
        a distinct value for each logical link.";
    reference
      "IEEE Std 802.3.1, dot3ExtPkgOptIfHighInputPower";
  }
  leaf out-trx-power-signal-detect {
    type boolean;
    config false;
    description
      "This object indicates whether a valid optical signal was
        detected (when read as 'true') or not
        (when read as 'false') at the output from the
        optical transceiver.

        This object is applicable for an OLT and an ONU. It has
        a distinct value for each logical link.";
    reference
      "IEEE Std 802.3.1, dot3ExtPkgOptIfTransmitAlarm";
  }
  leaf out-trx-power {
    type power-level;
    config false;
    description
      "This object reflects the value of the output power, as
        measured at the optical transceiver, expressed in units
        of 0.1 dBm.

        At the ONU, the measurement is performed in a burst-mode
        manner for each outgoing data burst.

        At the OLT, the measurement is performed in a continuous
        manner.

        This object is applicable for an OLT and an ONU. It has
        a distinct value for each logical link.";
    reference
      "IEEE Std 802.3.1, dot3ExtPkgOptIfOutputPower";
  }
  leaf out-trx-power-low-15-minutes-bin {
    type power-level;
    config false;
    description
      "This object reflects the lowest value of the output
        power during the period of the last 15 minutes, as
        measured at the optical transceiver, and expressed in
        units of 0.1 dBm.

        At the ONU, the measurement is performed in a burst-mode
        manner and stored in a rolling 15-minutes' long
        observation bin.

        At the OLT, the measurement is the average power for
```

```
        each incoming data burst, and stored in a rolling
        15-minutes' long observation bin.

        This object is applicable for an OLT and an ONU. It has
        a distinct value for each logical link.";
     reference
       "IEEE Std 802.3.1, dot3ExtPkgOptIfLowOutputPower";
   }
   leaf out-trx-power-high-15-minutes-bin {
     type power-level;
     config false;
     description
       "This object reflects the highest value of the output
        power during the period of the last 15 minutes, as
        measured at the optical transceiver, and expressed in
        units of 0.1 dBm.

        At the ONU, the measurement is performed in a burst-mode
        manner and stored in a rolling 15-minutes' long
        observation bin.

        At the OLT, the measurement is the average power for
        each incoming data burst, and stored in a rolling
        15-minutes' long observation bin.

        This object is applicable for an OLT and an ONU. It has
        a distinct value for each logical link.";
     reference
       "IEEE Std 802.3.1, dot3ExtPkgOptIfHighOutputPower";
   }
   leaf trx-data-reliable {
     if-feature "trx-power-level-reporting-supported";
     type boolean;
     config false;
     description
       "This object indicates whether data contained in
        individual counters in 'statistics-trx' container are
        reliable (when read as 'true') or not
        (when read as 'false').

        This object is applicable for an OLT and an ONU. It has
        a distinct value for each logical link.";
     reference
       "IEEE Std 802.3.1, dot3ExtPkgOptIfSuspectedFlag";
   }
 }
 container monitoring-trx {
   if-feature "trx-power-level-reporting-supported";
   description
     "This container defines a set of optical transceiver
      statistics counters of an EPON interface as defined in
      IEEE Std 802.3, Clause 60 and Clause 75.";
   reference
     "IEEE Std 802.3.1, dot3ExtPkgOptIfEntry";
   leaf in-trx-power-signal-detect {
     type boolean;
     config false;
     description
```

```
    "This object indicates whether a valid optical signal was
     detected (when read as 'true') or not
     (when read as 'false')at the input to the
     optical transceiver.

     This object is applicable for an OLT and an ONU. It has
     a distinct value for each logical link.";
  reference
    "IEEE Std 802.3.1, dot3ExtPkgOptIfSignalDetect";
}
leaf in-trx-power {
  type power-level;
  config false;
  description
    "This object reflects the value of the input power, as
     measured at the optical transceiver, expressed in units
     of 0.1 dBm.

     At the ONU, the measurement is performed in a continuous
     manner.

     At the OLT, the measurement is performed in a burst-mode
     manner, for each incoming data burst.

     This object is applicable for an OLT and an ONU. It has
     a distinct value for each logical link.";
  reference
    "IEEE Std 802.3.1, dot3ExtPkgOptIfInputPower";
}
leaf in-trx-power-low-15-minutes-bin {
  type power-level;
  config false;
  description
    "This object reflects the lowest value of the input power
     during the period of the last 15 minutes, as measured at
     the optical transceiver, and expressed in units of
     0.1 dBm.

     At the ONU, the measurement is performed in a continuous
     manner and stored in a rolling 15-minutes' long
     observation bin.

     At the OLT, the measurement is the average power for
     each incoming data burst, and stored in a rolling
     15-minutes' long observation bin.

     This object is applicable for an OLT and an ONU.
     It has a distinct value for each logical link.";
  reference
    "IEEE Std 802.3.1, dot3ExtPkgOptIfLowInputPower";
}
leaf in-trx-power-high-15-minutes-bin {
  type power-level;
  config false;
  description
    "This object reflects the highest value of the input
     power during the period of the last 15 minutes,
     as measured at the optical transceiver,
```

```
       and expressed in units of 0.1 dBm.

       At the ONU, the measurement is performed in a continuous
       manner and stored in a rolling 15-minutes' long
       observation bin.

       At the OLT, the measurement is the average power for
       each incoming data burst, and stored in a rolling
       15-minutes' long observation bin.

       This object is applicable for an OLT and an ONU. It has
       a distinct value for each logical link.";
    reference
      "IEEE Std 802.3.1, dot3ExtPkgOptIfHighInputPower";
  }
  leaf out-trx-power-signal-detect {
    type boolean;
    config false;
    description
      "This object indicates whether a valid optical signal was
       detected (when read as 'true') or not
       (when read as 'false') at the output from the
       optical transceiver.

       This object is applicable for an OLT and an ONU. It has
       a distinct value for each logical link.";
    reference
      "IEEE Std 802.3.1, dot3ExtPkgOptIfTransmitAlarm";
  }
  leaf out-trx-power {
    type power-level;
    config false;
    description
      "This object reflects the value of the output power, as
       measured at the optical transceiver, expressed in units
       of 0.1 dBm.

       At the ONU, the measurement is performed in a burst-mode
       manner for each outgoing data burst.

       At the OLT, the measurement is performed in a continuous
       manner.

       This object is applicable for an OLT and an ONU. It has
       a distinct value for each logical link.";
    reference
      "IEEE Std 802.3.1, dot3ExtPkgOptIfOutputPower";
  }
  leaf out-trx-power-low-15-minutes-bin {
    type power-level;
    config false;
    description
      "This object reflects the lowest value of the output
       power during the period of the last 15 minutes, as
       measured at the optical transceiver, and expressed in
       units of 0.1 dBm.

       At the ONU, the measurement is performed in a burst-mode
```

```
             manner and stored in a rolling 15-minutes' long
             observation bin.

             At the OLT, the measurement is the average power for
             each incoming data burst, and stored in a rolling
             15-minutes' long observation bin.

             This object is applicable for an OLT and an ONU. It has
             a distinct value for each logical link.";
          reference
            "IEEE Std 802.3.1, dot3ExtPkgOptIfLowOutputPower";
        }
        leaf out-trx-power-high-15-minutes-bin {
          type power-level;
          config false;
          description
            "This object reflects the highest value of the output
             power during the period of the last 15 minutes, as
             measured at the optical transceiver, and expressed in
             units of 0.1 dBm.

             At the ONU, the measurement is performed in a burst-mode
             manner and stored in a rolling 15-minutes' long
             observation bin.

             At the OLT, the measurement is the average power for
             each incoming data burst, and stored in a rolling
             15-minutes' long observation bin.

             This object is applicable for an OLT and an ONU. It has
             a distinct value for each logical link.";
          reference
            "IEEE Std 802.3.1, dot3ExtPkgOptIfHighOutputPower";
        }
        leaf trx-data-reliable {
          if-feature "trx-power-level-reporting-supported";
          type boolean;
          config false;
          description
            "This object indicates whether data contained in
             individual counters in 'statistics-trx' container are
             reliable (when read as 'true') or not
             (when read as 'false').

             This object is applicable for an OLT and an ONU. It has
             a distinct value for each logical link.";
          reference
            "IEEE Std 802.3.1, dot3ExtPkgOptIfSuspectedFlag";
        }
      }
      container statistics-pon-fec {
        when "(../fec-capability = 'supported') and
              (../fec-mode = 'enabled-Tx-Rx')";
        if-feature "fec-supported";
        config false;
        description
          "This container defines a set of FEC-related statistics
           counters of an EPON interface, as defined in
```

```
       IEEE Std 802.3, Clause 65 and Clause 76.

       Discontinuities in the value of this counter can occur at
       re-initialization of the management system, and at other
       times as indicated by the value of the
       'discontinuity-time' leaf defined in the ietf-interfaces
       YANG module (IETF RFC 8343).";
    reference
      "IEEE Std 802.3.1, dot3OmpEmulationStatEntry";
    leaf fec-code-group-violations {
      type yang:counter64;
      units "code-group";
      config false;
      description
        "For 1G-EPON this is a count of the number of events that
         cause the PHY to indicate ’Data reception error’ or
         ’Carrier Extend Error’ on the GMII (see IEEE Std 802.3,
         Table 35-1). The contents of this counter is undefined
         when FEC is operating. For 10G-EPON this object is not
         applicable.

         This object is applicable for an OLT and an ONU. At the
         OLT, it has a distinct value for each logical link.";
      reference
        "IEEE Std 802.3, 30.5.1.1.14";
    }
    leaf fec-buffer-head-coding-violations {
      type yang:counter64;
      units "code-group";
      config false;
      description
        "For 1G-EPON PHY, this object represents the count of the
         number of invalid code-groups received directly from the
         link when FEC is enabled. When FEC is disabled this
         counter stops counting.

         For 10G-EPON PHYs, this object is set to zero.

         This object is applicable for an OLT and an ONU. It has
         a distinct value for each logical link.";
      reference
        "IEEE Std 802.3.1, dot3EponFecBufferHeadCodingViolation";
    }
    leaf fec-code-word-corrected-errors {
      type yang:counter64;
      units "code-group";
      config false;
      description
        "For 1G-EPON or 10G-EPON PHYs, this object represents a
         count of corrected FEC blocks.

         This counter increments by one for each received
         FEC block that contained detected errors and was
         corrected by the FEC function in the PHY.

         This object is applicable for an OLT and an ONU. It has
         a distinct value for each logical link.";
      reference
```

```
        "IEEE Std 802.3, 30.5.1.1.17";
    }
    leaf fec-code-word-uncorrected-errors {
      type yang:counter64;
      units "code-group";
      config false;
      description
        "For 1G-EPON or 10G-EPON PHYs, this object represents a
         count of uncorrectable FEC blocks.

         This counter increments by one for each received FEC
         block that contained detected errors and was not
         corrected by the FEC function in the PHY.

         This object is applicable for an OLT and an ONU. It has
         a distinct value for each logical link.";
      reference
        "IEEE Std 802.3, 30.5.1.1.18";
    }
  }
  container mpcp-logical-link-admin-actions {
    description
      "Container of actions.";
    action state-change-action-type {
      description
        "Request a state change on the interface.";
      input {
        leaf state-change-action-type {
          type identityref {
            base state-change-action-type;
          }
          description
            "Type of interface state change requested.";
        }
      }
    }
    action reset-action-type {
      description
        "Request a reset-action of the interface.";
      input {
        leaf reset-action-type {
          type identityref {
            base reset-action-type;
          }
          description
            "Type of reset action requested of the interface.";
        }
      }
    }
    action register-type {
      description
        "Request a registration action.";
      input {
        leaf register-type {
          type identityref {
            base register-type;
          }
          description
```

```
          "Type of registration action requested of the
           interface.";
      }
    }
  }
}
list mpcp-queues {
  key "mpcp-queue-index";
  description
    "An instance of this object for each value of
     'mpcp-queue-index' is created when a new logical link is
     registered and deleted when the logical link is
     deregistered.

     All instances of this object in the ONU associated with
     the given logical link are then mapped to a REPORT MPCPDU,
     when generated.

     +------------------------------------+
     |           Destination Address      |
     +------------------------------------+
     |           Source Address           |
     +------------------------------------+
     |           Length/Type              |
     +------------------------------------+
     |           OpCode                   |
     +------------------------------------+
     |           TimeStamp                |
     +------------------------------------+
     |           Number of Queue Sets     |
     +------------------------------------+   -|
     |           Report bitmap            |    |
     +------------------------------------+    |
     |           Queue 0 report           |    |
     +------------------------------------+    | repeated
     |           Queue 1 report           |    | for every
     +------------------------------------+    | Queue Set
     |           Queue 2 report           |    |
     +------------------------------------+    |
     |           Queue 3 report           |    |
     +------------------------------------+    |
     |           Queue 4 report           |    |
     +------------------------------------+    |
     |           Queue 5 report           |    |
     +------------------------------------+    |
     |           Queue 6 report           |    |
     +------------------------------------+    |
     |           Queue 7 report           |    |
     +------------------------------------+   -|
     |           Pad/reserved             |
     +------------------------------------+
     |           FCS                      |
     +------------------------------------+

     The 'Queue N report' field reports the current occupancy
     of each upstream transmission queue associated with the
     given logical link.
```

```
   The 'Number of Queue Sets' field defines the number of
   reported 'Queue N report' sets.

   For each Queue Set, the 'Report bitmap' field defines
   which upstream transmission queues are present in the
   REPORT MPCPDU.
   Although the REPORT MPCPDU can report current occupation
   for up to 8 upstream transmission queues in a single
   REPORT MPCPDU, the actual number is flexible.
   The 'mpcp-queue-group' grouping has a variable size that
   is limited by value of
   'mpcp-maximum-queue-count-per-report' object, allowing
   ONUs report the occupancy of fewer upstream transmission
   queues, as needed.

   This object is applicable for an OLT and an ONU.
   At the OLT, this object has a distinct value for each
   logical link and every queue. At the ONU, it has a
   distinct value for every queue.";
reference
  "IEEE Std 802.3.1, dot3ExtPkgQueueEntry";
leaf mpcp-queue-index {
  type uint8 {
    range "0 .. 7" {
      description
        "This object indicates the identity (index) of a
         queue in the ONU. It can have a value between
         0 and 7, limited by the value stored in the
         'mpcp-maximum-queue-count-per-report' object.";
      reference
        "See 'mpcp-maximum-queue-count-per-report' object";
    }
  }
  description
    "An object represents the index of an upstream
     transmission queue storing subscriber packets.
     The size (occupancy) of the upstream transmission queue
     identified by this object is then reported within
     REPORT MPCPDU, defined in IEEE Std 802.3, Clause 64
     and Clause 77.

     This object indicates the identity (index) of a queue in
     the ONU. It can have a value between 0 and 7, limited by
     the value stored in the
     'mpcp-maximum-queue-count-per-report' object.

     This object is applicable for an OLT and an ONU.
     It has a distinct value for each logical link and each
     queue.
     At the ONU, it has a distinct value for each queue.";
  reference
    "IEEE Std 802.3.1, dot3QueueIndex";
}
leaf mpcp-queue-threshold-count {
  type uint8 {
    range "0 .. 7" {
      description
        "This object indicates the identity (index) of a
```

```
           queue in the ONU. It can have a value between
           0 and 7, limited by the value stored in the
           'mpcp-maximum-queue-count-per-report' object.";
        reference
          "See 'mpcp-queue-threshold-count-max' object";
      }
    }
    description
      "This object reflects the number of reporting thresholds
       for the specific upstream transmission queue, reflected
       in the REPORT MPCPDU, as defined in IEEE Std 802.3,
       Clause 64 and Clause 77.

       Each 'Queue set' provides information for the specific
       upstream transmission queue occupancy of frames below
       the matching reporting threshold.

       A read of this object reflects the number of reporting
       thresholds for the specific upstream transmission queue.

       This object is applicable for an OLT and an ONU. It has
       a distinct value for each logical link and each queue.
       At the ONU, it has a distinct value for each queue.";
    reference
      "IEEE Std 802.3.1, dot3ExtPkgObjectReportNumThreshold";
  }
  leaf mpcp-queue-threshold-count-max {
    type uint8 {
      range "0 .. 7" {
        description
          "This object can have a value between 0 and 7.";
      }
    }
    config false;
    description
      "This object reflects the maximum number of reporting
       thresholds for the specific upstream transmission queue,
       reflected in the REPORT MPCPDU, as defined in
       IEEE Std 802.3, Clause 64 and Clause 77.

       A read of this object reflects the maximum number of
       reporting thresholds for the specific upstream
       transmission queue.

       This object is applicable for an OLT and an ONU. It has
       a distinct value for each logical link and each queue.
       At the ONU, it has a distinct value for each queue.";
    reference
      "IEEE Std 802.3.1,
       dot3ExtPkgObjectReportMaximumNumThreshold";
  }
  list mpcp-queue-thresholds {
    when '../mpcp-queue-threshold-count > 0';
    key "mpcp-queue-set-index";
    max-elements 7;
    description
      "An instance of this object for each value of
       'mpcp-queue-index' is created when a new logical link is
```

registered and deleted when the logical link is
deregistered.

All instances of this object in the ONU associated with
the given logical link are then mapped to a
REPORT MPCPDU, when generated.

```
+-----------------------------------+
|          Destination Address      |
+-----------------------------------+
|          Source Address           |
+-----------------------------------+
|          Length/Type              |
+-----------------------------------+
|          OpCode                   |
+-----------------------------------+
|          TimeStamp                |
+-----------------------------------+
|          Number of Queue Sets     |
+-----------------------------------+   -|
|          Report bitmap            |    |
+-----------------------------------+    |
|          Queue 0 report           |    |
+-----------------------------------+    | repeated for
|          Queue 1 report           |    | every
+-----------------------------------+    | Queue Set
|          Queue 2 report           |    |
+-----------------------------------+    |
|          Queue 3 report           |    |
+-----------------------------------+    |
|          Queue 4 report           |    |
+-----------------------------------+    |
|          Queue 5 report           |    |
+-----------------------------------+    |
|          Queue 6 report           |    |
+-----------------------------------+    |
|          Queue 7 report           |    |
+-----------------------------------+   -|
|          Pad/reserved             |
+-----------------------------------+
|          FCS                      |
+-----------------------------------+
```

The 'Queue N report' field reports the current occupancy
of each upstream transmission queue associated with the
given logical link.

The 'Number of Queue Sets' field defines the number of
reported 'Queue N report' sets.

For each Queue Set, the 'Report bitmap' field defines
which upstream transmission queues are present in the
REPORT MPCPDU. Although the REPORT MPCPDU can report
current occupation for up to 8 upstream transmission
queues in a single REPORT MPCPDU, the actual number is
flexible.

The 'mpcp-queue-group' grouping has a variable size that

```
           is limited by value of
           'mpcp-maximum-queue-count-per-report' object, allowing
           ONUs to report the occupancy of fewer upstream
           transmission queues, as needed.

           This object is applicable for an OLT and an ONU. It has
           a distinct value for each logical link and every queue.
           At the ONU, it has a distinct value for every queue.";
        reference
          "IEEE Std 802.3.1, dot3ExtPkgQueueSetsEntry";
        leaf mpcp-queue-set-index {
          type uint8 {
            range "0 .. 7" {
              description
                "This object indicates the identity (index) of a
                 queue in the ONU. It can have a value between
                 0 and 7, limited by the value stored in the
                 'mpcp-maximum-queue-count-per-report' object.";
              reference
                "See 'mpcp-maximum-queue-count-per-report' object";
            }
          }
          description
            "This object represents the index of the Queue Set for
             the'mpcp-queue-set-group' grouping. The size
             (occupancy) of the upstream transmission queues
             belonging to the given Queue Set is then reported
             within REPORT MPCPDU, definedin IEEE Std 802.3,
             Clause 64 and Clause 77.

             This object can have a value between 0 and 7, limited
            \t by the value stored in the
             'mpcp-queue-threshold-count-max' object.";
          reference
            "IEEE Std 802.3.1, dot3QueueSetIndex";
        }
        leaf mpcp-queue-set-threshold {
          type uint64;
          units "TQ";
          default "0";
          description
            "This object defines the value of a reporting threshold
             for each Queue Set stored in REPORT MPCPDU defined in
             IEEE Std 802.3, Clause 64 and Clause 77.

             The number of Queue Sets for each upstream
             transmission queue is defined in the
             'mpcp-queue-threshold-count' object.

             Within REPORT MPCPDU, each Queue Set provides
             information on the current upstream transmission queue
             occupancy for frames below the matching threshold.

             The value stored in this object is expressed in the
             units of Time quanta (TQ), where 1 TQ = 16 ns.

             A read of this object provides the current threshold
             value for the specific upstream transmission queue.
```

```
          This object is applicable for an OLT and an ONU. At
          the OLT, it has a distinct value for each logical link,
          each queue, and each Queue Set.

          At the ONU, it has a distinct value for each queue and
          each Queue Set.";
      reference
        "IEEE Std 802.3.1, dot3ExtPkgObjectReportThreshold";
    }
  }
  leaf in-mpcp-queue-frames {
    type yang:counter64;
    config false;
    description
      "A count of the number of times a frame reception event
       results in a frame being queued in (for ONUs) or
       received from (for OLTs) the corresponding queue. This
       object is incremented by one for each frame written to
       (in the case of the ONU) or received for (in case of the
       OLT) the associated queue.

       The queue index matches the queue number in
       REPORT MPCPDU, as defined in IEEE Std 802.3, Clause 64
       and Clause 77.

       This object is applicable for an OLT and an ONU.
       At the OLT,it has a distinct value for each logical link
       and each queue. At the ONU, it has a distinct value for
       each queue.";
    reference
      "IEEE Std 802.3.1, dot3ExtPkgStatRxFramesQueue";
  }
  leaf out-mpcp-queue-frames {
    when "../../mpcp-mode = 'onu'";
    type yang:counter64;
    config false;
    description
      "This object reflects the number of frame transmission
       events from the corresponding upstream transmission
       queue. This object is incremented by one for each frame
       transmitted, when it is output from the associated
       queue.

       The queue index matches the queue number in
       REPORT MPCPDU, as defined in IEEE Std 802.3, Clause 64
       and Clause 77.

       This object is applicable for an ONU only. At the ONU,
       it has a distinct value for each queue.";
    reference
      "IEEE Std 802.3.1, dot3ExtPkgStatTxFramesQueue";
  }
  leaf mpcp-queue-frames-drop {
    when "../../mpcp-mode = 'onu'";
    type yang:counter64;
    config false;
    description
```

```
          "This object reflects the number of frame drop events
           from the corresponding upstream transmission queue.
           This object is incremented by one for each frame dropped
           in the associated queue.

           The queue index matches the queue number in
           REPORT MPCPDU, as defined in IEEE Std 802.3, Clause 64
           and Clause 77.

           This object is applicable for an ONU only. At the ONU,
           it has a distinct value for each queue.";
        reference
          "IEEE Std 802.3.1, dot3ExtPkgStatDroppedFramesQueue";
      }
    }
    list multicast-IDs {
      key "multicast-ID";
      description
        "Multicast-IDs list of multicast IDs
         to be recognized by the device.";
      leaf multicast-ID {
        type uint32;
        description
          "Multicast-IDs to be recognized by the device.";
        reference
          "IEEE Std 802.3, 30.3.5.1.25";
      }
    }
    leaf fec-capability {
      type fec-capability;
      config false;
      description
        "This object is used to identify whether the given
         interface is capable of supporting FEC or not.";
    }
    leaf mpcp-mode {
      type mpcp-mode;
      config false;
      description
        "This object is used to identify the operational state of
         the MultiPoint MAC Control sublayer as defined in
         IEEE Std 802.3, Clause 64 and Clause 77.

         Reading 'olt' for an OLT (controller) mode and 'onu' for
         an ONU (client) mode.

         This object is used to identify the operational mode for
         the MPCP objects.

         This object is applicable for an OLT, with the same value
         for all logical links, and for an ONU.";
      reference
        "IEEE Std 802.3, 30.3.5.1.3";
    }
    leaf mpcp-sync-time {
      type uint64;
      units "TQ (16ns)";
      config false;
```

```
  description
    "This object reports the 'sync lock time' of the OLT
     receiver in units of Time Quanta (TQ; 1 TQ = 16 ns; see
     IEEE Std 802.3, Clause 64 and Clause 77).

     The value returned is equal to [sync lock time ns]/16,
     rounded up to the nearest TQ. If this value exceeds
     4,294,967,295 TQ, the value 4,294,967,295 TQ is returned.

     This object is applicable for an OLT, with distinct values
     for all logical links, and for an ONU.";
  reference
    "IEEE Std 802.3.1, dot3MpcpSyncTime";
}
leaf mpcp-logical-link-id {
  type mpcp-supported;
  config false;
  description
    "This object is used to identify the operational state of
     the MultiPoint MAC Control sublayer as defined in
     IEEE Std 802.3, Clause 64 and Clause 77.

     Reading 'olt' for an OLT (controller) mode and 'onu' for
     an ONU (client) mode.

     This object is used to identify the operational mode for
     the MPCP objects.

     This object is applicable for an OLT, with the same value
     for all logical links, and for an ONU.";
  reference
    "IEEE Std 802.3, 30.3.5.1.3";
}
leaf mpcp-remote-mac-address {
  type ieee:mac-address;
  config false;
  description
    "This object identifies the source_address parameter of the
     last MPCPDUs passed to the MAC Control. This value is
     updated on reception of a valid frame with:

     1) a destination Field equal to the multicast address
     assigned for MAC Control as specified in
     IEEE Std 802.3, Annex 31A;

     2) the lengthOrType field value equal to the Type assigned
     for MAC Control as specified in IEEE Std 802.3, Annex 31A;

     3) an MPCP Control opcode value equal to the subtype
     assigned for MPCP as specified in IEEE Std 802.3,
     Annex 31A.

     This object is applicable for an OLT and an ONU. It has a
     distinct value for each logical link.

     The value reflects the MAC address of the remote entity
     and therefore the OLT holds a value for each LLID, which
     is the MAC address of the ONU.
```

```
     The ONU has a single value that is the OLT MAC address.";
  reference
    "IEEE Std 802.3, 30.3.5.1.5";
}
leaf mpcp-logical-link-state {
  type mpcp-logical-link-state;
  config false;
  description
    "This object identifies the registration state of the
     MultiPoint MAC Control sublayer as defined in
     IEEE Std 802.3, Clause 64 and Clause 77.

     When this object has the enumeration 'unregistered', the
     interface is unregistered and may be used for registering
     a link partner.

     When this object has the enumeration 'registering',
     the interface is in the process of registering a
             link-partner.

     When this object has the enumeration 'registered', the
     interface has an established link-partner.

     This object is applicable for an OLT and an ONU. It has a
     distinct value for each logical link.";
  reference
    "IEEE Std 802.3, 30.3.5.1.6";
}
leaf mpcp-elapsed-time-out {
  type uint64;
  units "TQ (16ns)";
  config false;
  description
    "This object reports the interval from the last MPCP frame
     transmission in increments of Time Quanta
     (TQ; 1 TQ = 16 ns;
     see IEEE Std 802.3, Clause 64 and Clause 77).

     The value returned is equal to [interval from last MPCP
     frame transmission on this EPON interface, expressed
     in ns]/16. If this value exceeds 4,294,967,295 TQ, the
     value 4,294,967,295 TQ is returned.

     This object is applicable for an OLT and an ONU. It has a
     distinct value for each logical link.";
  reference
    "IEEE Std 802.3, 30.3.5.1.19";
}
leaf mpcp-elapsed-time-in {
  type uint64;
  units "TQ (16ns)";
  config false;
  description
    "This object reports the interval from the last MPCP frame
     reception in increments of Time Quanta (TQ; 1 TQ = 16 ns;
     see IEEE Std 802.3, Clause 64 and Clause 77).
```

```
         The value returned is equal to [interval from last MPCP
         frame reception on this EPON interface, expressed in
         ns]/16. If this value exceeds 4,294,967,295 TQ, the value
         4,294,967,295 TQ is returned.

         This object is applicable for an OLT and an ONU. It has a
         distinct value for each logical link.";
      reference
        "IEEE Std 802.3, 30.3.5.1.20";
    }
    leaf mpcp-round-trip-time {
      when "../ompe-mode = 'olt'";
      type uint16;
      units "TQ (16ns)";
      config false;
      description
        "This object reports the MPCP round trip time in increments
         of Time Quanta (TQ; 1 TQ = 16 ns; see IEEE Std 802.3,
         Clause 64 and Clause 77).

         The value returned is equal to [round trip time in ns]/16.
         If this value exceeds 65,535 TQ, the value 65,535 TQ is
         returned.

         This object is applicable for an OLT. It has a distinct
         value for each logical link.";
      reference
        "IEEE Std 802.3, 30.3.5.1.21";
    }
    leaf mpcp-maximum-grant-count {
      when "../ompe-mode = 'onu'";
      type uint8;
      config false;
      description
        "This object reports the maximum number of grants that an
         ONU can store for handling. The maximum number of grants
         that an ONU can store for handling has a range of
         0 to 255.

         This object is applicable for an ONU and has a distinct
         value for each logical link.";
      reference
        "IEEE Std 802.3, 30.3.5.1.24";
    }
    leaf mpcp-logical-link-count {
      type mpcp-llid-count;
      units "LLID";
      config false;
      description
        "This object reflects the number of logical links
         registered on the OLT / ONU. The LLID field, as defined in
         the IEEE Std 802.3, Clause 65 and Clause 76, is a 2-byte
         register (15-bit field and a broadcast bit) limiting the
         number of logical links to 32,768.

         This object is initialized to the value of 0 when the
         OLT / ONU is powered up.
```

```
         This object is applicable for an OLT and an ONU. It has
                 the same value for all logical links.";
      reference
        "IEEE Std 802.3.1, dot3ExtPkgObjectNumberOfLLIDs";
    }
    leaf mpcp-maximum-queue-count-per-report {
      when "../ompe-mode = 'olt'";
      type mpcp-maximum-queue-count-per-report;
      config false;
      description
        "This object reflects the maximum number of queues (0-7)
         that can be accepted by the OLT in a single REPORT MPCPDU,
         as defined in IEEE Std 802.3, Clause 64 and Clause 77.

         This object is applicable for an OLT and has a distinct
         value for each logical link.";
      reference
        "IEEE Std 802.3.1, dot3ExtPkgObjectReportMaximumNumQueues";
    }
    leaf ompe-mode {
      type ompe-mode;
      config false;
      description
        "This object indicates the mode of operation of the
         Reconciliation Sublayer for Point-to-Point Emulation (see
         IEEE Std 802.3, 65.1 or 76.2 as appropriate).

         The value of 'unknown' is assigned in initialization; true
         state or type is not yet known.

         The value of 'olt' is assigned when the sublayer is
         operating in OLT mode.

         The value of 'onu' is assigned when the sublayer is
         operating in ONU mode.

         This object is applicable for an OLT and an ONU. It has
         the same value for each logical link.";
      reference
        "IEEE Std 802.3, 30.3.7.1.2";
    }
  }
}
module ieee802-ethernet-pse-2 {
  yang-version 1.1;
  namespace "urn:ieee:std:802.3:yang:ieee802-ethernet-pse-2";
  prefix ieee802-pse-2024;

  import ietf-interfaces {
    prefix if;
    reference
      "IETF RFC 8343";
  }
  import ietf-yang-types {
    prefix yang;
    reference
      "IETF RFC 6991";
  }
```

```
  import ieee802-ethernet-interface {
    prefix ieee802-eth-if;
  }

  organization
    "IEEE 802.3 Ethernet Working Group
     Web URL: http://www.ieee802.org/3/";
  contact
    "Web URL: http://www.ieee802.org/3/";
  description
    "This module contains YANG definitions for configuring and
     managing ports with Power Over Ethernet featurefeatures defined by
     IEEE Std 802.3-2022.
     This module is based on,and superscedes ieee802-ethernet-pse.yang
     from IEEE Std 802.3.2-2019.";

  revision 2024-12-23 {
    description
      "Initial revision.";
    reference
      "IEEE Std IEEE Std 802.3.2-202x, unless dated explicitly";
  }

  feature multi-pair-pse {
    description
      "This object indicates the support for IEEE Std 802.3
          Clause 33 and/or 145.";
  }

  feature single-pair-pse {
    description
      "This object indicates the support for IEEE Std 802.3
          Clause 104.";
  }

  typedef multi-pair-detection-state {
    type enumeration {
      enum disabled {
        description
          "PSE disabled.";
      }
      enum searching {
        description
          "PSE is searching.";
      }
      enum deliveringPower {
        description
          "PSE is delivering power.";
      }
      enum fault {
        description
          "PSE fault detected.";
      }
      enum test {
        description
          "PSE test mode.";
      }
      enum otherFault {
```

```
        description
          "PSE implementation specific fault detected.";
      }
    }
    description
      "Detection state of a multi-pair PSE.";
    reference
      "IEEE Std 802.3, 30.9.1.1.5";
  }

  typedef single-pair-detection-state {
    type enumeration {
      enum unknown {
        description
          "True detection state unknown.";
      }
      enum disabled {
        description
          "PSE is disabled.";
      }
      enum searching {
        description
          "PSE is searching.";
      }
      enum deliveringPower {
        description
          "PSE is delivering power.";
      }
      enum sleep {
        description
          "PSE is in sleep state.";
      }
      enum idle {
        description
          "PSE is idle.";
      }
      enum error {
        description
          "PSE error.";
      }
    }
    description
      "Detection state of a single-pair PSE.";
    reference
      "IEEE Std 802.3, 30.15.1.1.3";
  }

  typedef multi-pair-power-class {
    type enumeration {
      enum class0 {
        description
          "Class 0";
      }
      enum class1 {
        description
          "Class 1";
      }
      enum class2 {
```

```
        description
          "Class 2";
      }
      enum class3 {
        description
          "Class 3";
      }
      enum class4 {
        description
          "Class 4";
      }
      enum class5 {
        description
          "Class 5";
      }
      enum class6 {
        description
          "Class 6";
      }
      enum class7 {
        description
          "Class 7";
      }
      enum class8 {
        description
          "Class 8";
      }
    }
    description
      "Multi-pair PoE power class.";
    reference
      "IEEE Std 802.3, 30.9.1.1.6 aPSEPowerClassification ";
  }

  typedef pse-support {
    type enumeration {
      enum two-pair {
        description
          "PSE port supports IEEE Std 802.3, Clause 33.";
      }
      enum four-pair {
        description
          "PSE port supports IEEE Std 802.3, Clause 33 and 145.";
      }
      enum single-pair {
        description
          "PSE port supports IEEE Std 802.3, Clause 104.";
      }
    }
    description
      "PSE support.";
  }

  typedef multi-pair-powering-pairs {
    type enumeration {
      enum signal {
        description
          "Using signal pairs.";
```

```
        }
        enum spare {
          description
            "Using spare pairs.";
        }
        enum both {
          description
            "Using signal and spare pairs.";
        }
      }
      description
        "Powering pairs.";
  }

  typedef single-pair-pd-type {
    type enumeration {
      enum unknown {
        description
          "Unknown PD type";
      }
      enum typeA {
        description
          "TypeA PD";
      }
      enum typeB {
        description
          "TypeB PD";
      }
      enum typeC {
        description
          "TypeC PD";
      }
      enum typeD {
        description
          "TypeD PD";
      }
      enum typeE {
        description
          "TypeE PD";
      }
      enum typeF {
        description
          "TypeF PD";
      }
    }
    description
      "Single-pair PoE PD type.";
    reference
      "IEEE Std 802.3, 30.15.1.1.5 aPoDLPSEDetectedPDType";
  }

  typedef single-pair-pse-type {
    type enumeration {
      enum unknown {
        description
          "Unknown PSE type.";
      }
      enum typeA {
```

```
          description
            "TypeA PSE";
        }
        enum typeB {
          description
            "TypeB PSE";
        }
        enum typeC {
          description
            "TypeC PSE";
        }
        enum typeD {
          description
            "TypeD PSE";
        }
        enum typeE {
          description
            "TypeE PSE";
        }
        enum typeF {
          description
            "TypeF PSE";
        }
      }
      description
        "Single-pair PoE PSE type.";
      reference
        "IEEE Std 802.3, 30.15.1.1.5 aPoDLPSEDetectedPDType";
  }

  typedef single-pair-power-class {
    type enumeration {
      enum class0 {
        description
          "Class 0";
      }
      enum class1 {
        description
          "Class 1";
      }
      enum class2 {
        description
          "Class 2";
      }
      enum class3 {
        description
          "Class 3";
      }
      enum class4 {
        description
          "Class 4";
      }
      enum class5 {
        description
          "Class 5";
      }
      enum class6 {
        description
```

```
            "Class 6";
        }
        enum class7 {
          description
            "Class 7";
        }
        enum class8 {
          description
            "Class 8";
        }
        enum class9 {
          description
            "Class 9 ";
        }
        enum unknown {
          description
            "Initializing, true Power Class not yet known.";
        }
      }
      description
        "Single-pair PoE power class.";
      reference
        "IEEE Std 802.3, 30.9.1.1.6 aPSEPowerClassification";
  }

  augment "/if:interfaces/if:interface/ieee802-eth-if:ethernet" {
    description
      "Augments ethernet interface configuration model with
       nodes specific to Multi-pair and Single-pair PoE.";
    container pse {
      description
        "Multi-pair and Single-pair PoE port configuration.

         Discontinuities in the values of counters in this container
         can occur at re-initialization of the  management system,
         and at other times as indicated by the value of the
         'discontinuity-time' leaf  defined in the ietf-interfaces
          YANG module (IETF RFC 8343).";
      reference
        "IEEE Std 802.3, 30.9.1 multi-pair PoE PSE and
                IEEE Std 802.3, 30.15.1 single-pair PoE PSE";
      leaf supported-pse-type {
        type pse-support;
        config false;
        description
          "The PSE types are supported.";
      }
      /* Multi-pair objects */
      container multi-pair {
        if-feature "multi-pair-pse";
        presence "PSE port supports IEEE Std 802.3 Clause 33
                         and/or Clause 145.";
        description
          "PSE port configuration in IEEE Std 802.3, 30.9.1.";
        leaf pse-enable {
          type boolean;
          default "false";
          description
```

```
            "When true enables the PSE function on the port,
             when false disables the PSE function on the port.";
          reference
            "IEEE Std 802.3, 30.9.1.2.1 acPSEAdminControl";
        }
        leaf pse-state {
          type boolean;
          config false;
          description
            "The PSE enabled state on the port.";
          reference
            "IEEE Std 802.3, 30.9.1.1.2 aPSEAdminState";
        }
        leaf multi-pair-powering-pairs {
          type multi-pair-powering-pairs;
          description
            "The PSE pairs in use. If the value of
             pairs-control-ability is true, this object is
             writeable.";
          reference
            "IEEE Std 802.3, 30.9.1.1.4 aPSEPowerPairs";
        }
        leaf pairs-control-ability {
          type boolean;
          default "true";
          config false;
          description
            "Can the PSE Port control switching the
             power sourcing pins.";
          reference
            "IEEE Std 802.3, 30.9.1.1.3 aPSEPowerPairsControlAbility";
        }
        leaf detection-status {
          type multi-pair-detection-state;
          config false;
          status obsolete;
          description
            "The operational status of the port PD detection.";
          reference
            "IEEE Std 802.3, 30.9.1.1.5 aPSEPowerDetectionStatus";
        }
        leaf classifications {
          when "../detection-status  = 'deliveringPower'" {
            description
              "This node only applies when the detection status is
               deliveringPower.";
          }
          type multi-pair-power-class;
          config false;
          description
            "The power class of the detected PD.";
          reference
            "IEEE Std 802.3, 30.9.1.1.6 aPSEPowerClassfication";
        }
        /* Multi-pair statistics */
        container statistics {
          config false;
          description
```

```
            "Multi-pair PoE Statistics information for the port.";
          leaf power-denied {
            type yang:counter64;
            description
              "The number of times the PSE denied power.";
            reference
              "IEEE Std 802.3, 30.9.1.1.14";
          }
          leaf invalid-signature {
            type yang:counter64;
            description
              "The number of times the PSE detected an
               invalid signature.";
            reference
              "IEEE Std 802.3, 30.9.1.1.11";
          }
          leaf mps-absent {
            type yang:counter64;
            description
              "The number of times the PSE
               lost the 'Maintain Power Signature'(MPS).";
            reference
              "IEEE Std 802.3, 30.9.1.1.20";
          }
          leaf overload {
            type yang:counter64;
            status deprecated;
            description
              "The number of times the PSE detected
               an output current overload.";
            reference
              "IEEE Std 802.3, 30.9.1.1.17";
          }
          leaf short {
            type yang:counter64;
            status deprecated;
            description
              "This object is deprecated as it's not defined in base
               standard.";
          }
          leaf cumulative-energy {
            type yang:counter64;
            units "millijoules";
            description
              "The cumulative energy supplied by the PSE.";
            reference
              "IEEE Std 802.3, 30.9.1.1.25";
          }
        }
        /* Multi-pair measurement */
        leaf actual-power {
          type uint32;
          units "milliwatts";
          config false;
          description
            "The actual power drawn by a PD over the port.";
          reference
            "IEEE Std 802.3, 30.9.1.1.23";
```

```
        }
        leaf power-accuracy {
          type int64;
          units "milliwatts";
          config false;
          description
            "The power accuracy of the port.";
          reference
            "IEEE Std 802.3, 30.9.1.1.24";
        }
      }
      /* Single-pair objects */
      container single-pair {
        if-feature "single-pair-pse";
        presence "PSE port supports IEEE Std 802.3 Clause 104.";
        description
          "Single-pair PSE configuration as defined in
           IEEE Std 802.3, 30.15.1.";
        leaf pse-enable {
          type boolean;
          default "false";
          description
            "When true enables the PSE function on the port,
             when false disables the PSE function on the port.";
          reference
            "IEEE Std 802.3, 30.15.1.1.2 aPoDLPSEAdminState";
        }
        leaf detection-status {
          type single-pair-detection-state;
          config false;
          description
            "Indicates the current status of the Single-pair PSE.";
          reference
            "IEEE Std 802.3,
             30.15.1.1.3 aPoDLPSEPowerDetectionStatus";
        }
        leaf single-pair-type {
          type single-pair-pse-type;
          config false;
          description
            "Single-pair PSE type.";
          reference
            "IEEE Std 802.3, 30.15.1.1.4 aPoDLPSEType";
        }
        leaf detected-pd-type {
          when "../detection-status = 'deliveringPower'" {
            description
              "This node only applies when the detection status is
               delivering power.";
          }
          type single-pair-pd-type;
          config false;
          description
            "Indicates the detected type of the single-pair PD.";
          reference
            "IEEE Std 802.3, 30.15.1.1.5 aPoDLPSEDetectedPDType";
        }
        leaf pd-power-class {
```

```
          when "../detection-status = 'deliveringPower'" {
            description
              "This node only applies when the detection status is
               delivering power.";
          }
          type single-pair-power-class;
          config false;
          description
            "The power class of the detected PD.";
          reference
            "IEEE Std 802.3,
             30.15.1.1.6 aPoDLPSEDetectedPDPowerClass";
        }
        /* Single-pair statistics */
        container statistics {
          config false;
          description
            "Single-pair PoE Statistics information for the port.";
          leaf power-denied {
            type yang:counter64;
            description
              "The number of times the PSE denied power.";
            reference
              "IEEE Std 802.3,
                \t  30.15.1.1.9 aPoDLPSEPowerDeniedCounter";
          }
          leaf invalid-signature {
            type yang:counter64;
            description
              "The number of times the PSE detected an
               invalid signature.";
            reference
              "IEEE Std 802.3, 30.15.1.1.7
               aPoDLPSEInvalidSignatureCounter";
          }
          leaf invalid-class {
            type yang:counter64;
            description
              "The number of times PD classification timed out
               or did not get valid class information from SCCP.";
            reference
              "IEEE Std 802.3,
               30.15.1.1.8 aPoDLPSEInvalidClassCounter";
          }
          leaf overload {
            type yang:counter64;
            description
              "The number of times the PSE detected
               an output current overload.";
            reference
              "IEEE Std 802.3, 30.15.1.1.10 aPoDLPSEOverLoadCounter";
          }
          leaf fvs-absence {
            type yang:counter64;
            description
              "The number of times the PSE
               lost the 'Maintain Full Voltage Signature'(MFVS).";
            reference
```

```
              "IEEE Std 802.3, 30.15.1.1.11
               aPoDLPSEMaintainFullVoltageSignatureAbsentCounter";
          }
          leaf cumulative-energy {
            type yang:counter64;
            units "millijoules";
            description
              "The cumulative energy supplied by the PSE as
               measured at the MDI.";
            reference
              "IEEE Std 802.3, 30.15.1.1.14 aPoDLPSECumulativeEnergy";
          }
        }
        /* Single-pair measurement */
        leaf actual-power {
          type uint32;
          units "milliwatts";
          config false;
          description
            "The actual power drawn by a PD over the port.";
          reference
            "IEEE Std 802.3, 30.15.1.1.12 aPoDLPSEActualPower";
        }
        leaf power-accuracy {
          type int64;
          units "milliwatts";
          config false;
          description
            "The power accuracy of the port.";
          reference
            "IEEE Std 802.3, 30.15.1.1.13 aPoDLPSEPowerAccuracy";
        }
      }
    }
  }
}
module ieee802-ethernet-pse {
  yang-version 1.1;
  namespace "urn:ieee:std:802.3:yang:ieee802-ethernet-pse";
  prefix ieee802-pse;

  import ietf-interfaces {
    prefix if;
    reference
      "IETF RFC 8343";
  }
  import ietf-yang-types {
    prefix yang;
    reference
      "IETF RFC 6991";
  }
  import ieee802-ethernet-interface {
    prefix ieee802-eth-if;
  }

  organization
    "IEEE 802.3 Ethernet Working Group
     Web URL: http://www.ieee802.org/3/";
```

```
  contact
    "Web URL: http://www.ieee802.org/3/";
  description
    "This module is deprecated and superceeded by
         ieee802-ethernet-pse-2.
         This module contains YANG definitions for configuring and
     managing ports with Power Over Ethernet feature defined by
     IEEE 802.3. It provides functionality roughly equivalent to
     that of the POWER-ETHERNET-MIB defined in IETF RFC 3621.";

  revision 2024-07-1712-23 {
    description
      "Updates underDeprecated in IEEE Std 802.3.2-202x, Draft D3.0";1,
superceeded by
          ieee802-ethernet-pse-2";
    reference
      "IEEE Std IEEE Std 802.3.2-202x, unless dated explicitly";
  }

  identity pse-type {
    description
      "Base type for PSE.";
  }

  identity all {
    base powering-pairs;
    description
      "All pairs are in use.";
  }

  identity four-pair {
    base pse-type;
    description
      "PSE support IEEE Std 802.3, Clause 145.";
  }

  identity two-pair {
    base pse-type;
    description
      "PSE supports IEEE Std 802.3, Clause 33.";
  }

  identity single-pair {
    base pse-type;
    description
      "PSE support IEEE Std 802.3, Clause 104.";
  }

  identity powering-pairs {
    description
      "Base type for powering pairs.";
  }

  identity signal {
    base powering-pairs;
    description
      "The signal pairs are in use.";
  }
```

```
  identity spare {
    base powering-pairs;
    description
      "The spare pairs are in use.";
  }

  typedef multi-pair-detection-state {
    type enumeration {
      enum disabled {
        value 1;
        description
          "PSE disabled.";
      }
      enum searching {
        value 2;
        description
          "PSE is searching.";
      }
      enum deliveringPower {
        value 3;
        description
          "PSE is delivering power.";
      }
      enum fault {
        value 4;
        description
          "PSE fault detected.";
      }
      enum test {
        value 5;
        description
          "PSE test mode.";
      }
      enum otherFault {
        value 6;
        description
          "PSE implementation specific fault detected.";
      }
    }
    status deprecated;
    description
      "      "Deprecated in IEEE Std 802.3.2-202x, Draft D3.1,
superceeded by
           ieee802-ethernet-pse-2
           Detection state of a multi-pair PSE.";
    reference
      "IEEE Std 802.3, 30.9.1.1.5";
  }

  typedef single-pair-detection-state {
    type enumeration {
      enum unknown {
        value 1;
        description
          "True detection state unknown.";
      }
      enum disabled {
```

```
        value 2;
        description
          "PoDL PSE is disabled.";
      }
      enum searching {
        value 3;
        description
          "PoDL PSE is searching.";
      }
      enum deliveringPower {
        value 4;
        description
          "PoDL PSE is delivering power.";
      }
      enum sleep {
        value 5;
        description
          "PoDL PSE is in sleep state.";
      }
      enum idle {
        value 6;
        description
          "PoDL PSE is idle.";
      }
      enum error {
        value 7;
        description
          "PoDL PSE error.";
      }
    }
    status deprecated;
    description
      "      "Deprecated in IEEE Std 802.3.2-202x, Draft D3.1,
superceeded by
          ieee802-ethernet-pse-2
       Detection state of a PoDL PSE.";
    reference
      "IEEE Std 802.3, 30.15.1.1.3";
  }

  typedef power-class {
    type enumeration {
      enum class0 {
        value 1;
        description
          "Class 0";
      }
      enum class1 {
        value 2;
        description
          "Class 1";
      }
      enum class2 {
        value 3;
        description
          "Class 2";
      }
      enum class3 {
```

```
          value 4;
          description
            "Class 3";
        }
        enum class4 {
          value 5;
          description
            "Class 4";
        }
        enum class5 {
          value 6;
          description
            "Class 5 (for PoDL-only)";
        }
        enum class6 {
          value 7;
          description
            "Class 6 (for PoDL-only)";
        }
        enum class7 {
          value 8;
          description
            "Class 7 (for PoDL-only)";
        }
        enum class8 {
          value 9;
          description
            "Class 8 (for PoDL-only)";
        }
        enum class9 {
          value 10;
          description
            "Class 9 (for PoDL-only)";
        }
        enum unknown {
          value 11;
          description
            "Initializing, true Power Class not yet known
             (only for PoDL PSE).";
        }
      }
      status deprecated;
      description
        "      "Deprecated in IEEE Std 802.3.2-202x, Draft D3.1,
superceeded by
          ieee802-ethernet-pse-2
          The power class.";
      reference
        "IEEE Std 802.3, 30.9.1.1.6 aPSEPowerClassification and
         IEEE Std 802.3, 30.15.1.1.6 aPoDLPSEDetectedPDPowerClass.";
    }

    augment "/if:interfaces/if:interface/ieee802-eth-if:ethernet" {
      status deprecated;
      description
        "      "Deprecated in IEEE Std 802.3.2-202x, Draft D3.1,
superceeded by
           ieee802-ethernet-pse-2
```

```
           Augments ethernet interface configuration model with
       nodes specific to DTE Power via MDI devices and ports";
    container pse {
      description
        "DTE Power via MDI port configuration";
      reference
        "IEEE Std 802.3, 30.9.1 PoE PSE & IEEE Std 802.3, 30.15.1
         PoDL PSE";
      leaf supported-pse-type {
        type identityref {
          base ieee802-pse:pse-type;
        }
        config false;
        description
          "PSE supports one or more of IEEE Std 802.3 Clause 33,
           Clause 104, or Clause 145.";
      }
      container multi-pair {
        presence "PSE port supports IEEE Std 802.3, Clause 33.";
        status deprecated;
        description
          "~~PSE~~Deprecated in IEEE Std 802.3.2-202x, Draft D3.1,
superceeded by
           ieee802-ethernet-pse-2PSE port configuration in IEEE Std
802.3,
           30.9.1.";
        leaf pse-enable {
          type boolean;
          default "false";
          description
            "When true enables the PSE function on the interface,
             when false disables the PSE function on the
             interface.";
          reference
            "IEEE Std 802.3, 30.9.1.1.2 aPSEAdminState";
        }
        leaf powering-pairs {
          type identityref {
            base powering-pairs;
          }
          description
            "Describes or controls the PSE pairs in use. If the
             value of pairs-control-ability is true, this object
             is writeable.";
          reference
            "IEEE Std 802.3, 30.9.1.1.4 aPSEPowerPairs";
        }
        leaf pairs-control-ability {
          type boolean;
          default "true";
          config false;
          description
            "Describes the ability to control switching the
             power sourcing pins of the PSE.";
          reference
            "IEEE Std 802.3, 30.9.1.1.3
             aPSEPowerPairsControlAbility";
        }
```

```
        leaf detection-status {
          type multi-pair-detection-state;
          config false;
          status deprecated;
          description
            "Deprecated in IEEE Std 802.3.2-202x, Draft D3.1, superceeded
by
             ieee802-ethernet-pse-2
             Describes the operational status of the port
             PD detection.";
          reference
            "IEEE Std 802.3, 30.9.1.1.5 aPSEPowerDetectionStatus";
        }
        leaf classifications {
          when "../detection-status = 'deliveringPower'" {
            description
              "Deprecated in IEEE Std 802.3.2-202x, Draft D3.1,
superceeded by
               ieee802-ethernet-pse-2
               This node only applies when the detection status is
               delivering power.";
          }
          type power-class;
          config false;
          status deprecated;
          description
            "The power class of the PSE port.";
          reference
            "IEEE Std 802.3, 30.9.1.1.6 aPSEPowerClassfication";
        }
        container statistics {
          config false;
          status deprecated;
          description
            "~~statistics~~Deprecated in IEEE Std 802.3.2-202x, Draft D3.1,
superceeded by
             ieee802-ethernet-pse-2statistics information of the multi-
pair
             port.";
          leaf power-denied {
            type yang:counter64;
            description
              "This counter is incremented when the PSE state
               diagram enters the POWER_DENIED state, per
               IEEE Std 802.3, Figure 33-9.";
            reference
              "IEEE Std 802.3, 30.9.1.1.14";
          }
          leaf invalid-signature {
            type yang:counter64;
            description
              "This counter is incremented when the PSE state
               diagram enters the SIGNATURE_INVALID state per
               IEEE Std 802.3, Figure 33-9.";
            reference
              "IEEE Std 802.3, 30.9.1.1.11";
          }
          leaf mps-absent {
```

```
        type yang:counter64;
        description
          "This counter is incremented when the PSE
           transitions directly from the POWER_ON state to the
           IDLE state due to tmpdo_timer_done being asserted,
           per IEEE Std 802.3, Figure 33-9.";
        reference
          "IEEE Std 802.3, 30.9.1.1.20";
      }
      leaf overload {
        type yang:counter64;
        status deprecated;
        description
          "This counter is incremented when the PSE state
           diagram enters the ERROR_DELAY state due to the
           ovld_detected variable being TRUE, per
           IEEE Std 802.3, Figure 33-9.";
        reference
          "IEEE Std 802.3, 30.9.1.1.17";
      }
      leaf short {
        type yang:counter64;
        status deprecated;
        description
          "This Yang object is deprecated as its not defined in
           base standard.
           This counter is incremented when the PSE state
           diagram enters the ERROR_DELAY state due to the
           short_detected variable being TRUE, per
           IEEE Std 802.3, Figure 33-9.";
        reference
          "IEEE Std 802.3, 30.9.1.1.10 aPSEShortCounter";
      }
      leaf cumulative-energy {
        type yang:counter64;
        units "millijoules";
        description
          "The cumulative energy supplied by the PSE as
           measured at the MDI in millijoules.";
        reference
          "IEEE Std 802.3, 30.9.1.1.25";
      }
    }
    leaf actual-power {
      type decimal64 {
        fraction-digits 4;
      }
      units "milliwatts";
      config false;
      description
        "The actual power drawn by a PD over the port.";
      reference
        "IEEE Std 802.3, 30.9.1.1.23";
    }
    leaf power-accuracy {
      type int64;
      units "milliwatts";
      config false;
```

```
          description
            "An integer value indicating the accuracy
             associated with power-accuracy in +/- milliwatts.";
          reference
            "IEEE Std 802.3, 30.9.1.1.24";
        }
      }
      container single-pair {
        presence "PSE port working in PoDL.";
        status deprecated;
        description
          "Deprecated in IEEE Std 802.3.2-202x, Draft D3.1, superceeded
by
           ieee802-ethernet-pse-2
           PoDL PSE configuration as defined in
           IEEE Std 802.3, 30.15.1.";
        leaf pse-enable {
          type boolean;
          default "false";
          status deprecated;
          description
            "Deprecated in IEEE Std 802.3.2-202x, Draft D3.1, superceeded
by
             ieee802-ethernet-pse-2
             When true enables the PSE function on the interface,
             when false disables the PSE function on the
             interface.";
          reference
            "IEEE Std 802.3, 30.15.1.1.2 aPoDLPSEAdminState";
        }
        leaf detection-status {
          type single-pair-detection-state;
          config false;
          status deprecated;
          description
            "Deprecated in IEEE Std 802.3.2-202x, Draft D3.1, superceeded
by
             ieee802-ethernet-pse-2
             Indicates the current status of the PoDL PSE.";
          reference
            "IEEE Std 802.3, 30.15.1.1.3
             aPoDLPSEPowerDetectionStatus";
        }
        leaf podl-type {
          type enumeration {
            enum unknown {
              description
                "Unknown PSE type.";
            }
            enum typeA {
              description
                "TypeA PSE";
            }
            enum typeB {
              description
                "TypeB PSE";
            }
            enum typeC {
```

```
        description
          "Type PSEC";
      }
      enum typeD {
        description
          "TypeD PSE";
      }
      enum typeE {
        description
          "TypeE PSE";
      }
      enum typeF {
        description
          "TypeF PSE";
      }
    }
    config false;
    description
      "PSE type specified in and
       IEEE Std 802.3,  30.15.1.1.4.";
  }
  leaf detected-pd-type {
    when "../detection-status = 'deliveringPower'" {
      description
        "This node only applies when the detection status is
         delivering power.";
    }
    type enumeration {
      enum unknown {
        description
          "Unknown PD type";
      }
      enum typeA {
        description
          "TypeA PD";
      }
      enum typeB {
        description
          "TypeB PD";
      }
      enum typeC {
        description
          "TypeC PD";
      }
      enum typeD {
        description
          "TypeD PD";
      }
      enum typeE {
        description
          "TypeE PD";
      }
      enum typeF {
        description
          "TypeF PD";
      }
    }
    config false;
```

```
            description
              "Indicates the Type of the detected PoDL PD as
               specified in IEEE Std 802.3, 104.5.1.";
            reference
              "IEEE Std 802.3, 30.15.1.1.5 aPoDLPSEDetectedPDType";
          }
          leaf pd-power-class {
            when "../detection-status = 'deliveringPower'" {
              description
                "This node only applies when the detection status is
                 delivering power.";
            }
            type power-class;
            config false;
            status deprecated;
            description
              "Deprecated in IEEE Std 802.3.2-202x, Draft D3.1, superceeded
by
               ieee802-ethernet-pse-2
               Power class of the PD detected on the PSE port.";
            reference
              "IEEE Std 802.3, 30.15.1.1.6
               aPoDLPSEDetectedPDPowerClass";
          }
          container statistics {
            config false;
            status deprecated;
            description
              "~~Statistics information of the single-pair~~Deprecated in IEEE
Std 802.3.2-202x, Draft D3.1, superceeded by
               ieee802-ethernet-pse-2
               PSE Discontinuities in the values of counters in this
               container can occur at re-initialization of the
               management system, and at other times as indicated by the
               value of the 'discontinuity-time' leaf  defined in the
               ietf-interfaces YANG module (IETF RFC 8343).";
            leaf power-denied {
              type yang:counter64;
              description
                "This counter is incremented when the PoDL PSE state
                 diagram variable power_available transitions from
                 true to false (see IEEE Std 802.3, 104.4.3.3).";
              reference
                "IEEE Std 802.3, 30.15.1.1.9
              ~~\t~~   aPoDLPSEPowerDeniedCounter";
            }
            leaf invalid-signature {
              type yang:counter64;
              description
                "This counter is incremented when the PSE state
                 diagram enters the SIGNATURE_INVALID state per
                 IEEE Std 802.3, Figure 33-9.";
              reference
                "IEEE Std 802.3, 30.15.1.1.7
                 aPoDLPSEInvalidSignatureCounter";
            }
            leaf invalid-class {
              type yang:counter64;
```

```
    description
      "This counter is incremented when the PoDL PSE state
       diagram variable tclass_timer_done transitions from
       false to true or when the valid_class variable
       transitions from true to false
       (see IEEE Std 802.3, 104.4.3.3).";
    reference
      "IEEE Std 802.3, 30.15.1.1.8
       aPoDLPSEInvalidClassCounter";
  }
  leaf overload {
    type yang:counter64;
    description
      "This counter is incremented when the PSE state
       diagram variable overload_held transitions from
       false to true (see IEEE Std 802.3, 104.4.3.3).";
    reference
      "IEEE Std 802.3, 30.15.1.1.10
       aPoDLPSEOverLoadCounter";
  }
  leaf fvs-absence {
    type yang:counter64;
    description
      "Maintain Full Voltage Signature absent counter.
       This counter is incremented when the PoDL PSE state
       diagram variable mfvs_timeout transitions from false
       to true (see IEEE Std 802.3, 104.4.3.3).";
    reference
      "IEEE Std 802.3, 30.15.1.1.11
       aPoDLPSEMaintainFullVoltageSignatureAbsentCounter";
  }
  leaf cumulative-energy {
    type yang:counter64;
    units "millijoules";
    description
      "A count of the cumulative energy supplied by the
       PoDL PSE, measured at the MDI, and expressed in
       units of millijoules.";
    reference
      "IEEE Std 802.3, 30.15.1.1.14
       aPoDLPSECumulativeEnergy";
  }
}
leaf actual-power {
  type decimal64 {
    fraction-digits 4;
  }
  units "milliwatts";
  config false;
  description
    "An integer value indicating present (actual) power
     being supplied by the PoDL PSE as measured at the MDI
     in milliwatts.";
  reference
    "IEEE Std 802.3, 30.15.1.1.12 aPoDLPSEActualPower";
}
leaf power-accuracy {
  type int64;
```

```
          units "milliwatts";
          config false;
          description
            "A signed integer value indicating the accuracy
             associated
             with power-accuracy in milliwatts.";
          reference
            "IEEE Std 802.3, 30.15.1.1.13 aPoDLPSEPowerAccuracy";
        }
      }
    }
  }
}
```